



US012386869B2

(12) **United States Patent**
Williams et al.

(10) **Patent No.:** **US 12,386,869 B2**

(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **IMAGE PROCESSING UTILIZING AN
ENTIGEN CONSTRUCT**

(71) Applicant: **entigenlogic LLC**, Schaumburg, IL
(US)

(72) Inventors: **Frank John Williams**, Rossmoor, CA
(US); **David Ralph Lazzara**,
Huntington Beach, CA (US); **Stephen
Emerson Sundberg**, Chicago, IL (US);
Ameeta Vasant Reed, Deerfield, IL
(US); **Dennis Arlen Roberson**, Chapin,
SC (US); **Thomas James MacTavish**,
Inverness, IL (US); **Karl Olaf
Knutson**, Palatine, IL (US); **Jessy
Thomas**, Palatine, IL (US); **Niklas
Josiah MacTavish**, Inverness, IL (US);
David Michael Corns, II, Elgin, IL
(US); **Andrew Chu**, St. Charles, IL
(US); **Theodore Mazurkiewicz**, Lake
Zurich, IL (US); **Gary W. Grube**,
Barrington Hills, IL (US)

(73) Assignee: **entigenlogic LLC**, Schaumburg, IL
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 384 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **17/970,112**

(22) Filed: **Oct. 20, 2022**

(65) **Prior Publication Data**

US 2023/0043805 A1 Feb. 9, 2023

Related U.S. Application Data

(63) Continuation-in-part of application No. 16/696,114,
filed on Nov. 26, 2019, now Pat. No. 11,544,320.

(Continued)

(51) **Int. Cl.**
G06V 10/00 (2022.01)
G06F 16/332 (2019.01)

(Continued)

(52) **U.S. Cl.**
CPC **G06F 16/3322** (2019.01); **G06F 40/247**
(2020.01); **G06N 5/04** (2013.01)

(58) **Field of Classification Search**
CPC **G06F 16/3322**; **G06F 40/247**; **G06F 40/30**;
G06N 5/022; **G06N 5/04**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,538,188 A 8/1985 Barker
6,021,220 A 2/2000 Anderholm

(Continued)

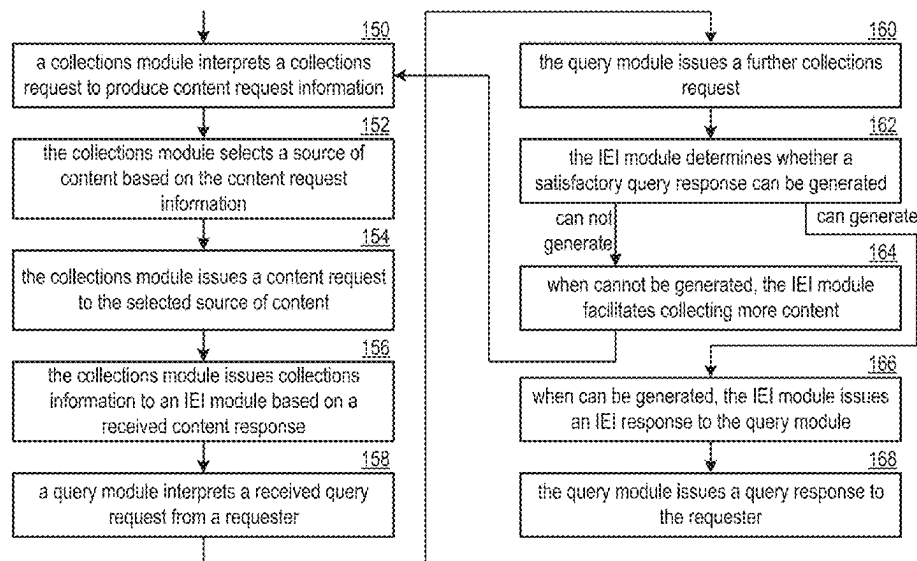
Primary Examiner — Amir Alavi

(74) *Attorney, Agent, or Firm* — Gary W. Grube

(57) **ABSTRACT**

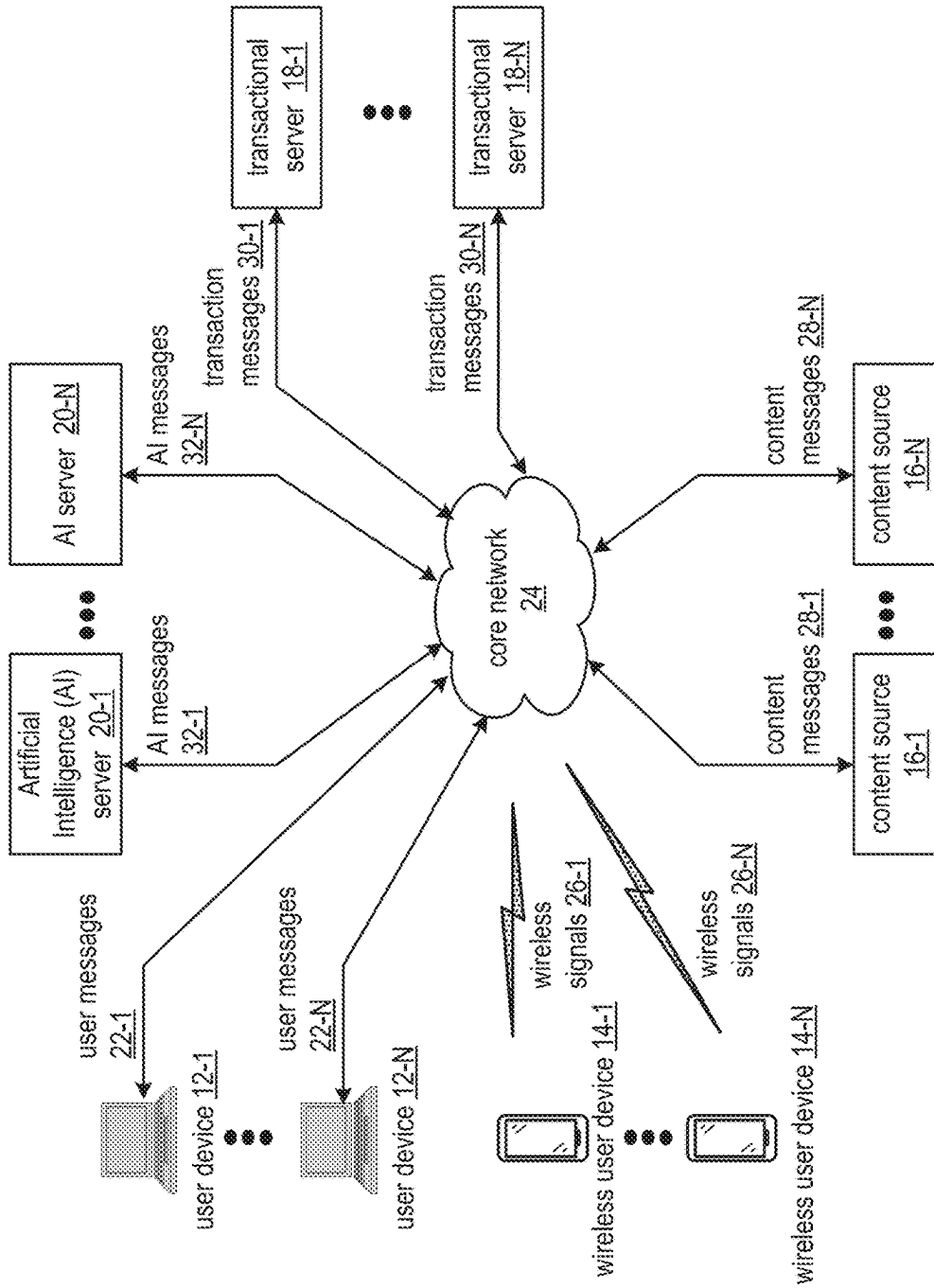
A method performed by a computing device includes deter-
mining a set of identigens for each word of a query of a topic
to produce sets of identigens. Each set of identigens repre-
sents one or more different meanings of a word of the query.
The method further includes interpreting, using identigen
pairing rules, the sets of identigens to determine a most
likely meaning interpretation of the query and produce an
excluding query entigen group with an excluding entigen.
The method further includes recovering a response entigen
group for the query from a knowledge database utilizing the
excluding query entigen group. The response entigen group
provides a response to the query.

18 Claims, 47 Drawing Sheets



Related U.S. Application Data			
(60)	Provisional application No. 62/772,655, filed on Nov. 29, 2018.	10,390,055 B1	8/2019 Wilms
		10,943,075 B2 *	3/2021 Williams G06F 16/313
(51)	Int. Cl. G06F 40/247 (2020.01) G06N 5/04 (2023.01)	11,100,324 B2 *	8/2021 Williams G06T 7/11
		11,232,260 B2 *	1/2022 Williams G06F 16/3344
(56)	References Cited	11,288,583 B2 *	3/2022 Williams G06F 16/9035
		11,544,320 B2 *	1/2023 Williams G06V 10/764
	U.S. PATENT DOCUMENTS	11,748,563 B2 *	9/2023 Grube G06F 40/237
		12,019,628 B2 *	6/2024 Williams G06F 16/90344
		12,050,869 B2 *	7/2024 Williams G06N 5/04
		12,147,908 B2 *	11/2024 Williams G06N 5/041
		12,182,188 B2 *	12/2024 Williams G06F 16/338
		2003/0097251 A1	5/2003 Yamada
		2008/0208569 A1	8/2008 Simpson
		2016/0147979 A1	5/2016 Kato
		2019/0362145 A1	11/2019 Williams
		2020/0034358 A1	1/2020 Wurzel
		2020/0175064 A1	6/2020 Williams
		2021/0182604 A1	6/2021 Anthony
		2021/0287012 A1	9/2021 Li

* cited by examiner



computing system 10

FIG. 1

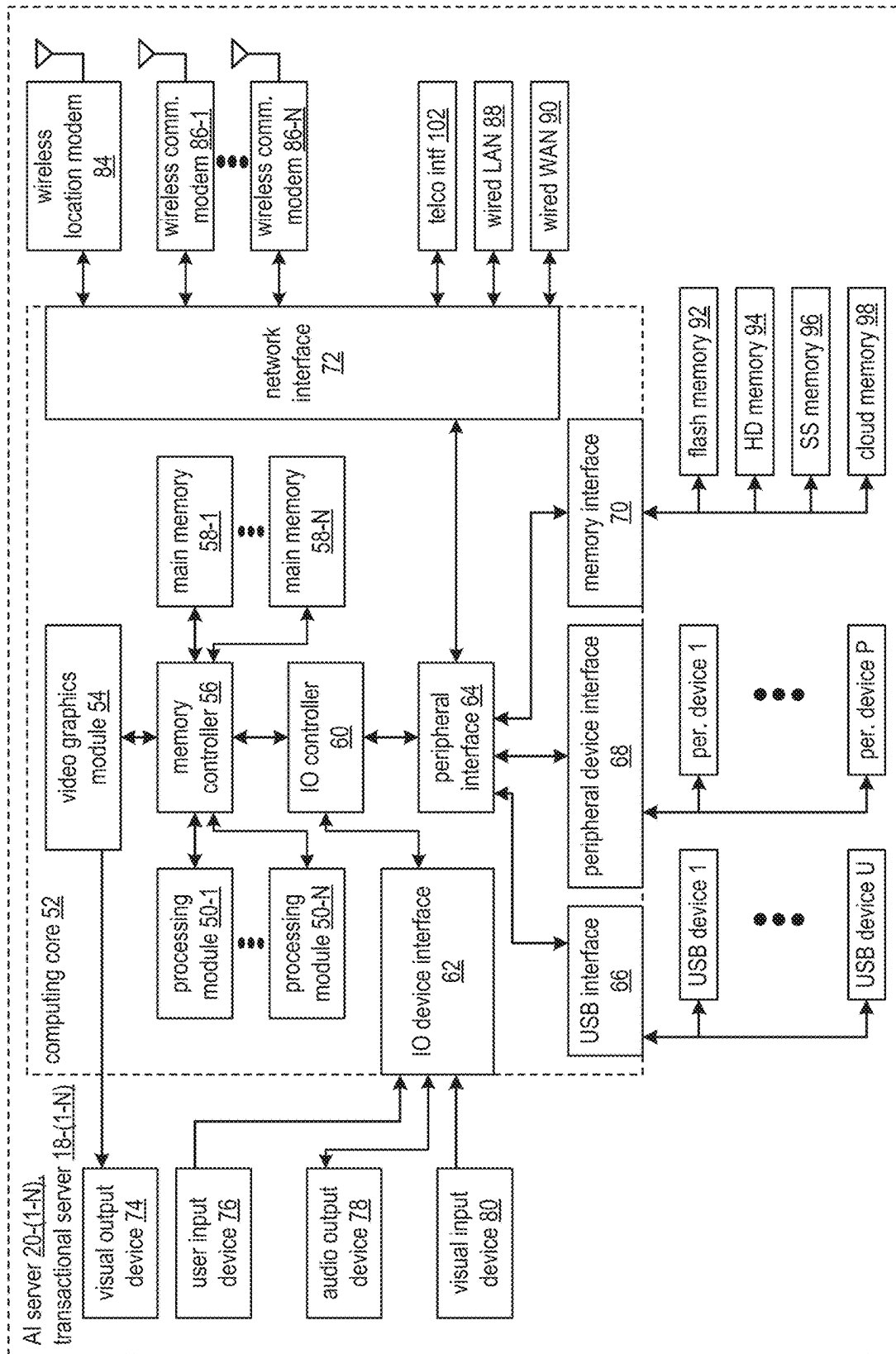


FIG. 2

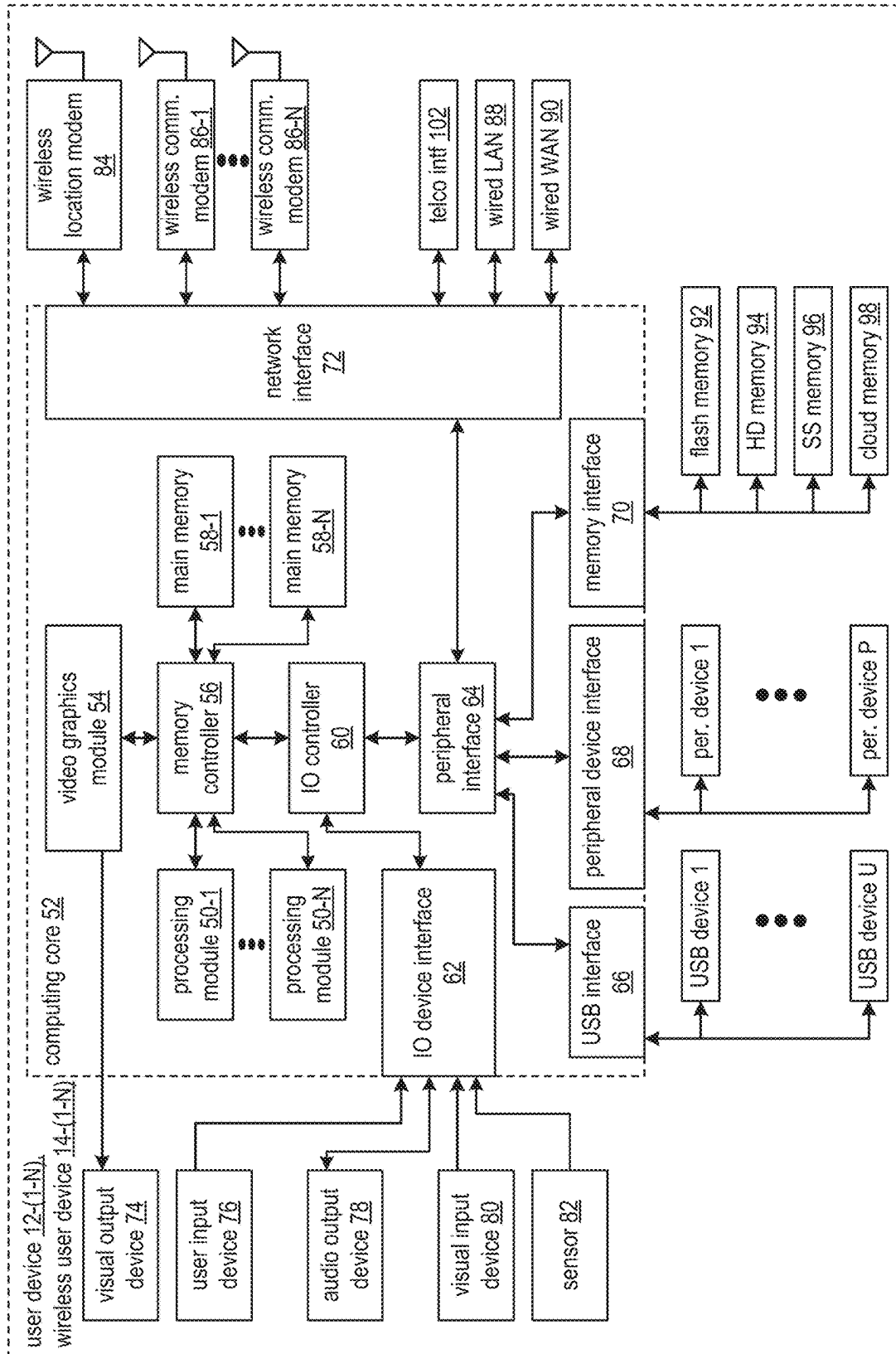


FIG. 3

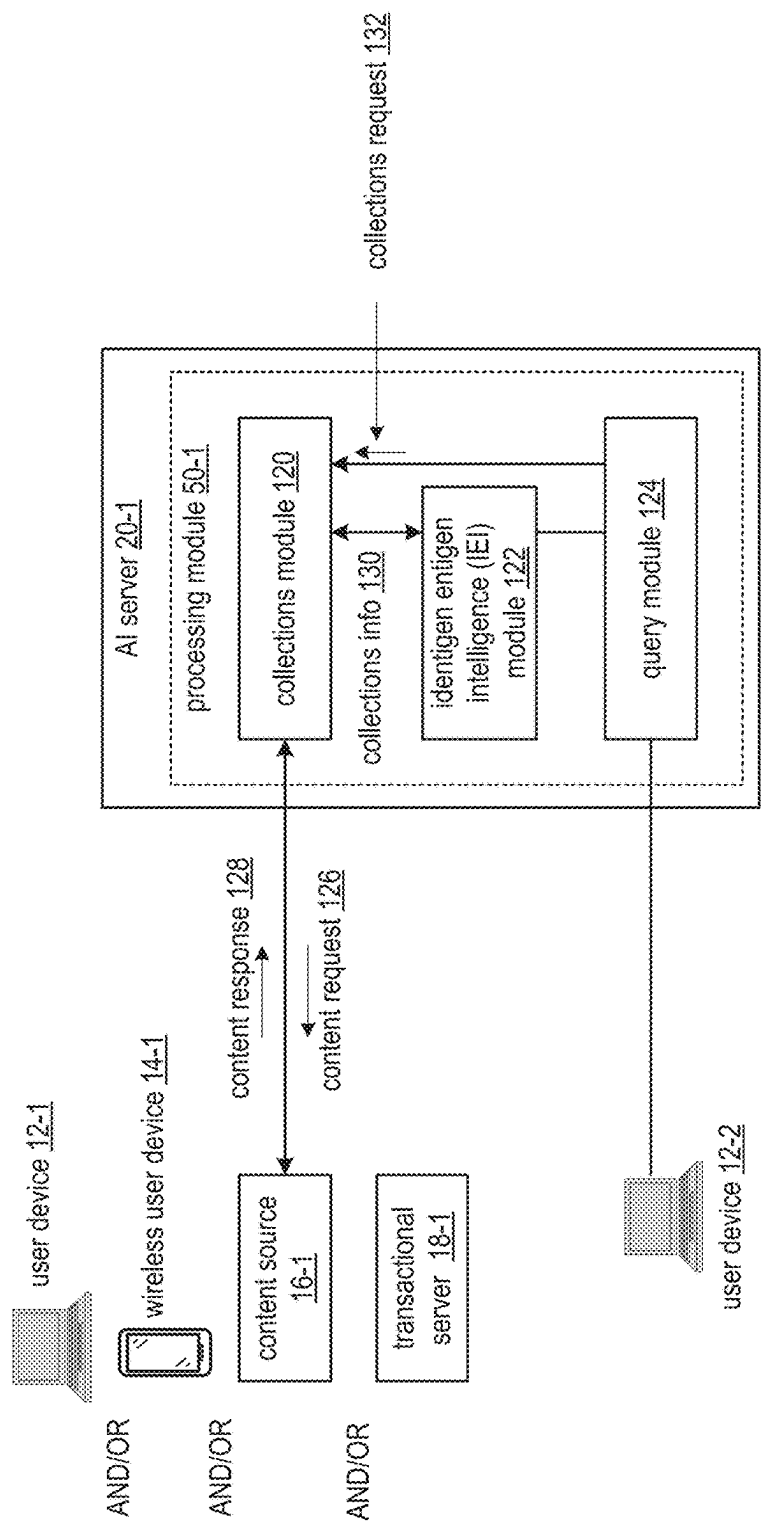


FIG. 4A

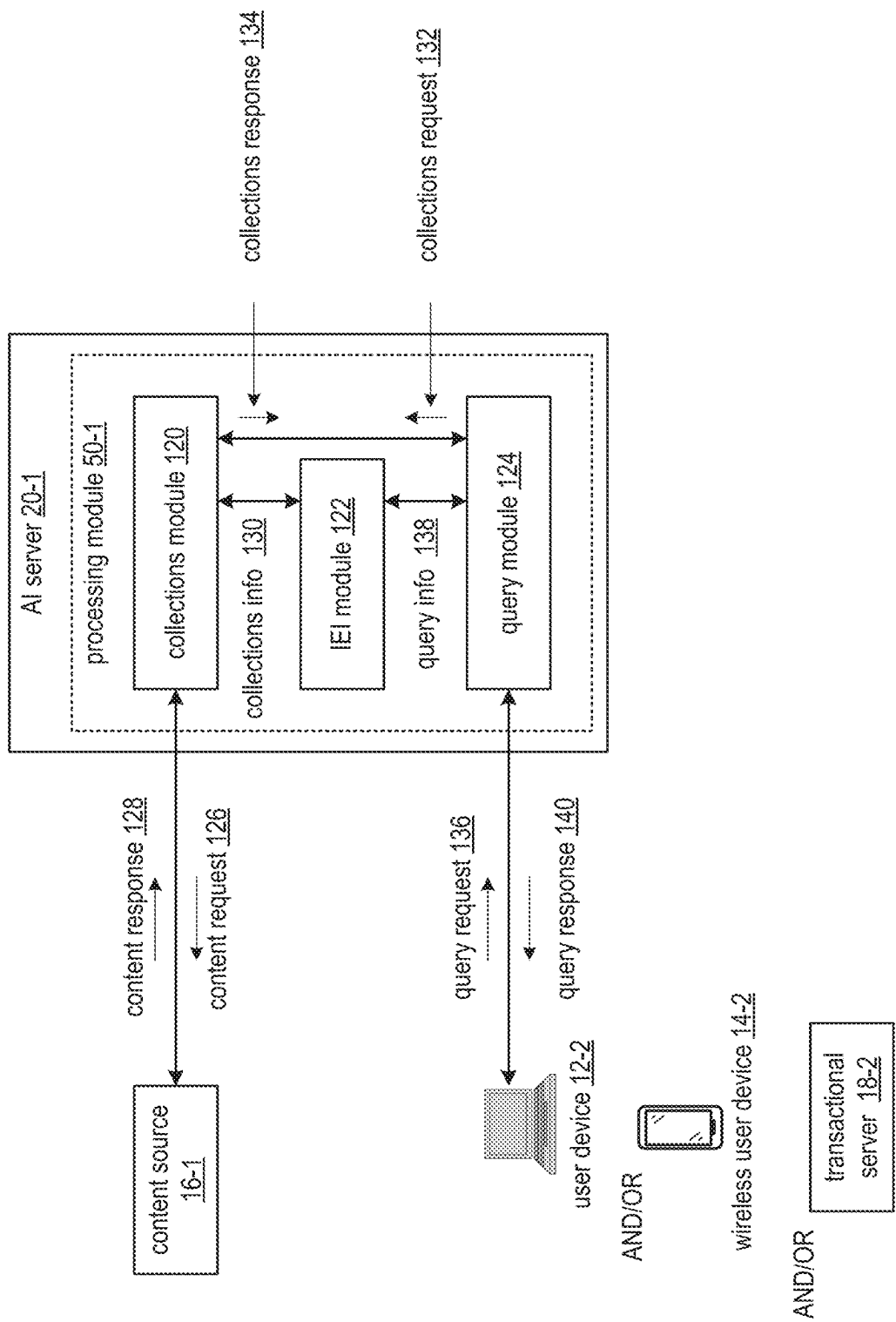


FIG. 4B

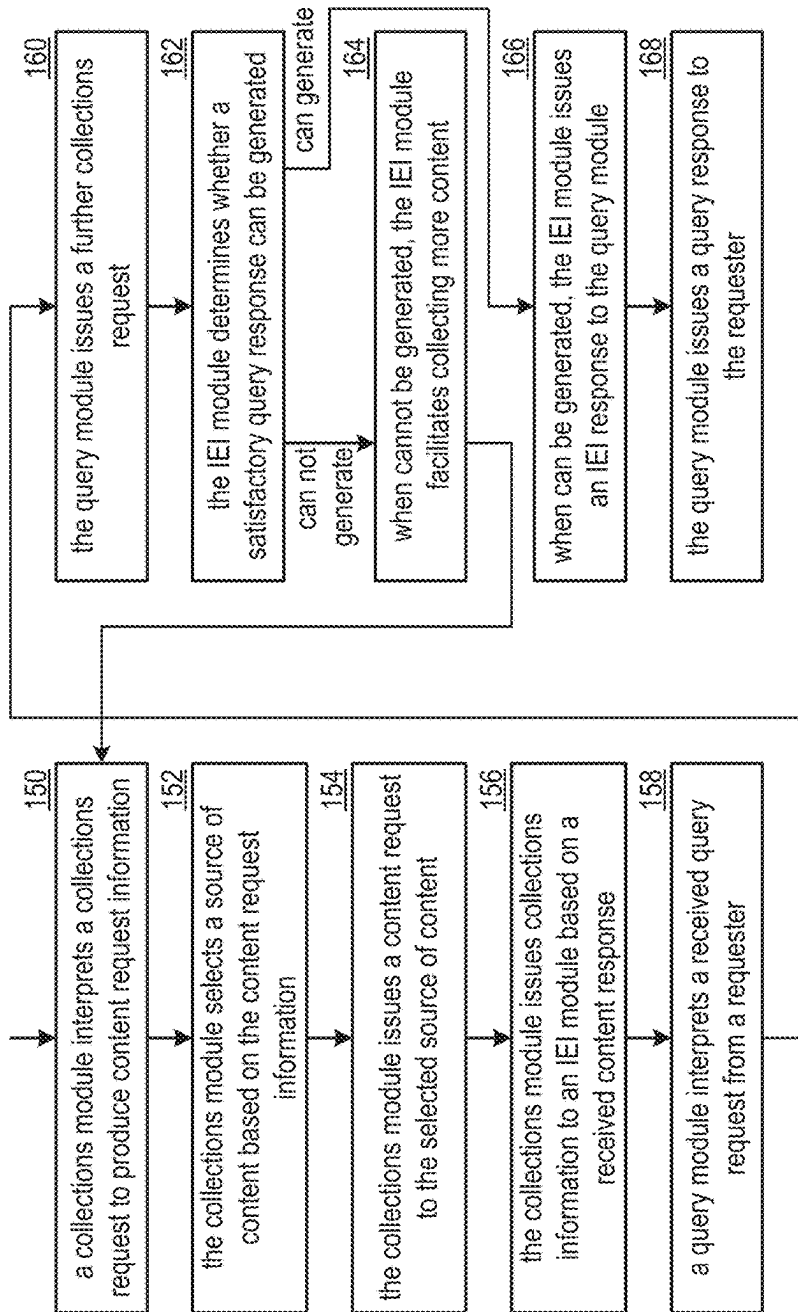


FIG. 4C

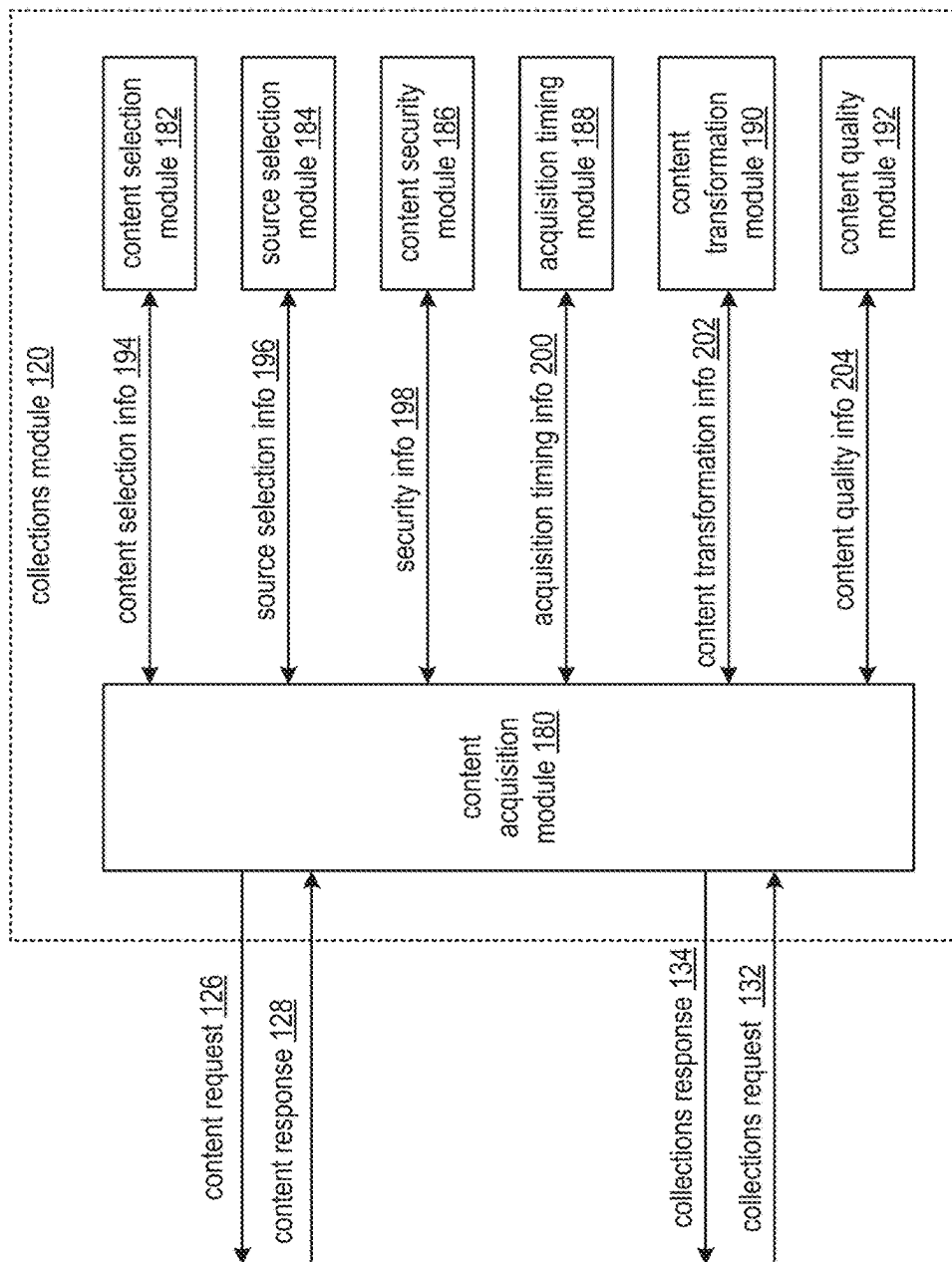
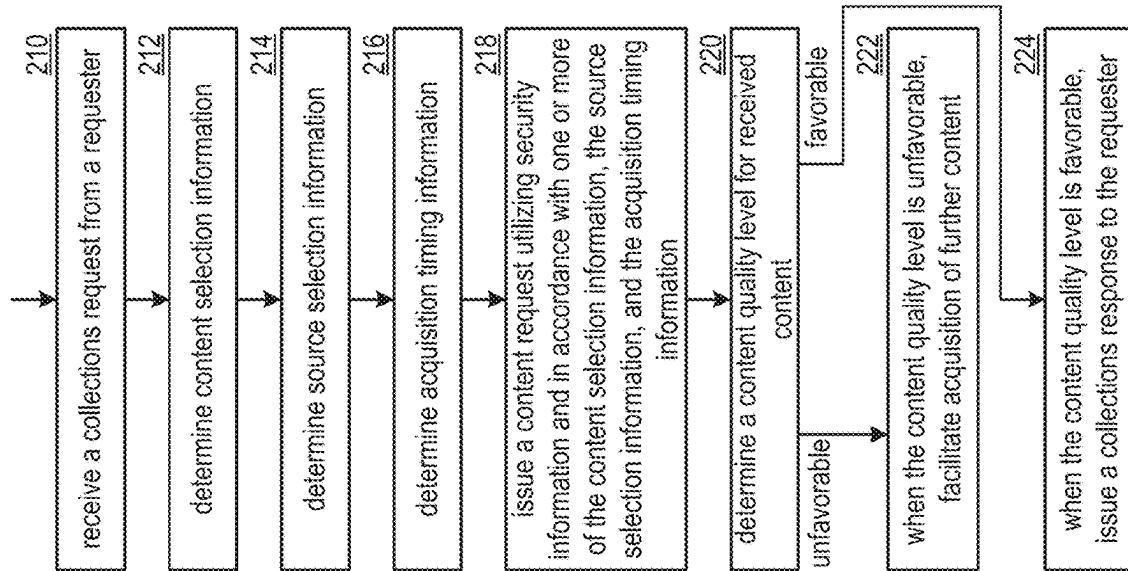


FIG. 5A

**FIG. 5B**

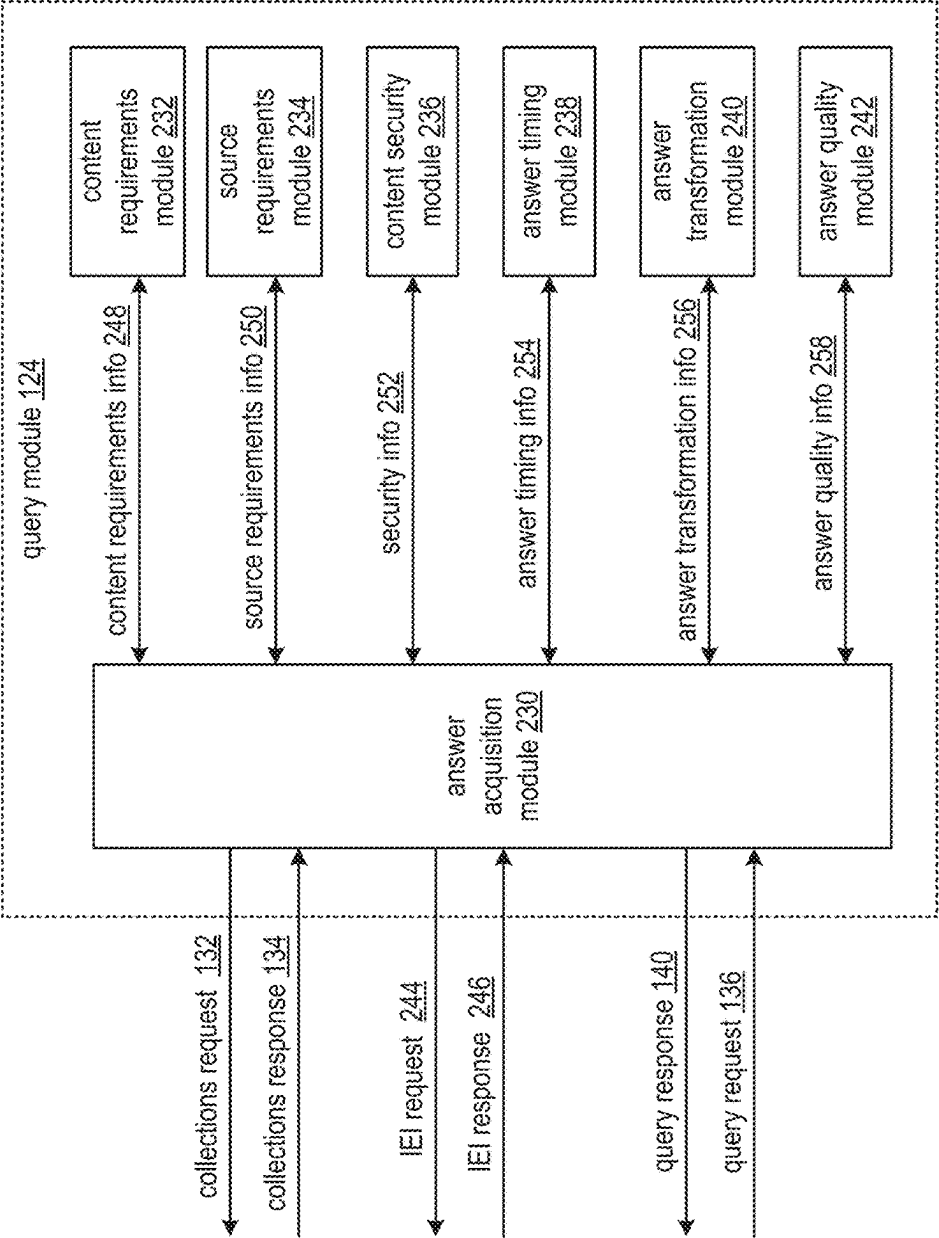


FIG. 5C

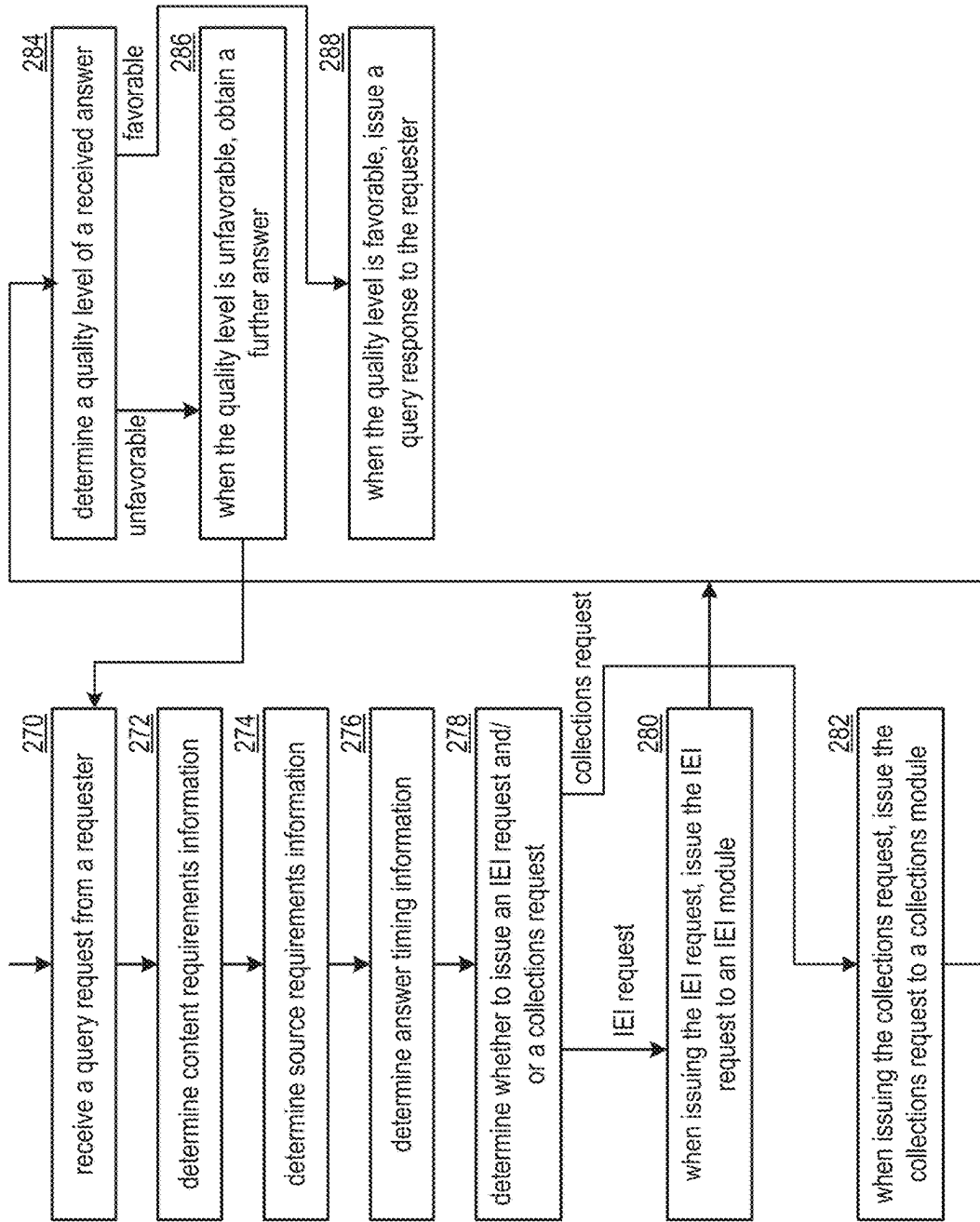


FIG. 5D

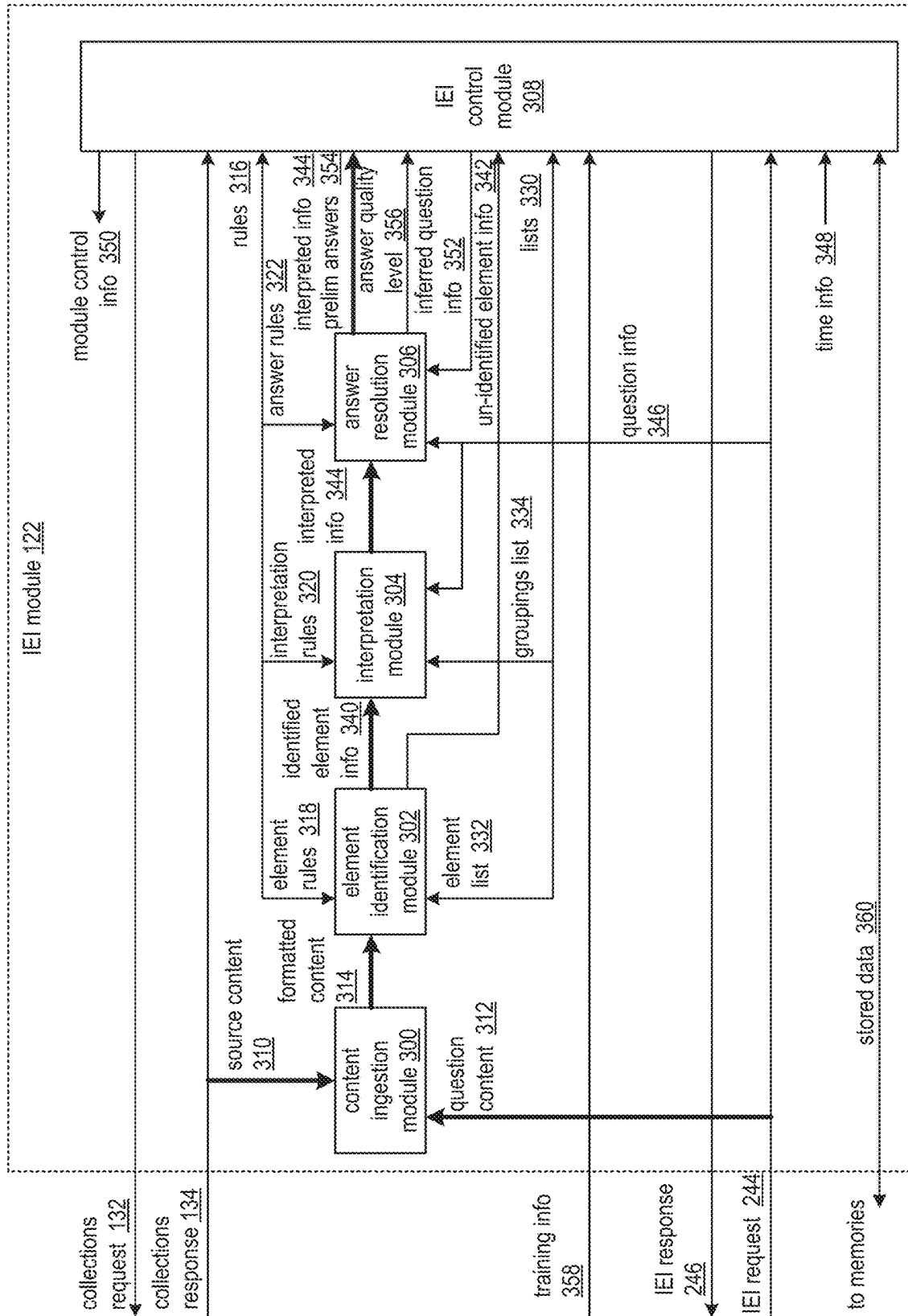
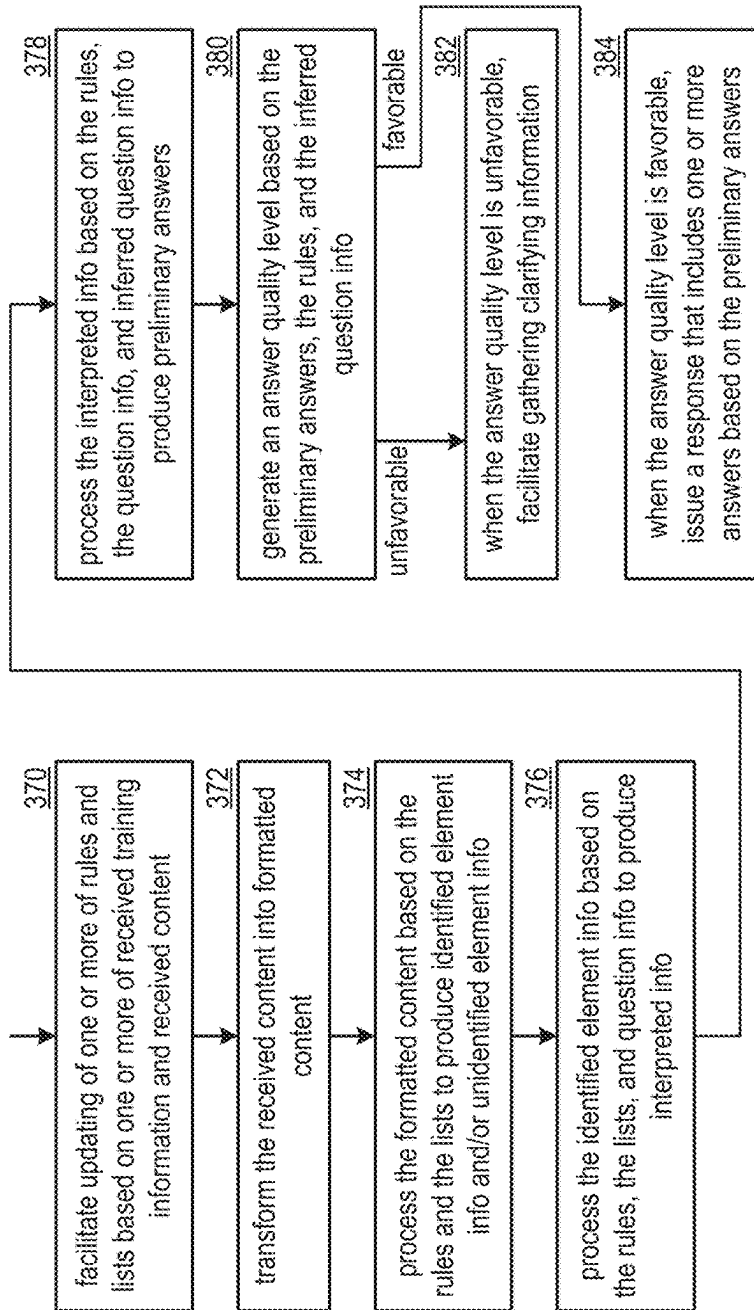


FIG. 5E

FIG. 5F

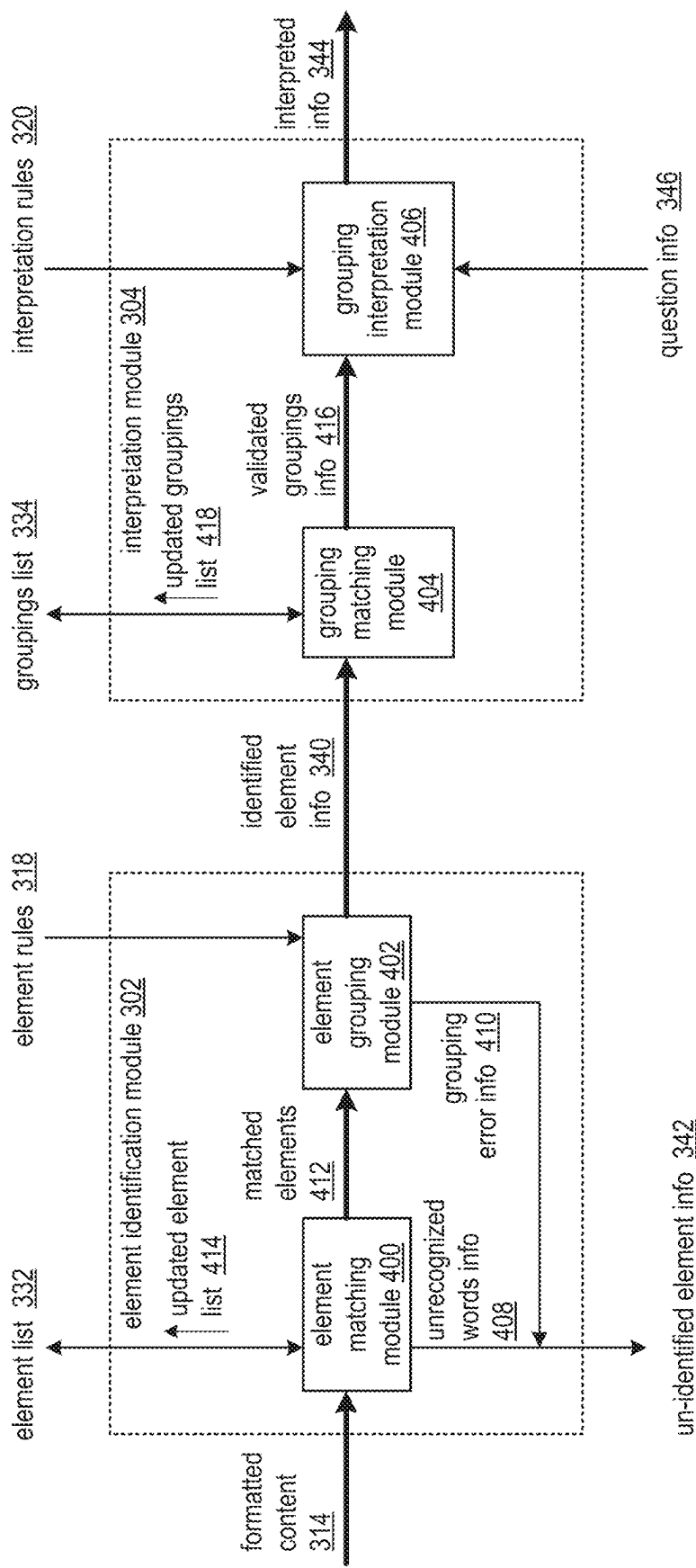
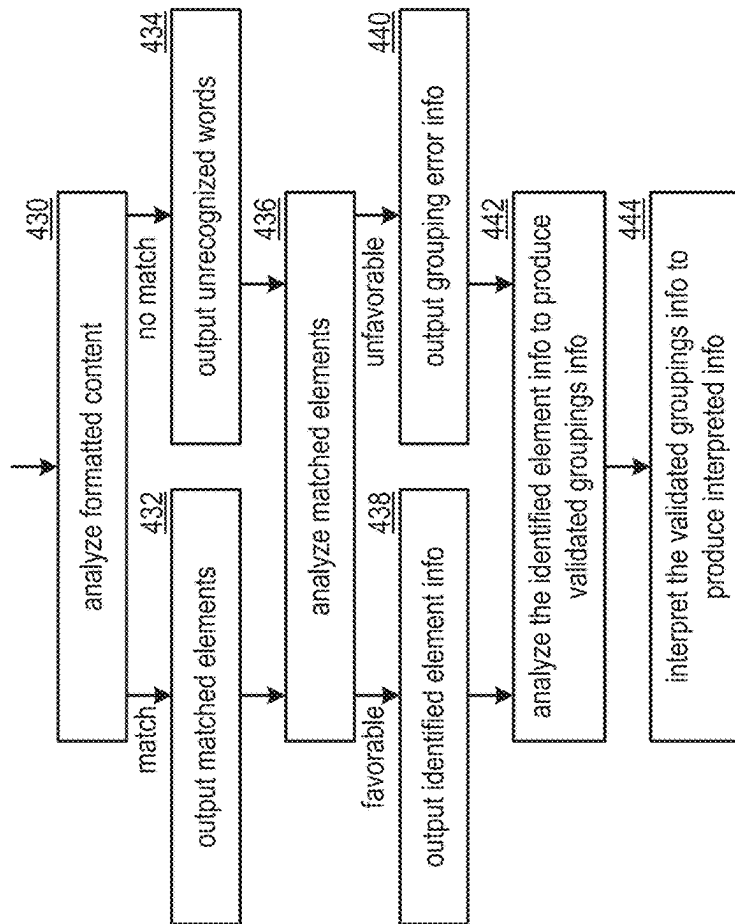


FIG. 6A

**FIG. 6B**

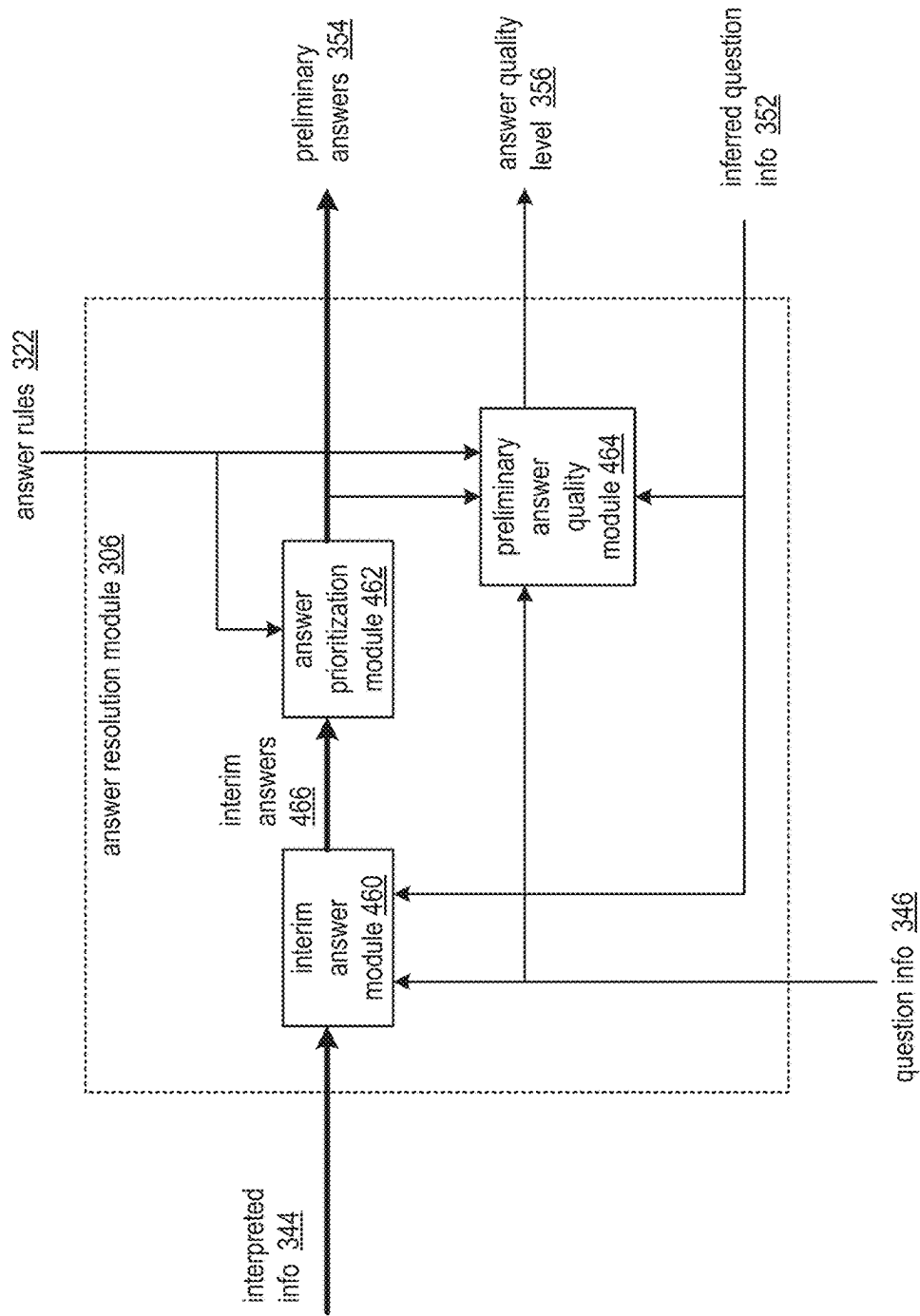
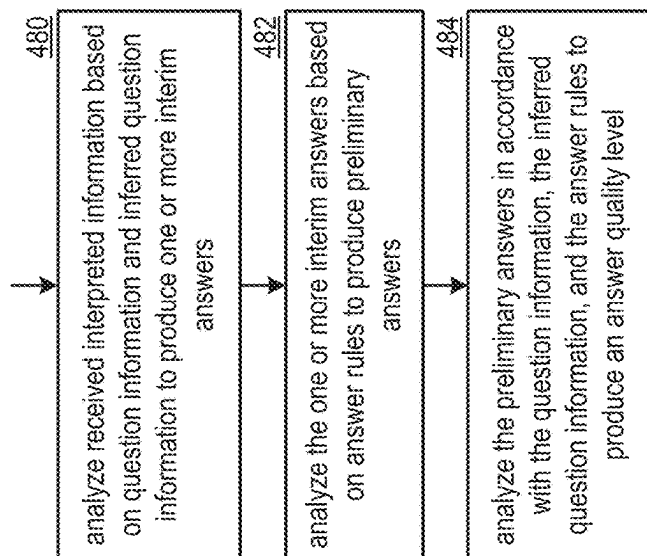


FIG. 6C

FIG. 6D

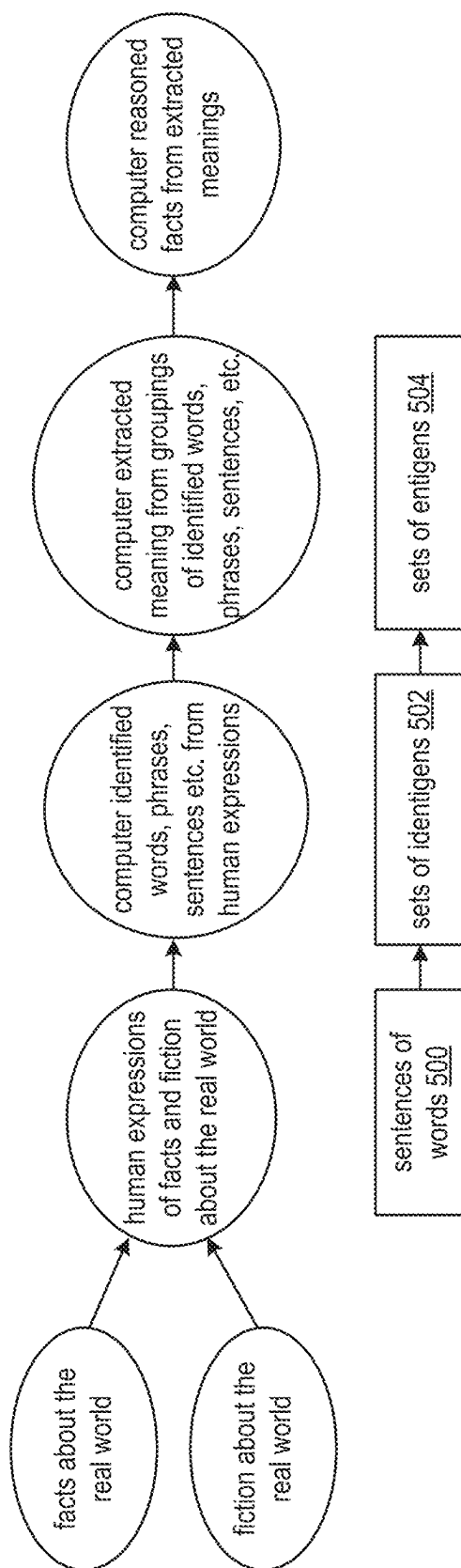


FIG. 7A

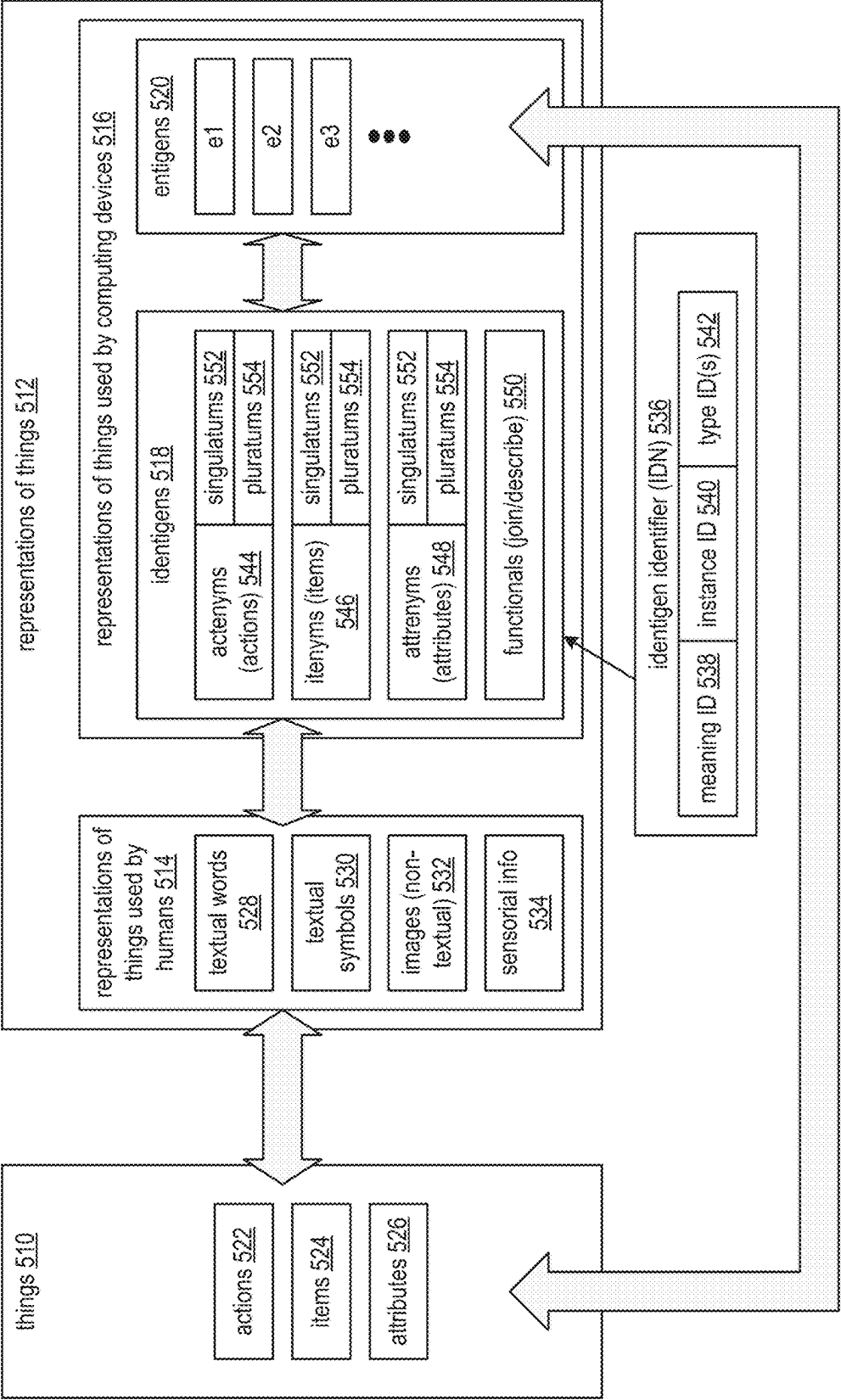


FIG. 7B

FIG. 7C

synonym words table 570			
textual words 572	identigen 518		entigens 520
	meaning ID 538	instance ID 540	
car	50	001	e1000
automobile		002	
auto		003	
bil (Swedish)		004	
carro (Spanish)		005	
bil (Danish)		006	

FIG. 7D

polysemous words table 576			
textual words 572	identigen 518		entigens 520
	meaning ID 538	instance ID 540	
duck (bird)	116937	001	e2000
duck (fabric)	116938	001	e2001
duck (to submerge)	112195	056	e2002
duck (to elude)	010654	051	e2003

words table 580									
textual words 572		identigen (IDN) 518			entigens (ENI)				
		meaning ID 538	instance ID 540	type ID 542					
pilot (flyer)		150	001	itenym (item)				520	
pilot (to fly)		291	001	actenym (action)				e717	
Tom (person)		457	001	itenym (item)				e4320	
Tom (male animal)		648	001	itenym (item)				e61	
tall (attribute - height)		823	001	aitrenym (attribute)				e930	
tall (attribute - unlikely)		399	001	aitrenym (attribute)				e90	
								e729	

→ apply rules to identigens of word strings to validate

pilot (to fly)	Tom (male an)	IDN 291.001	IDN 648.001	X	tall (attribute - height)	Tom (male an)	IDN 823.001	IDN 648.001	✓
pilot (flyer)	Tom (male an)	IDN 150.001	IDN 648.001	X	tall (attribute - unlikely)	Tom (male an)	IDN 399.001	IDN 648.001	X
pilot (to fly)	Tom (person)	IDN 291.001	IDN 457.001	X	tall (attribute - height)	Tom (person)	IDN 823.001	IDN 457.001	✓
pilot (flyer)	Tom (person)	IDN 150.001	IDN 457.001	✓	tall (attribute - unlikely)	Tom (person)	IDN 399.001	IDN 457.001	X

→ list valid groupings

groupings table 584									
grouping ID 586		word strings 588		identigens (IDN) 518			entigens (ENI) 520		
3001		pilot Tom		IDN 150.001	IDN 457.001		e717	e61	
3040		tall Tom		IDN 823.001	IDN 648.001		e90	e930	
3041		tall Tom		IDN 823.001	IDN 457.001		e90	e61	
3070		tall pilot Tom		IDN 823.001	IDN 150.001	IDN 457.001	e90	e717	e61

FIG. 7E

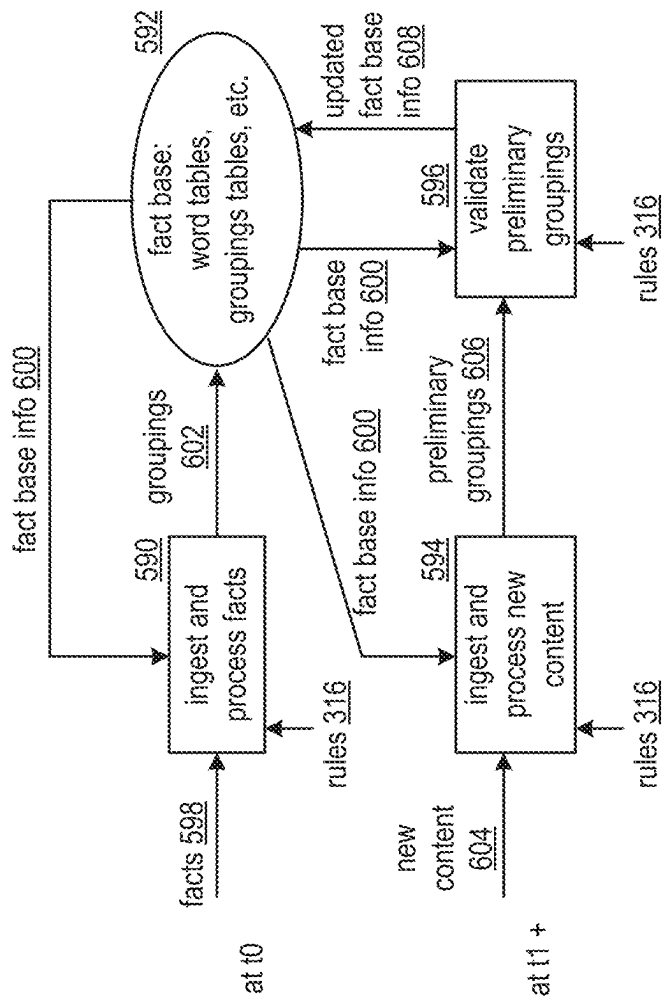


FIG. 8A

groupings table 620						
grouping (GRP) ID 586	word strings 588	IF string 622			THEN string 624	
		IDN string 626	ENI 628		IDN string 626	ENI 628
5493	If someone has a tumor, then someone is sick.	someone has tumor has	12aa 06aa	someone sick	12js	
5494	If someone is sick, then someone is possibly sad.	someone sick	12js	someone sad	12ja	
new knowledge inference 630						
5495	If someone has a tumor, then someone is possibly sad.	someone has	12aa	someone sad	12ja	
		tumor has	06aa	possibly sad	05b	

FIG. 8B

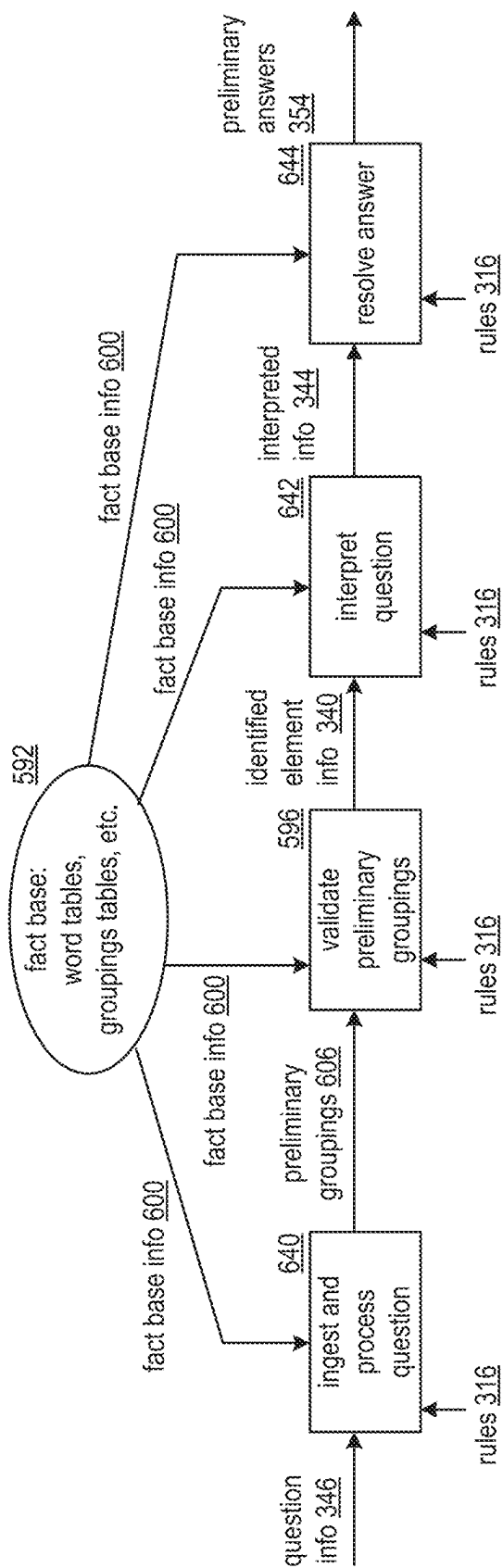


FIG. 8C

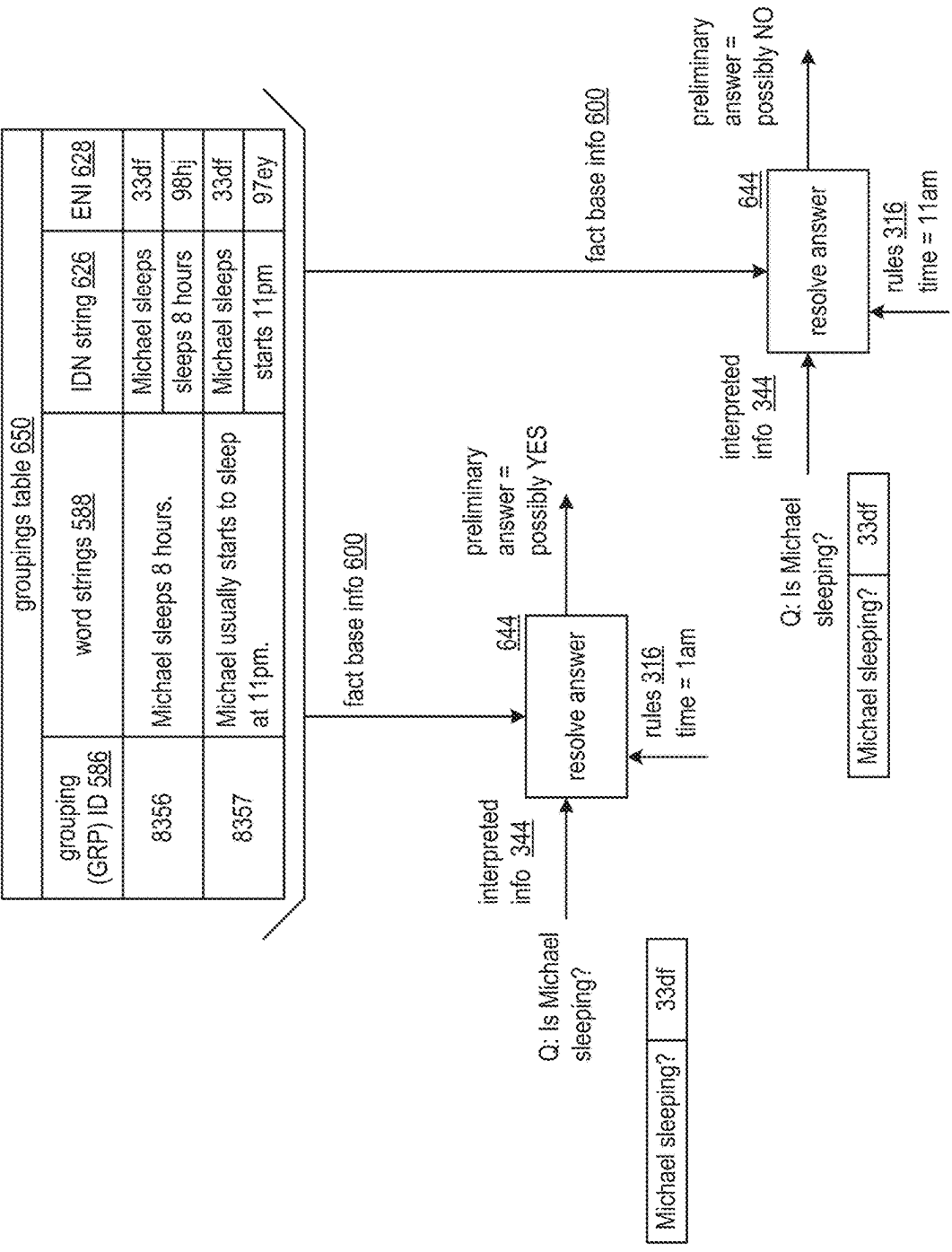


FIG. 8D

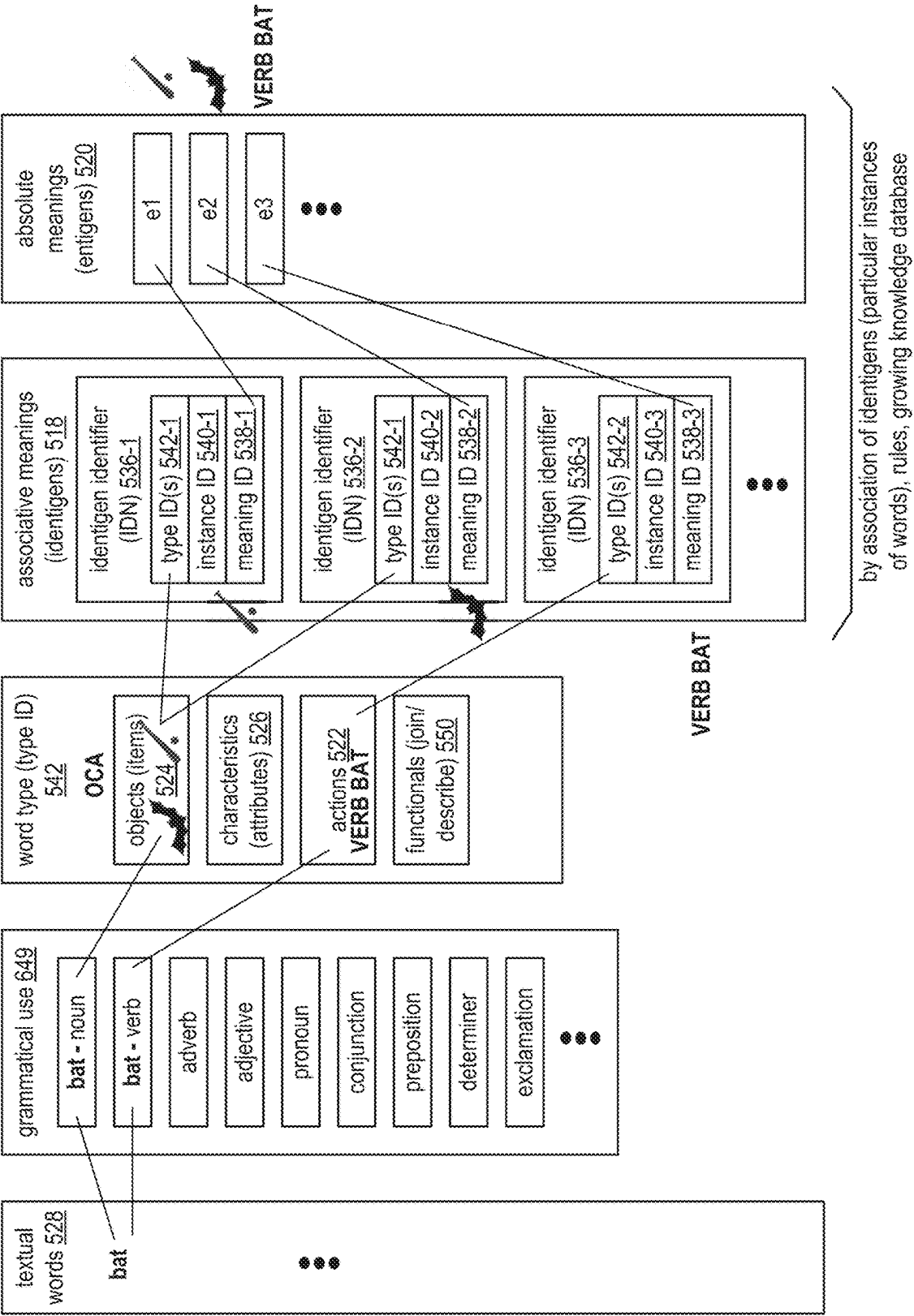


FIG. 8E

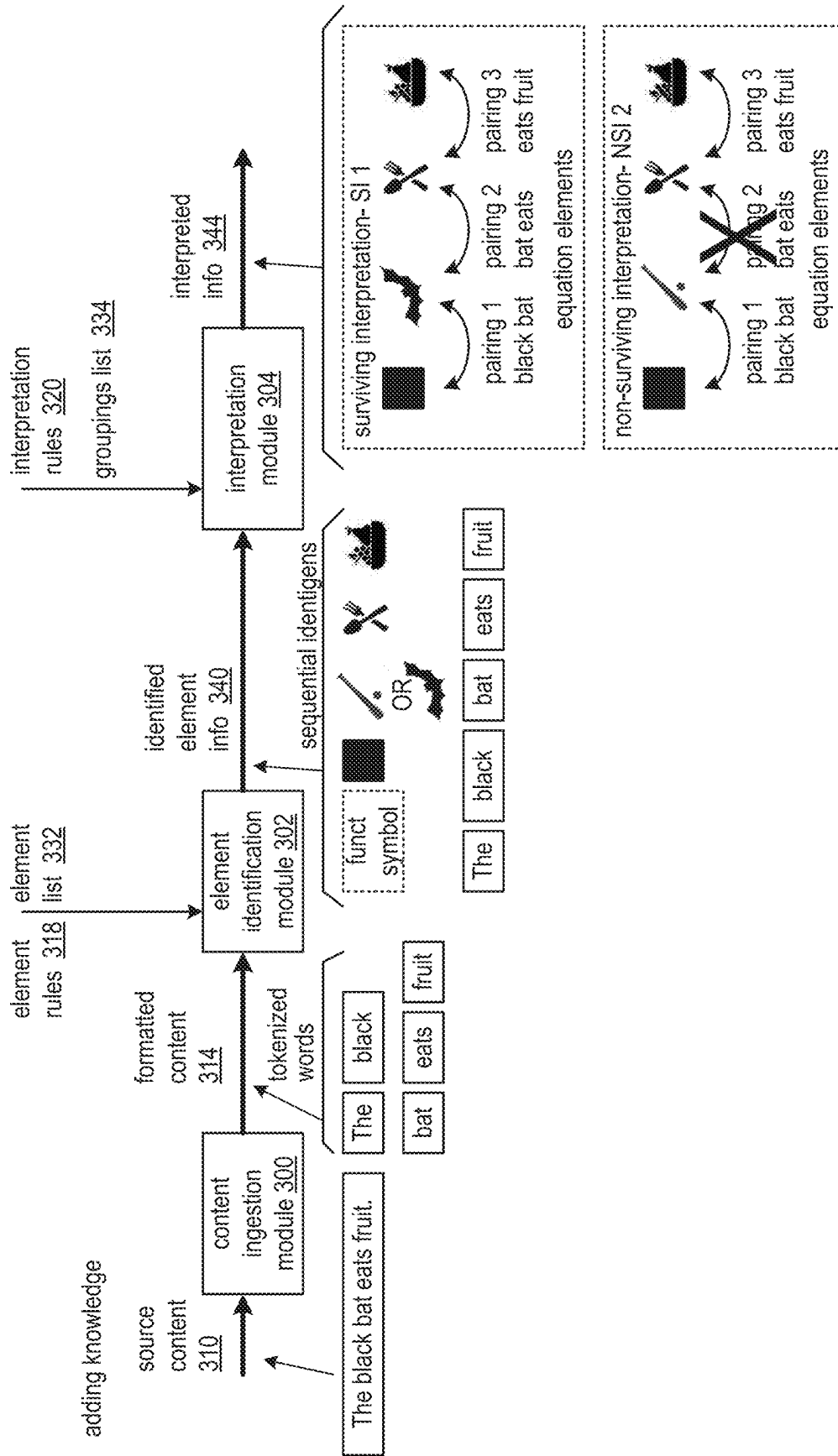
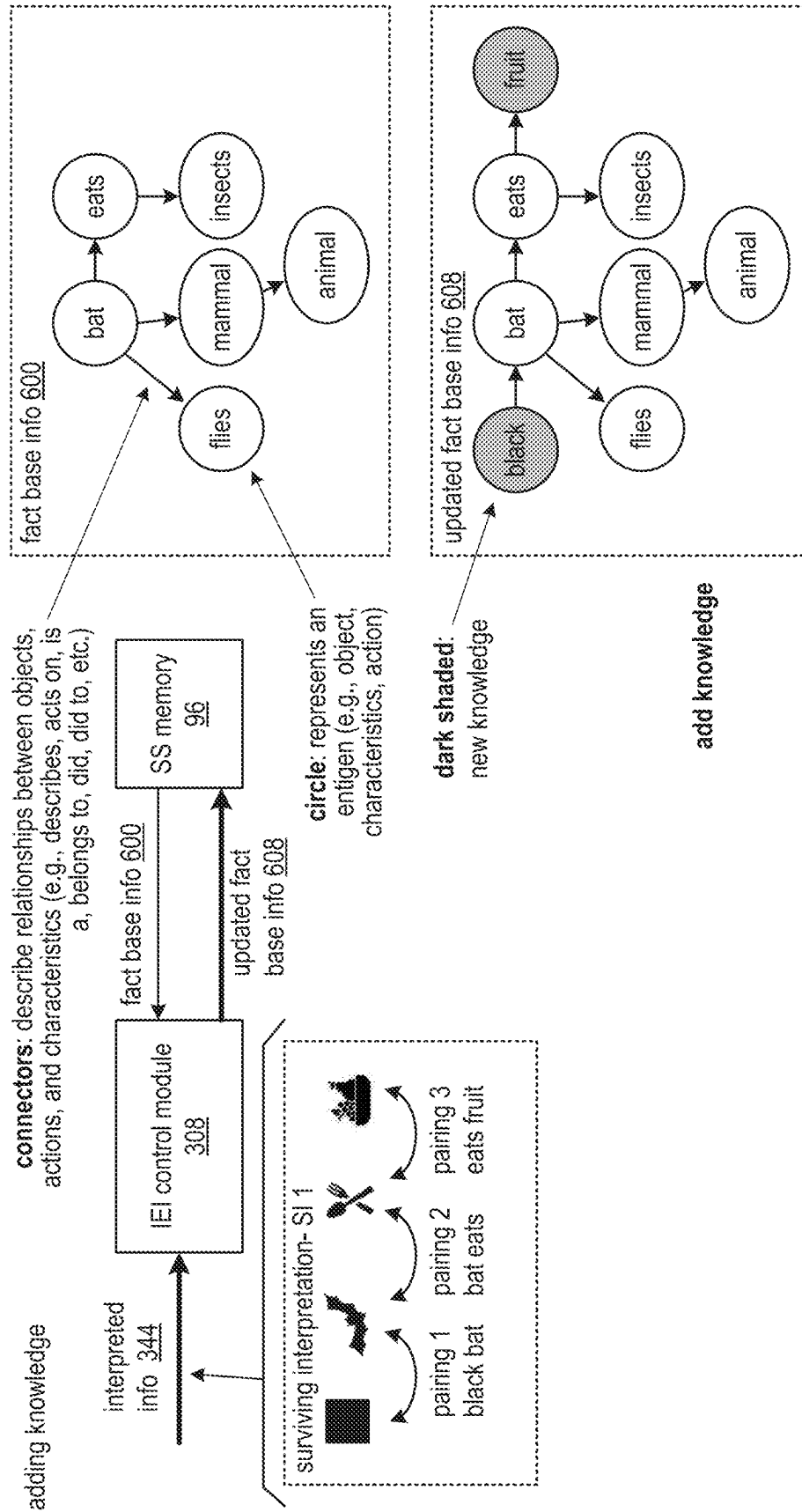
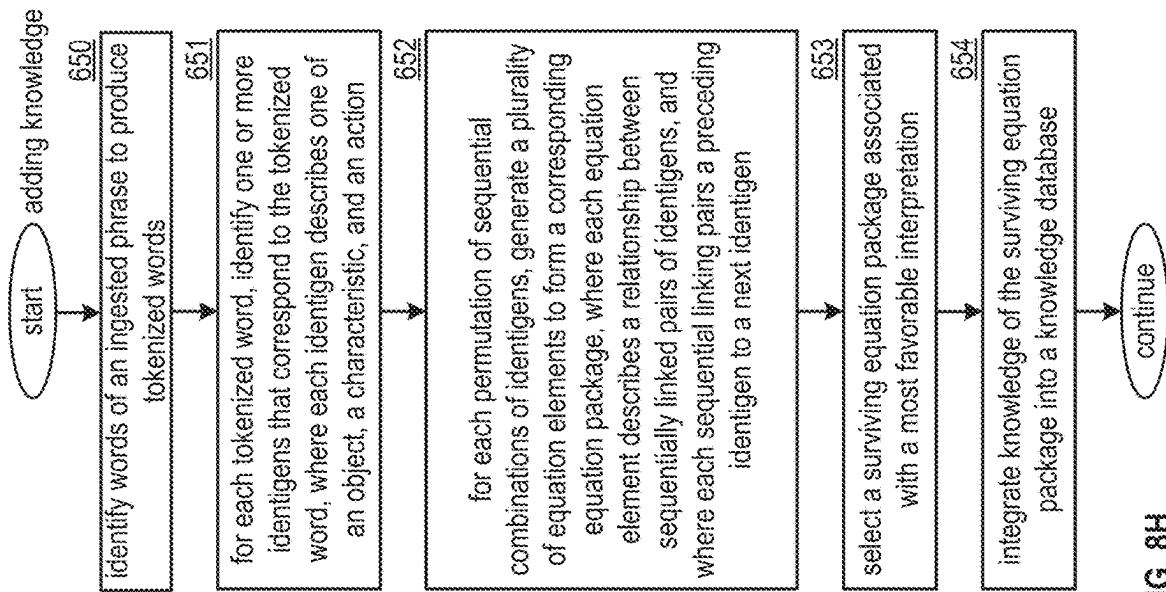
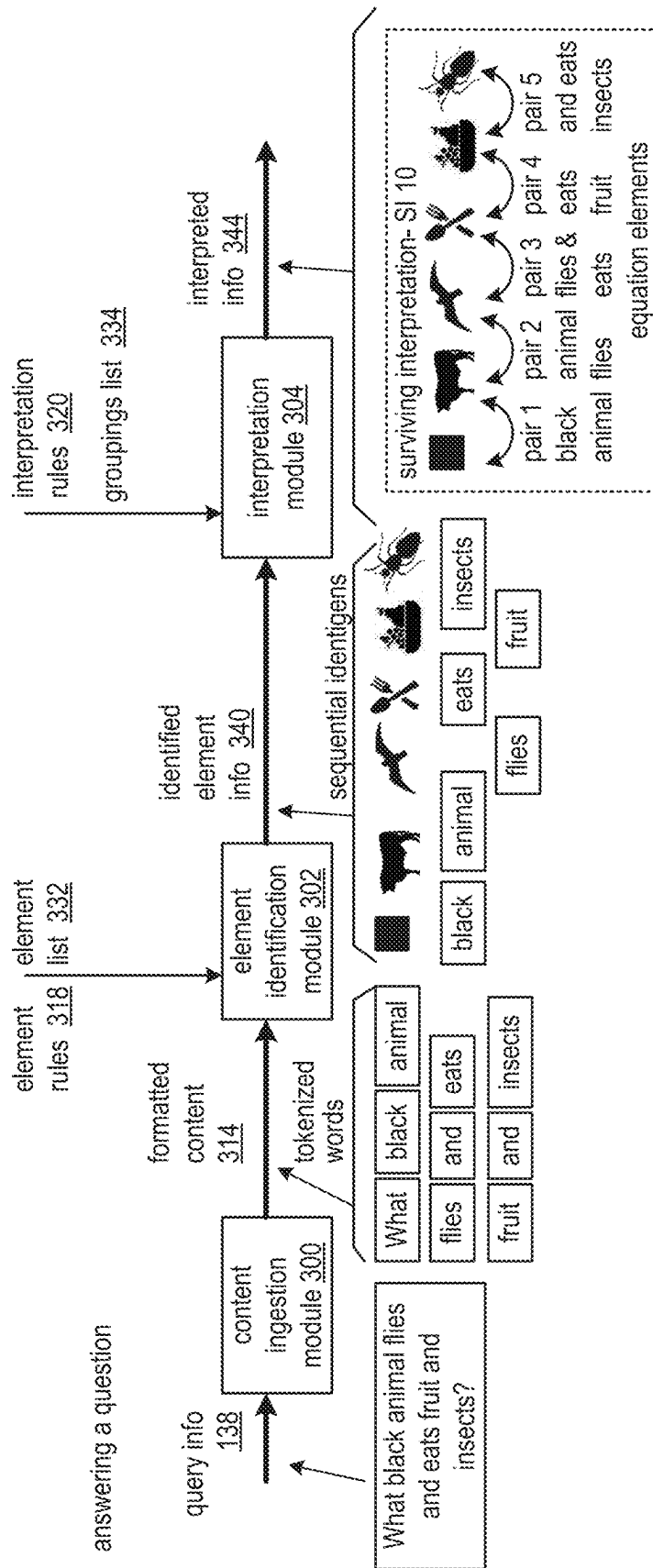


FIG. 8F







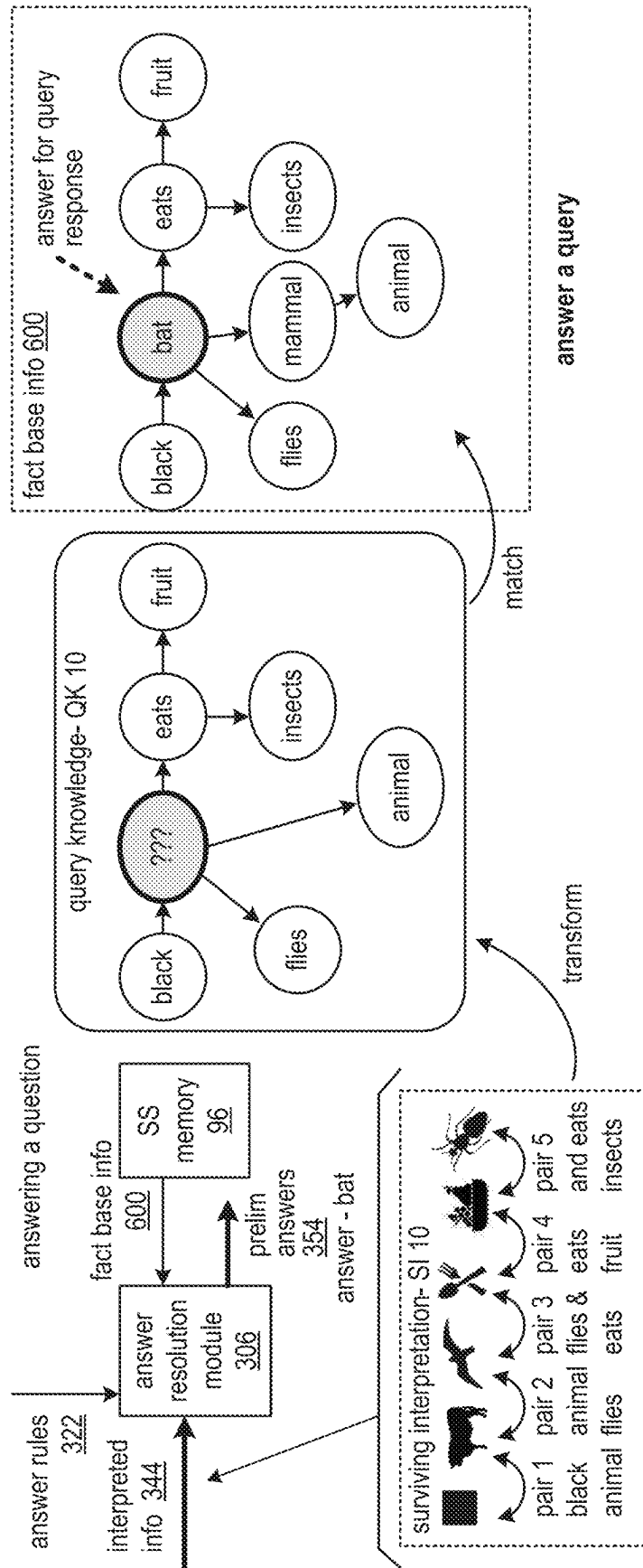
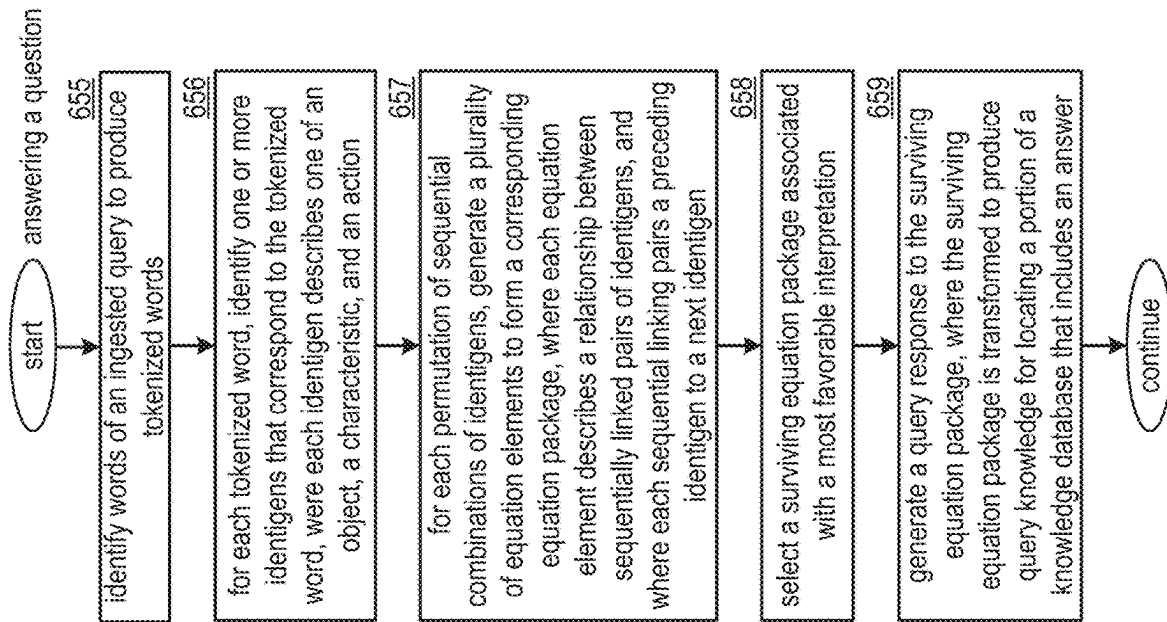


FIG. 8K

**FIG. 8L**

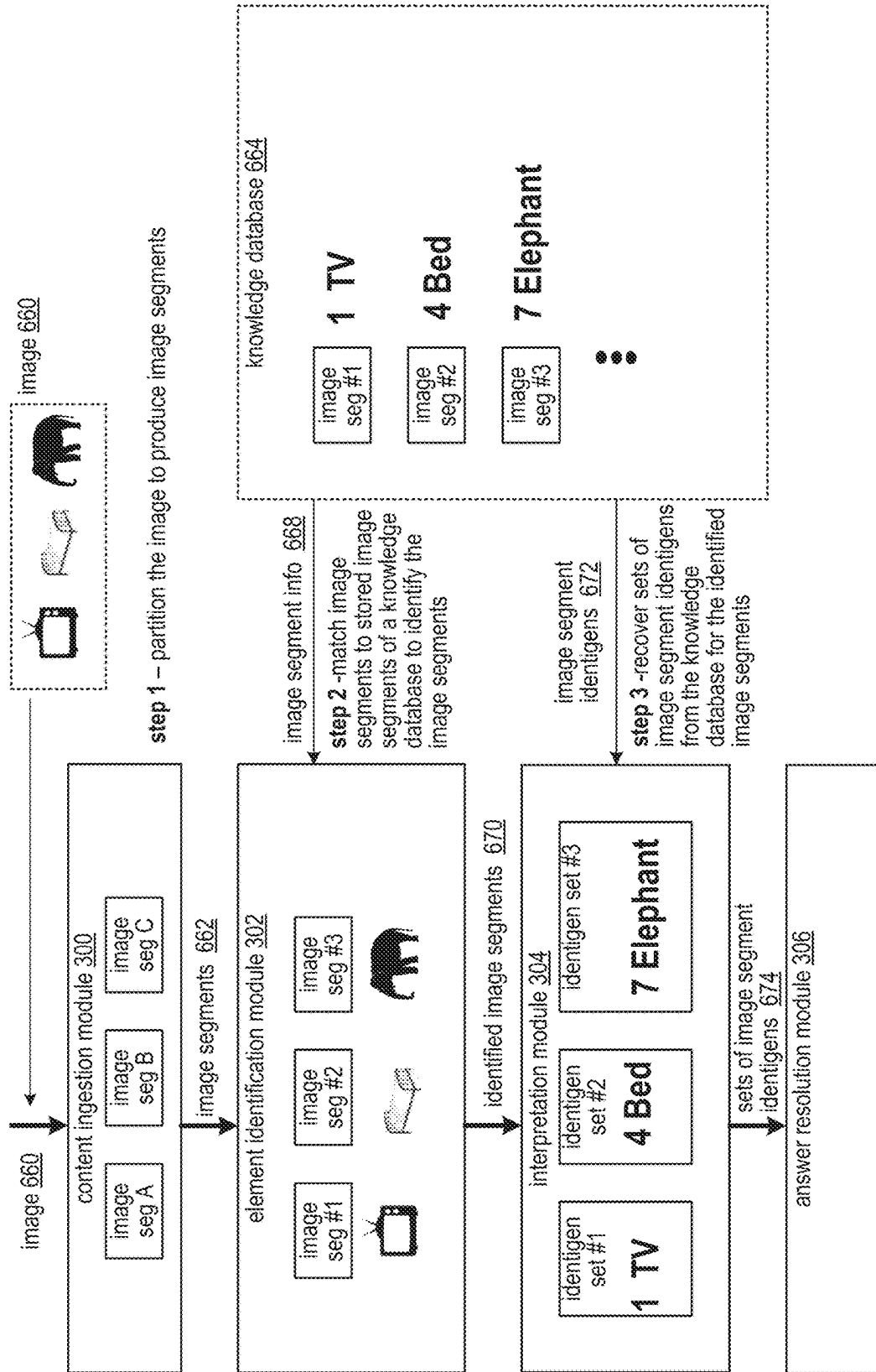


FIG. 9A

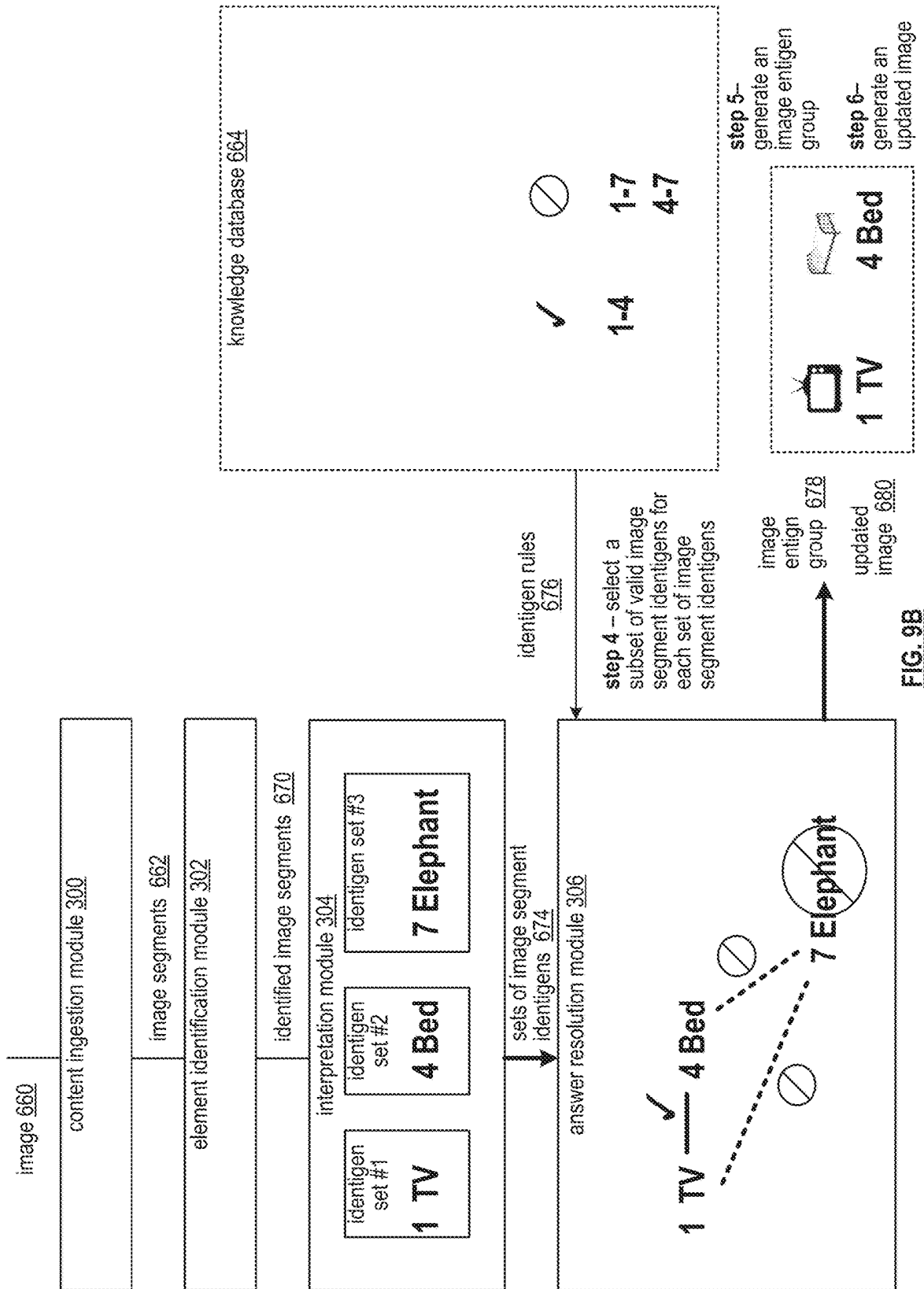


FIG. 9B

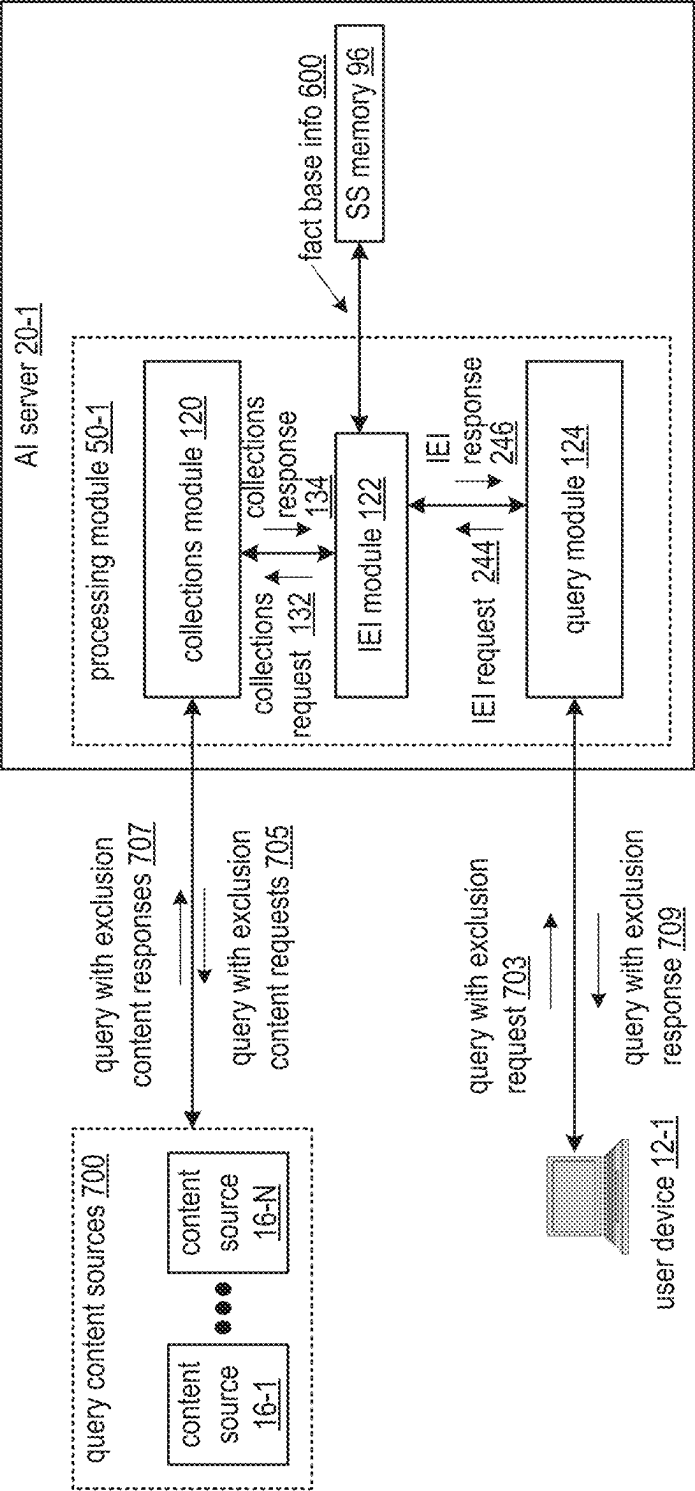


FIG. 10A

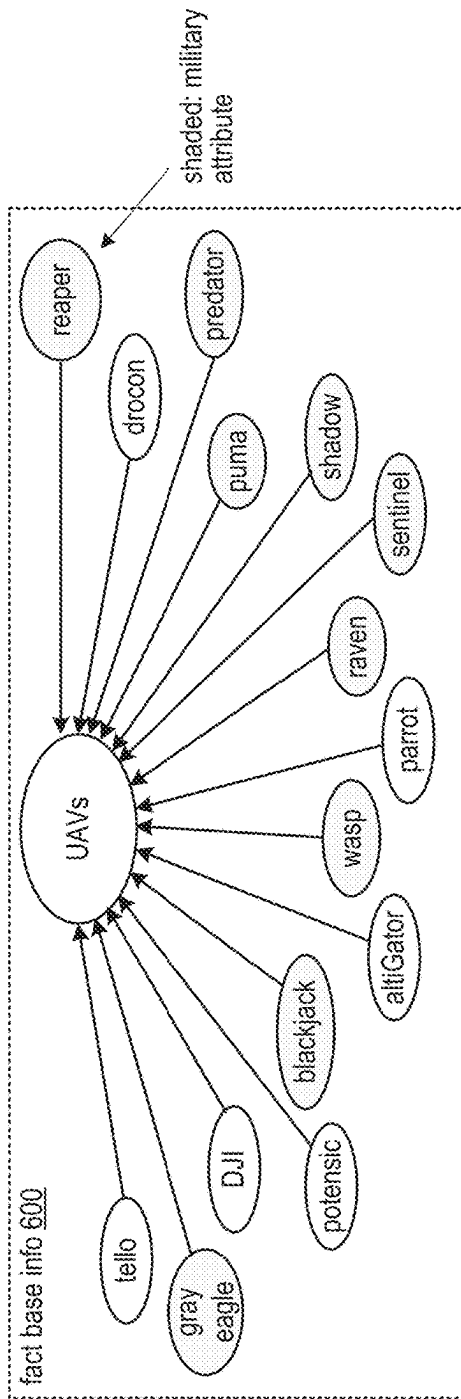


FIG. 10B

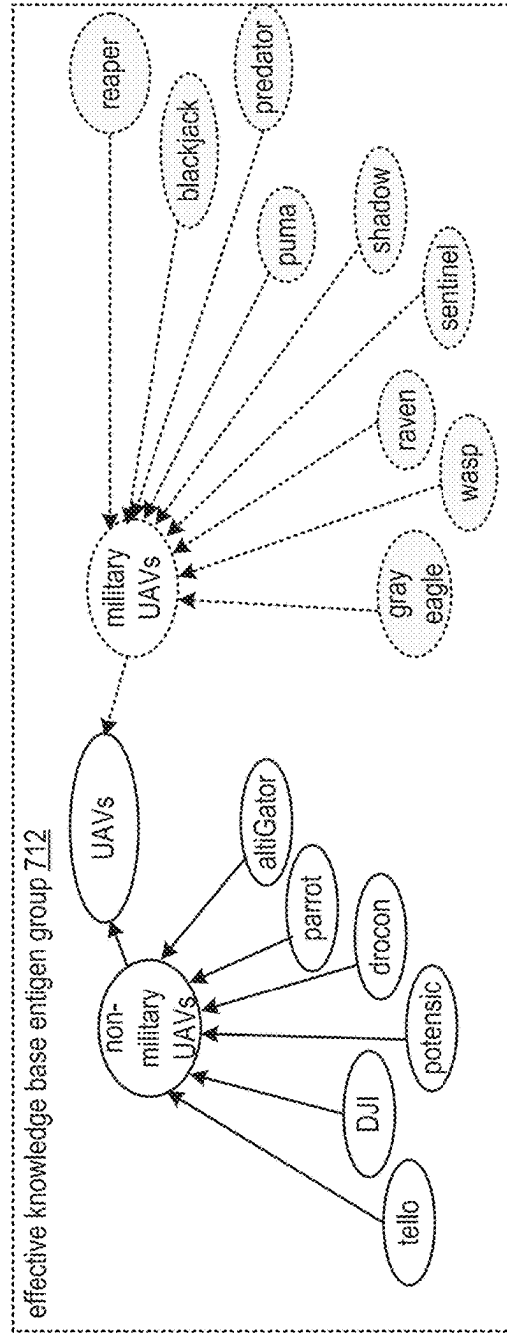
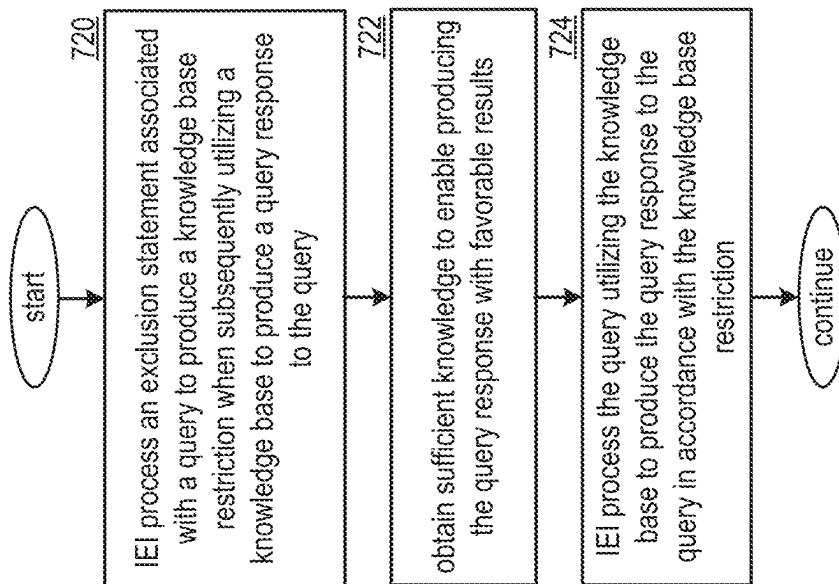
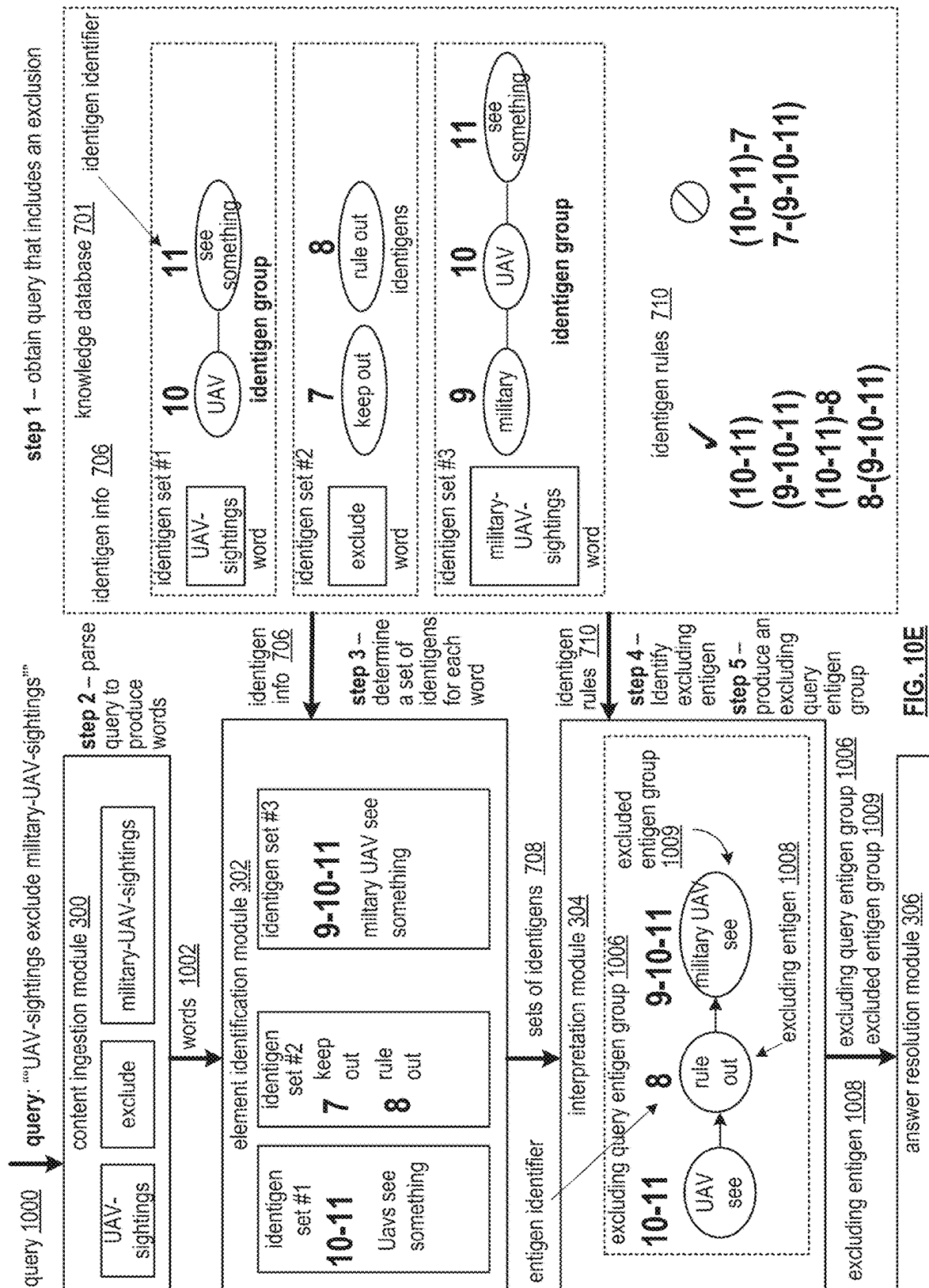
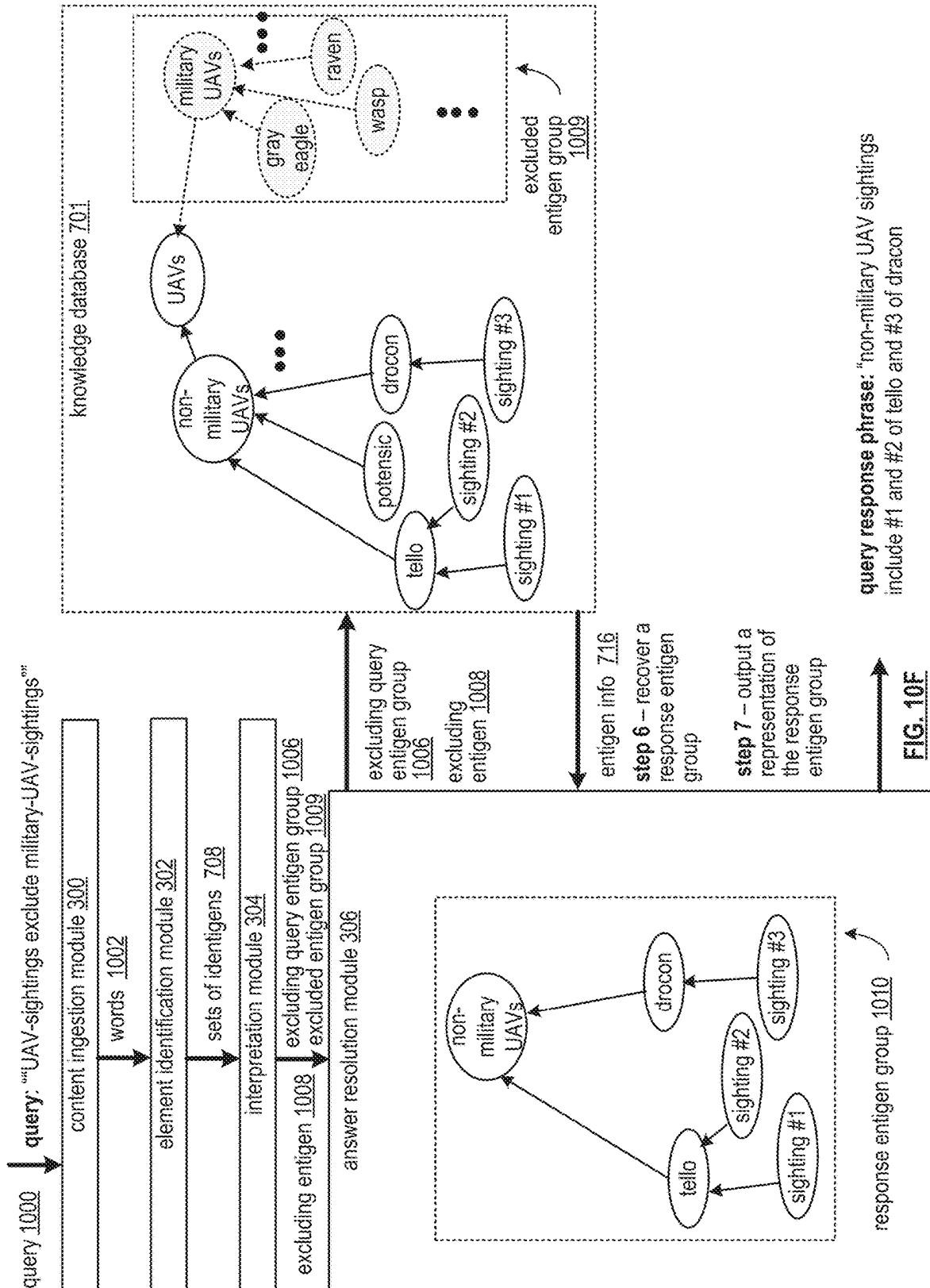


FIG. 10C

query with exclusion request 702
example:
"find UAV sightings around a military base that excludes military UAVs"

**FIG. 10D**





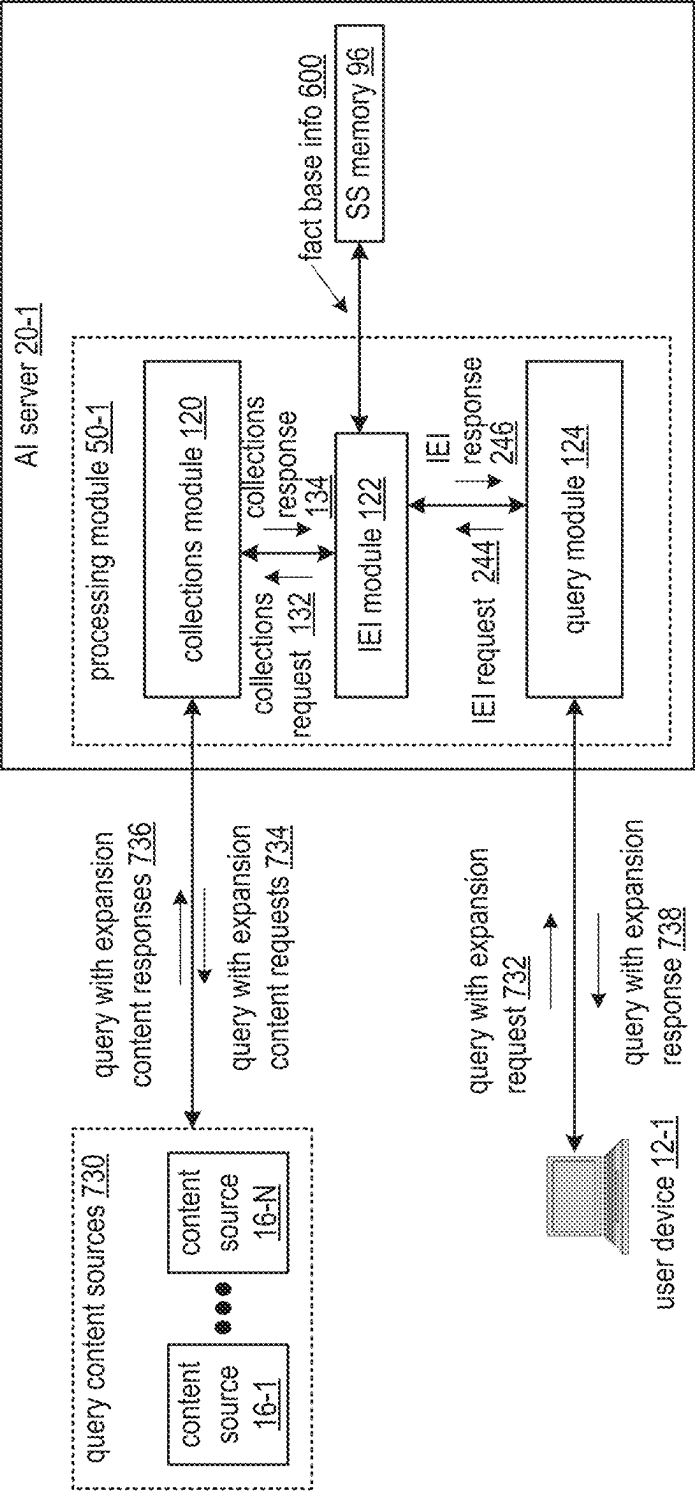


FIG. 11A

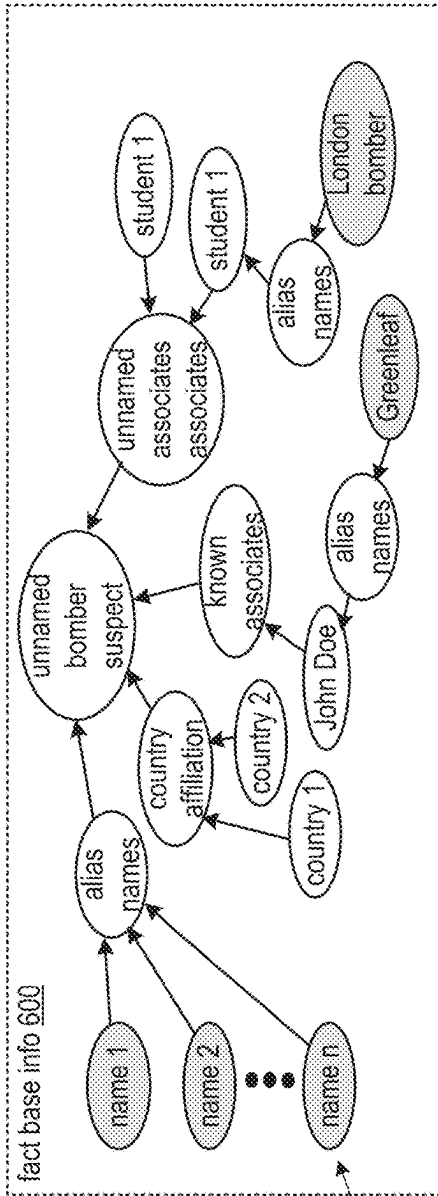


FIG. 11B

shaded: alias names of suspect or associates

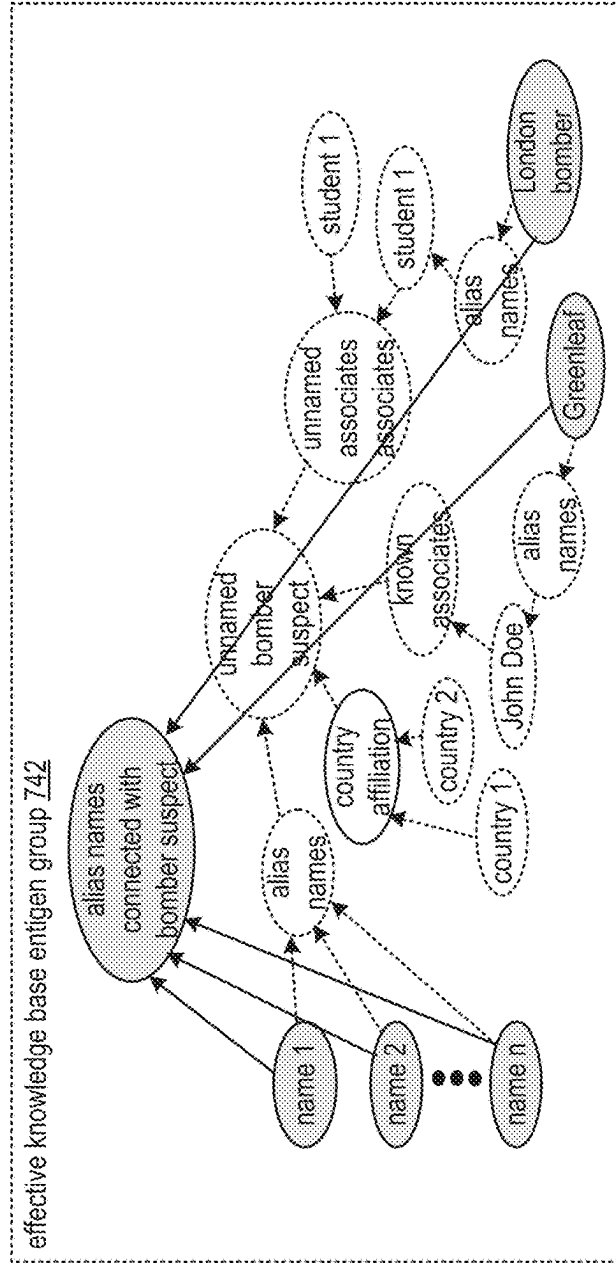
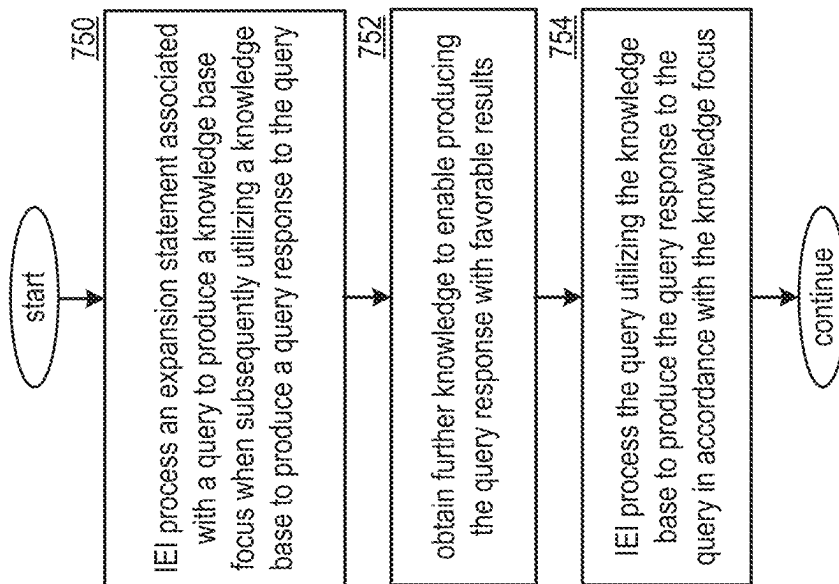


FIG. 11C

query with expansion request 732
example:
"locate unnamed bomber suspect utilizing alias names"

**FIG. 11D**

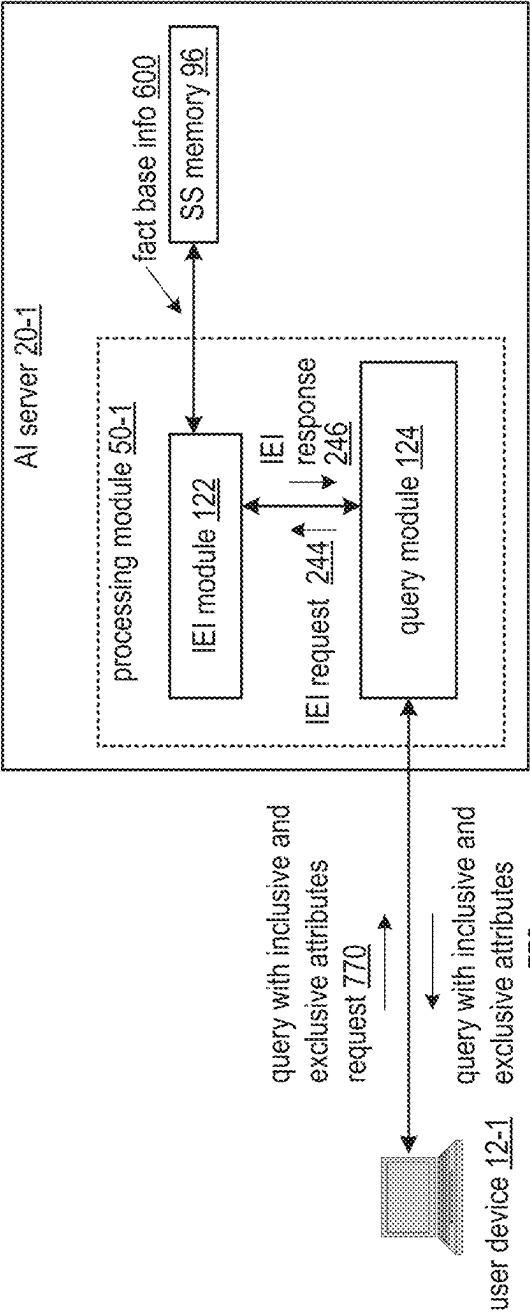


FIG. 12A

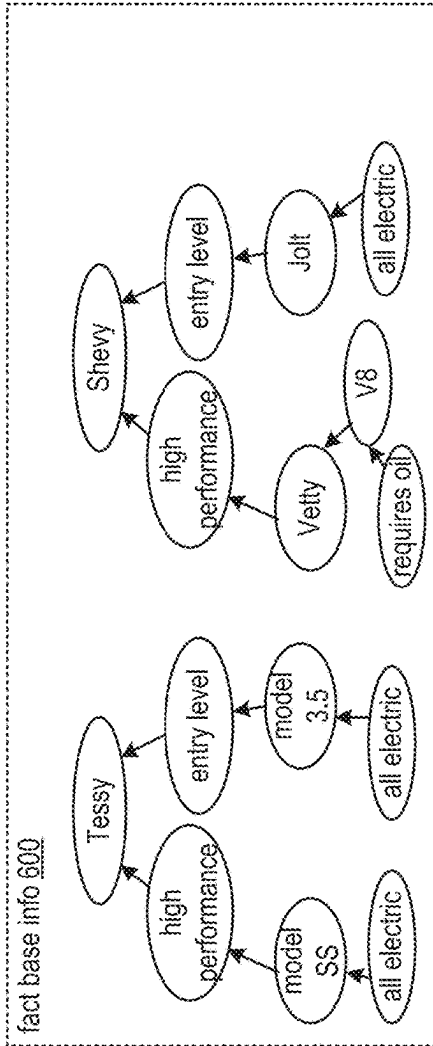


FIG. 12B

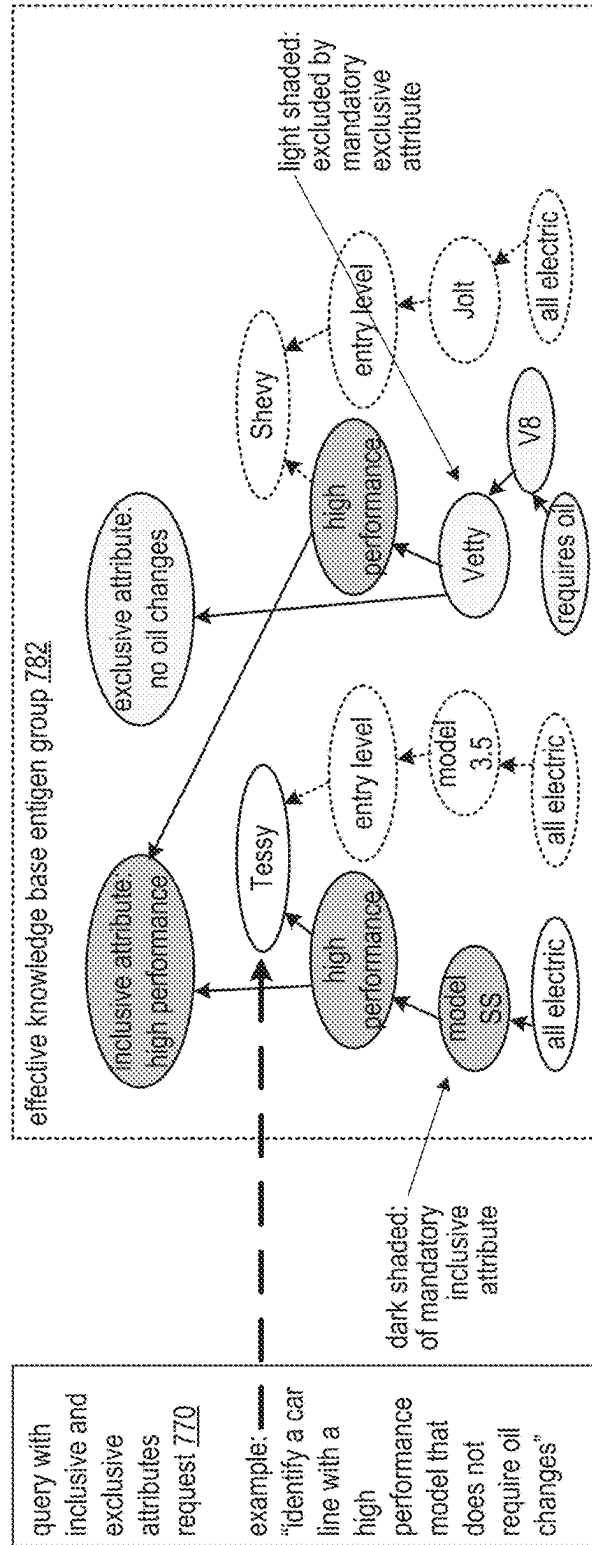
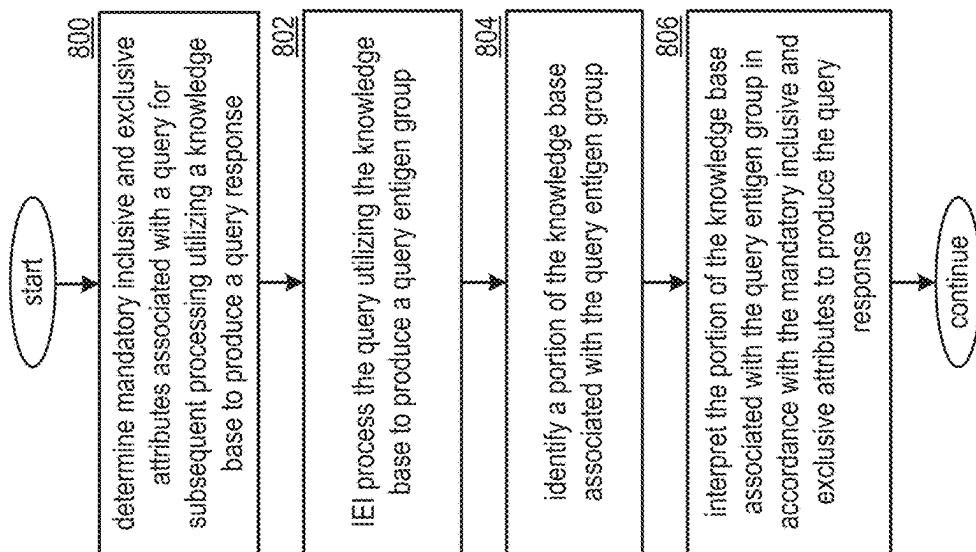
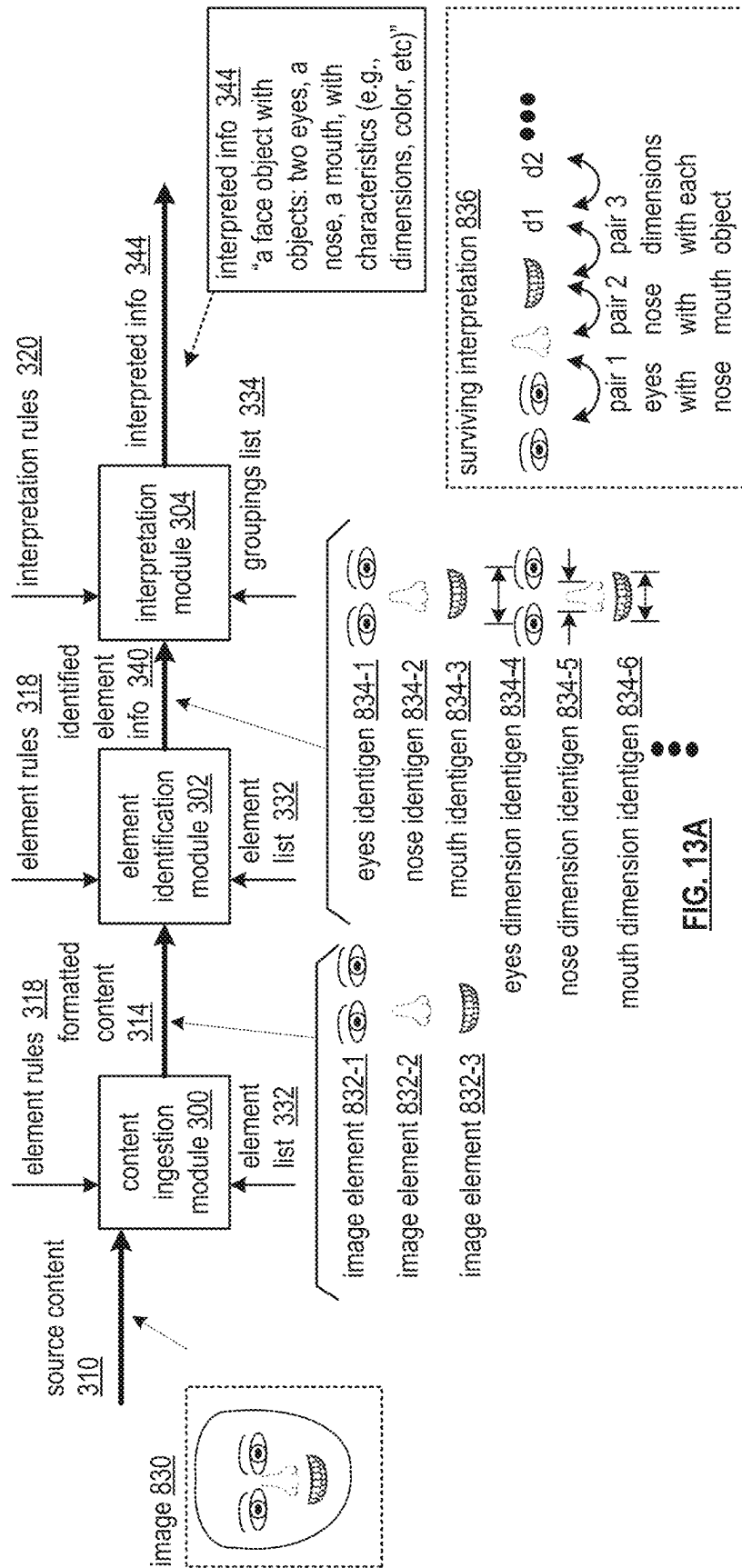


FIG. 12C

**FIG. 12D**



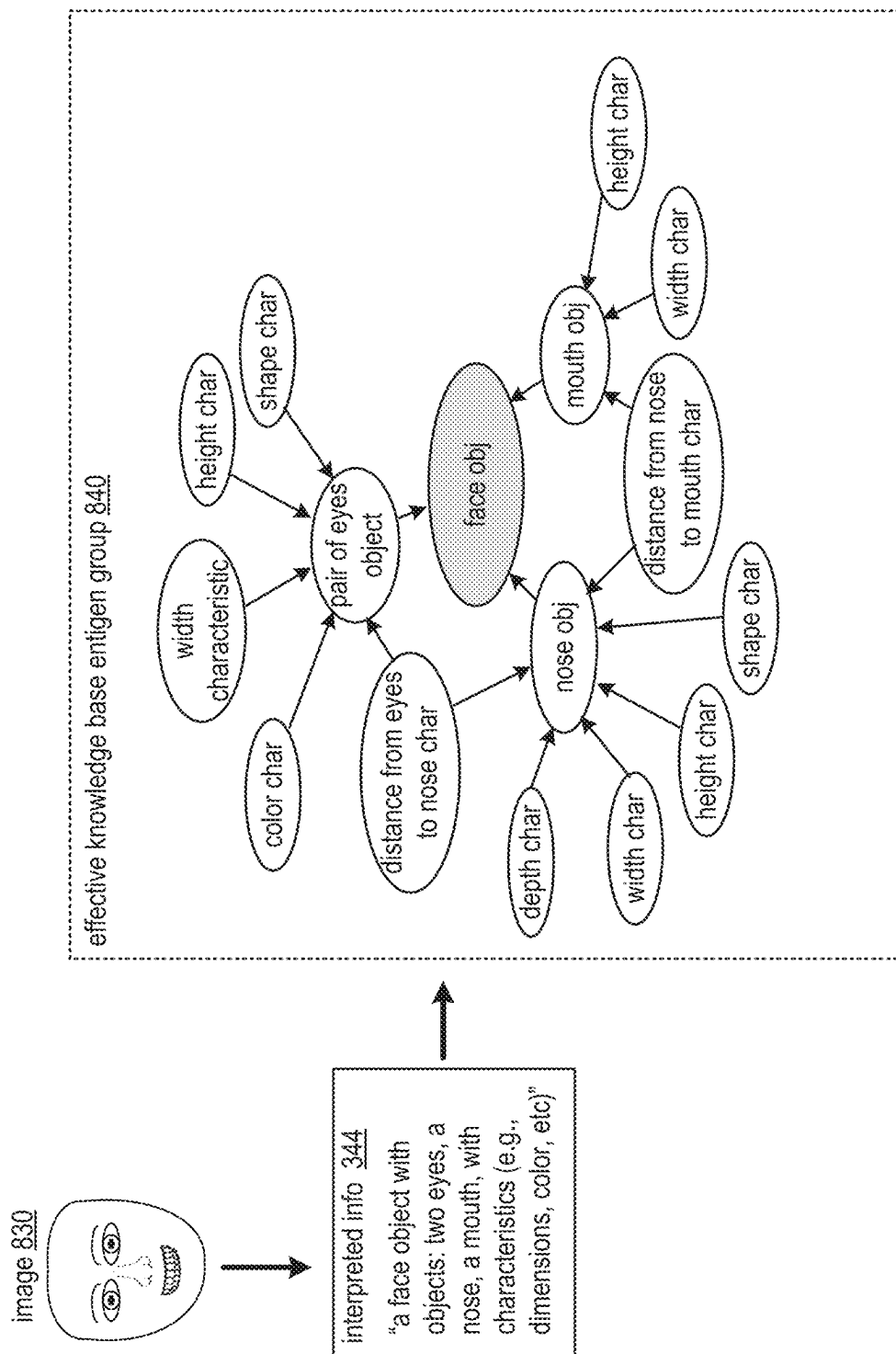
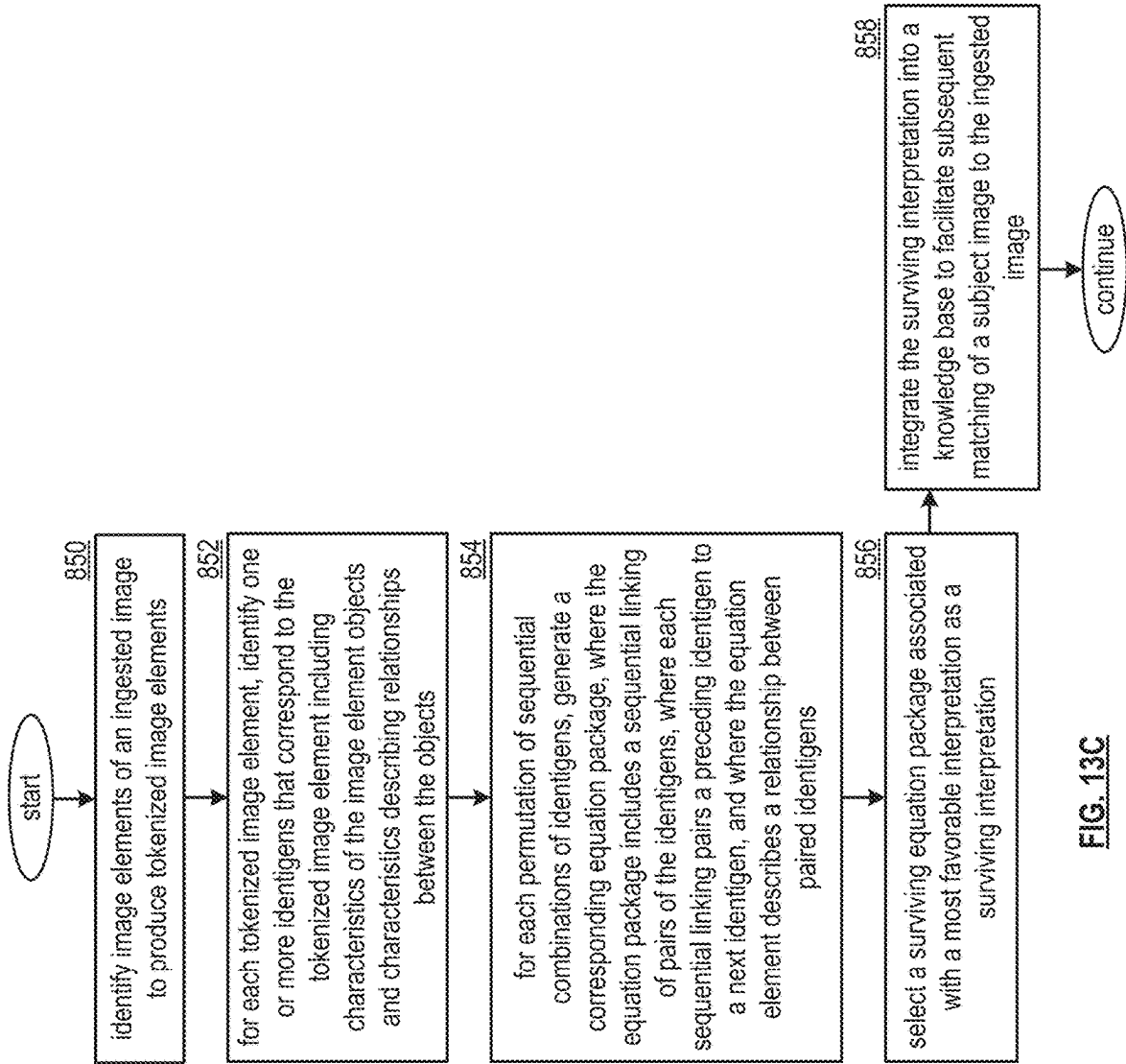


FIG. 13B



1

IMAGE PROCESSING UTILIZING AN ENTIGEN CONSTRUCT

CROSS REFERENCE TO RELATED PATENTS

The present U.S. Utility Patent Applications claims priority pursuant to 35 U.S.C. § 120 as a continuation in part of U.S. Utility application Ser. No. 16/696,114, entitled “IMAGE PROCESSING UTILIZING AN ENTIGEN CONSTRUCT” filed Nov. 26, 2019, allowed, which claims priority pursuant to 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/772,655, entitled “IMAGE PROCESSING UTILIZING AN ENTIGEN CONSTRUCT” filed Nov. 29, 2018, all of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility Patent Application for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC

Not Applicable

BACKGROUND OF THE INVENTION

Technical Field of the Invention

This invention relates to computing systems and more particularly to generating data representations of data and analyzing the data utilizing the data representations.

Description of Related Art

It is known that data is stored in information systems, such as files containing text. It is often difficult to produce useful information from this stored data due to many factors. The factors include the volume of available data, accuracy of the data, and variances in how text is interpreted to express knowledge. For example, many languages and regional dialects utilize the same or similar words to represent different concepts.

Computers are known to utilize pattern recognition techniques and apply statistical reasoning to process text to express an interpretation in an attempt to overcome ambiguities inherent in words. One pattern recognition technique includes matching a word pattern of a query to a word pattern of the stored data to find an explicit textual answer. Another pattern recognition technique classifies words into major grammatical types such as functional words, nouns, adjectives, verbs and adverbs. Grammar based techniques then utilize these grammatical types to study how words should be distributed within a string of words to form a properly constructed grammatical sentence where each word is forced to support a grammatical operation without necessarily identifying what the word is actually trying to describe.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

FIG. 1 is a schematic block diagram of an embodiment of a computing system in accordance with the present invention;

2

FIG. 2 is a schematic block diagram of an embodiment of various servers of a computing system in accordance with the present invention;

FIG. 3 is a schematic block diagram of an embodiment of various devices of a computing system in accordance with the present invention;

FIGS. 4A and 4B are schematic block diagrams of another embodiment of a computing system in accordance with the present invention;

FIG. 4C is a logic diagram of an embodiment of a method for interpreting content to produce a response to a query within a computing system in accordance with the present invention;

FIG. 5A is a schematic block diagram of an embodiment of a collections module of a computing system in accordance with the present invention;

FIG. 5B is a logic diagram of an embodiment of a method for obtaining content within a computing system in accordance with the present invention;

FIG. 5C is a schematic block diagram of an embodiment of a query module of a computing system in accordance with the present invention;

FIG. 5D is a logic diagram of an embodiment of a method for providing a response to a query within a computing system in accordance with the present invention;

FIG. 5E is a schematic block diagram of an embodiment of an intigen entigen intelligence (IEI) module of a computing system in accordance with the present invention;

FIG. 5F is a logic diagram of an embodiment of a method for analyzing content within a computing system in accordance with the present invention;

FIG. 6A is a schematic block diagram of an embodiment of an element identification module and an interpretation module of a computing system in accordance with the present invention;

FIG. 6B is a logic diagram of an embodiment of a method for interpreting information within a computing system in accordance with the present invention;

FIG. 6C is a schematic block diagram of an embodiment of an answer resolution module of a computing system in accordance with the present invention;

FIG. 6D is a logic diagram of an embodiment of a method for producing an answer within a computing system in accordance with the present invention;

FIG. 7A is an information flow diagram for interpreting information within a computing system in accordance with the present invention;

FIG. 7B is a relationship block diagram illustrating an embodiment of relationships between things and representations of things within a computing system in accordance with the present invention;

FIG. 7C is a diagram of an embodiment of a synonym words table within a computing system in accordance with the present invention;

FIG. 7D is a diagram of an embodiment of a polysemous words table within a computing system in accordance with the present invention;

FIG. 7E is a diagram of an embodiment of transforming words into groupings within a computing system in accordance with the present invention;

FIG. 8A is a data flow diagram for accumulating knowledge within a computing system in accordance with the present invention;

FIG. 8B is a diagram of an embodiment of a groupings table within a computing system in accordance with the present invention;

3

FIG. 8C is a data flow diagram for answering questions utilizing accumulated knowledge within a computing system in accordance with the present invention;

FIG. 8D is a data flow diagram for answering questions utilizing interference within a computing system in accordance with the present invention;

FIG. 8E is a relationship block diagram illustrating another embodiment of relationships between things and representations of things within a computing system in accordance with the present invention;

FIGS. 8F and 8G are schematic block diagrams of another embodiment of a computing system in accordance with the present invention;

FIG. 8H is a logic diagram of an embodiment of a method for processing content to produce knowledge within a computing system in accordance with the present invention;

FIGS. 8J and 8K are schematic block diagrams another embodiment of a computing system in accordance with the present invention;

FIG. 8L is a logic diagram of an embodiment of a method for generating a query response to a query within a computing system in accordance with the present invention;

FIGS. 9A and 9B are schematic block diagrams of another embodiment of a computing system illustrating an embodiment of a method for interpreting an image within the computing system in accordance with the present invention;

FIG. 10A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

FIGS. 10B and 10C are schematic block diagrams of an embodiment of a portion of knowledge base within a computing system in accordance with the present invention;

FIG. 10D is a logic diagram of an embodiment of a method for processing a query within a computing system in accordance with the present invention;

FIGS. 10E and 10F are schematic block diagrams of another embodiment of a computing system illustrating an embodiment of a method for generating a response to a query within the computing system in accordance with the present invention;

FIG. 11A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

FIGS. 11B and 11C are schematic block diagrams of another embodiment of a portion of knowledge base within a computing system in accordance with the present invention;

FIG. 11D is a logic diagram of an embodiment of another method for processing a query within a computing system in accordance with the present invention;

FIG. 12A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

FIGS. 12B and 12C are schematic block diagrams of another embodiment of a portion of knowledge base within a computing system in accordance with the present invention;

FIG. 12D is a logic diagram of an embodiment of another method for processing a query within a computing system in accordance with the present invention;

FIG. 13A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

FIG. 13B is schematic block diagram of another embodiment of a portion of knowledge base within a computing system in accordance with the present invention; and

4

FIG. 13C is a logic diagram of an embodiment of another method for processing an image within a computing system in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

FIG. 1 is a schematic block diagram of an embodiment of a computing system 10 that includes a plurality of user devices 12-1 through 12-N, a plurality of wireless user devices 14-1 through 14-N, a plurality of content sources 16-1 through 16-N, a plurality of transactional servers 18-1 through 18-N, a plurality of artificial intelligence (AI) servers 20-1 through 20-N, and a core network 24. The core network 24 includes at least one of the Internet, a public radio access network (RAN), and any private network. Hereafter, the computing system 10 may be interchangeably referred to as a data network, a data communication network, a system, a communication system, and a data communication system. Hereafter, the user device and the wireless user device may be interchangeably referred to as user devices, and each of the transactional servers and the AI servers may be interchangeably referred to as servers.

Each user device, wireless user device, transactional server, and AI server includes a computing device that includes a computing core. In general, a computing device is any electronic device that can communicate data, process data, and/or store data. A further generality of a computing device is that it includes one or more of a central processing unit (CPU), a memory system, a sensor (e.g., internal or external), user input/output interfaces, peripheral device interfaces, communication elements, and an interconnecting bus structure.

As further specific examples, each of the computing devices may be a portable computing device and/or a fixed computing device. A portable computing device may be an embedded controller, a smart sensor, a smart pill, a social networking device, a gaming device, a cell phone, a smart phone, a robot, a personal digital assistant, a digital music player, a digital video player, a laptop computer, a handheld computer, a tablet, a video game controller, an engine controller, a vehicular controller, an aircraft controller, a maritime vessel controller, and/or any other portable device that includes a computing core. A fixed computing device may be security camera, a sensor device, a household appliance, a machine, a robot, an embedded controller, a personal computer (PC), a computer server, a cable set-top box, a satellite receiver, a television set, a printer, a fax machine, home entertainment equipment, a camera controller, a video game console, a critical infrastructure controller, and/or any type of home or office computing equipment that includes a computing core. An embodiment of the various servers is discussed in greater detail with reference to FIG. 2. An embodiment of the various devices is discussed in greater detail with reference to FIG. 3.

Each of the content sources 16-1 through 16-N includes any source of content, where the content includes one or more of data files, a data stream, a tech stream, a text file, an audio stream, an audio file, a video stream, a video file, etc. Examples of the content sources include a weather service, a multi-language online dictionary, a fact server, a big data storage system, the Internet, social media systems, an email server, a news server, a schedule server, a traffic monitor, a security camera system, audio monitoring equipment, an information server, a service provider, a data aggregator, and airline traffic server, a shipping and logistics server, a banking server, a financial transaction server, etc. Alterna-

tively, or in addition to, one or more of the various user devices may provide content. For example, a wireless user device may provide content (e.g., issued as a content message) when the wireless user device is able to capture data (e.g., text input, sensor input, etc.).

Generally, an embodiment of this invention presents solutions where the computing system 10 supports the generation and utilization of knowledge extracted from content. For example, the AI servers 20-1 through 20-N ingest content from the content sources 16-1 through 16-N by receiving, via the core network 24 content messages 28-1 through 28-N as AI messages 32-1 through 32-N, extract the knowledge from the ingested content, and interact with the various user devices to utilize the extracted knowledge by facilitating the issuing, via the core network 24, user messages 22-1 through 22-N to the user devices 12-1 through 12-N and wireless signals 26-1 through 26-N to the wireless user devices 14-1 through 14-N.

Each content message 28-1 through 28-N includes a content request (e.g., requesting content related to a topic, content type, content timing, one or more domains, etc.) or a content response, where the content response includes real-time or static content such as one or more of dictionary information, facts, non-facts, weather information, sensor data, news information, blog information, social media content, user daily activity schedules, traffic conditions, community event schedules, school schedules, user schedules, airline records, shipping records, logistics records, banking records, census information, global financial history information, etc. Each AI message 32-1 through 32-N includes one or more of content messages, user messages (e.g., a query request, a query response that includes an answer to a query request), and transaction messages (e.g., transaction information, requests and responses related to transactions). Each user message 22-1 through 22-N includes one or more of a query request, a query response, a trigger request, a trigger response, a content collection, control information, software information, configuration information, security information, routing information, addressing information, presence information, analytics information, protocol information, all types of media, sensor data, statistical data, user data, error messages, etc.

When utilizing a wireless signal capability of the core network 24, each of the wireless user devices 14-1 through 14-N encodes/decodes data and/or information messages (e.g., user messages such as user messages 22-1 through 22-N) in accordance with one or more wireless standards for local wireless data signals (e.g., Wi-Fi, Bluetooth, ZigBee) and/or for wide area wireless data signals (e.g., 2G, 3G, 4G, 5G, satellite, point-to-point, etc.) to produce wireless signals 26-1 through 26-N. Having encoded/decoded the data and/or information messages, the wireless user devices 14-1 through 14-N and/or receive the wireless signals to/from the wireless capability of the core network 24.

As another example of the generation and utilization of knowledge, the transactional servers 18-1 through 18-N communicate, via the core network 24, transaction messages 30-1 through 30-N as further AI messages 32-1 through 32-N to facilitate ingesting of transactional type content (e.g., real-time crypto currency transaction information) and to facilitate handling of utilization of the knowledge by one or more of the transactional servers (e.g., for a transactional function) in addition to the utilization of the knowledge by the various user devices. Each transaction message 30-1 through 30-N includes one or more of a query request, a query response, a trigger request, a trigger response, a content message, and transactional information, where the

transactional information may include one or more of consumer purchasing history, crypto currency ledgers, stock market trade information, other investment transaction information, etc.

In another specific example of operation of the generation and utilization of knowledge extracted from the content, the user device 12-1 issues a user message 22-1 to the AI server 20-1, where the user message 22-1 includes a query request and where the query request includes a question related to a first domain of knowledge. The issuing includes generating the user message 22-1 based on the query request (e.g., the question), selecting the AI server 20-1 based on the first domain of knowledge, and sending, via the core network 24, the user message 22-1 as a further AI message 32-1 to the AI server 20-1. Having received the AI message 32-1, the AI server 20-1 analyzes the question within the first domain, generates further knowledge, generates a preliminary answer, generates a quality level indicator of the preliminary answer, and determines to gather further content when the quality level indicator is below a minimum quality threshold level.

When gathering the further content, the AI server 20-1 issues, via the core network 24, a still further AI message 32-1 as a further content message 28-1 to the content source 16-1, where the content message 28-1 includes a content request for more content associated with the first domain of knowledge and in particular the question. Alternatively, or in addition to, the AI server 20-1 issues the content request to another AI server to facilitate a response within a domain associated with the other AI server. Further alternatively, or in addition to, the AI server 20-1 issues the content request to one or more of the various user devices to facilitate a response from a subject matter expert.

Having received the content message 28-1, the contents or 16-1 issues, via the core network 24, a still further content message 28-1 to the AI server 20-1 as a yet further AI message 32-1, where the still further content message 28-1 includes requested content. The AI server 20-1 processes the received content to generate further knowledge. Having generated the further knowledge, the AI server 20-1 re-analyzes the question, generates still further knowledge, generates another preliminary answer, generates another quality level indicator of the other preliminary answer, and determines to issue a query response to the user device 12-1 when the quality level indicator is above the minimum quality threshold level. When issuing the query response, the AI server 20-1 generates an AI message 32-1 that includes another user message 22-1, where the other user message 22-1 includes the other preliminary answer as a query response including the answer to the question. Having generated the AI message 32-1, the AI server 20-1 sends, via the core network 24, the AI message 32-1 as the user message 22-1 to the user device 12-1 thus providing the answer to the original question of the query request.

FIG. 2 is a schematic block diagram of an embodiment of the AI servers 20-1 through 20-N and the transactional servers 18-1 through 18-N of the computing system 10 of FIG. 1. The servers include a computing core 52, one or more visual output devices 74 (e.g., video graphics display, touchscreen, LED, etc.), one or more user input devices 76 (e.g., keypad, keyboard, touchscreen, voice to text, a push button, a microphone, a card reader, a door position switch, a biometric input device, etc.), one or more audio output devices 78 (e.g., speaker(s), headphone jack, a motor, etc.), and one or more visual input devices 80 (e.g., a still image camera, a video camera, photocell, etc.).

The servers further include one or more universal serial bus (USB) devices (USB devices 1-U), one or more peripheral devices (e.g., peripheral devices 1-P), one or more memory devices (e.g., one or more flash memory devices 92, one or more hard drive (HD) memories 94, and one or more solid state (SS) memory devices 96, and/or cloud memory 98). The servers further include one or more wireless location modems 84 (e.g., global positioning satellite (GPS), Wi-Fi, angle of arrival, time difference of arrival, signal strength, dedicated wireless location, etc.), one or more wireless communication modems 86-1 through 86-N (e.g., a cellular network transceiver, a wireless data network transceiver, a Wi-Fi transceiver, a Bluetooth transceiver, a 315 MHz transceiver, a zig bee transceiver, a 60 GHz transceiver, etc.), a telco interface 102 (e.g., to interface to a public switched telephone network), and a wired local area network (LAN) 88 (e.g., optical, electrical), and a wired wide area network (WAN) 90 (e.g., optical, electrical).

The computing core 52 includes a video graphics module 54, one or more processing modules 50-1 through 50-N (e.g., which may include one or more secure co-processors), a memory controller 56 and one or more main memories 58-1 through 58-N (e.g., RAM serving as local memory). The computing core 52 further includes one or more input/output (I/O) device interfaces 62, an input/output (I/O) controller 60, a peripheral interface 64, one or more USB interfaces 66, one or more network interfaces 72, one or more memory interfaces 70, and/or one or more peripheral device interfaces 68.

The processing modules may be a single processing device or a plurality of processing devices where the processing device may further be referred to as one or more of a "processing circuit", a "processor", and/or a "processing unit". Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions.

The processing module, module, processing circuit, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network).

Further note that if the processing module, module, processing circuit, and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, and/or processing unit executes,

hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

Each of the interfaces 62, 66, 68, 70, and 72 includes a combination of hardware (e.g., connectors, wiring, etc.) and may further include operational instructions stored on memory (e.g., driver software) that are executed by one or more of the processing modules 50-1 through 50-N and/or a processing circuit within the interface. Each of the interfaces couples to one or more components of the servers. For example, one of the TO device interfaces 62 couples to an audio output device 78. As another example, one of the memory interfaces 70 couples to flash memory 92 and another one of the memory interfaces 70 couples to cloud memory 98 (e.g., an on-line storage system and/or on-line backup system). In other embodiments, the servers may include more or less devices and modules than shown in this example embodiment of the servers.

FIG. 3 is a schematic block diagram of an embodiment of the various devices of the computing system 10 of FIG. 1, including the user devices 12-1 through 12-N and the wireless user devices 14-1 through 14-N. The various devices include the visual output device 74 of FIG. 2, the user input device 76 of FIG. 2, the audio output device 78 of FIG. 2, the visual input device 80 of FIG. 2, and one or more sensors 82.

The sensor may be implemented internally and/or externally to the device. Example sensors includes a still camera, a video camera, servo motors associated with a camera, a position detector, a smoke detector, a gas detector, a motion sensor, an accelerometer, velocity detector, a compass, a gyro, a temperature sensor, a pressure sensor, an altitude sensor, a humidity detector, a moisture detector, an imaging sensor, and a biometric sensor. Further examples of the sensor include an infrared sensor, an audio sensor, an ultrasonic sensor, a proximity detector, a magnetic field detector, a biomaterial detector, a radiation detector, a weight detector, a density detector, a chemical analysis detector, a fluid flow volume sensor, a DNA reader, a wind speed sensor, a wind direction sensor, and an object detection sensor.

Further examples of the sensor include an object identifier sensor, a motion recognition detector, a battery level detector, a room temperature sensor, a sound detector, a smoke detector, an intrusion detector, a motion detector, a door position sensor, a window position sensor, and a sunlight detector. Still further sensor examples include medical category sensors including: a pulse rate monitor, a heart rhythm monitor, a breathing detector, a blood pressure monitor, a blood glucose level detector, blood type, an electrocardiogram sensor, a body mass detector, an imaging sensor, a microphone, body temperature, etc.

The various devices further include the computing core 52 of FIG. 2, the one or more universal serial bus (USB) devices (USB devices 1-U) of FIG. 2, the one or more peripheral devices (e.g., peripheral devices 1-P) of FIG. 2, and the one or more memories of FIG. 2 (e.g., flash memories 92, HD memories 94, SS memories 96, and/or cloud memories 98). The various devices further include the one or more wireless location modems 84 of FIG. 2, the one or more wireless communication modems 86-1 through 86-N of FIG. 2, the telco interface 102 of FIG. 2, the wired local area network (LAN) 88 of FIG. 2, and the wired wide area network (WAN) 90 of FIG. 2. In other embodiments,

the various devices may include more or less internal devices and modules than shown in this example embodiment of the various devices.

FIGS. 4A and 4B are schematic block diagrams of another embodiment of a computing system that includes one or more of the user device 12-1 of FIG. 1, the wireless user device 14-1 of FIG. 1, the content source 16-1 of FIG. 1, the transactional server 18-1 of FIG. 1, the user device 12-2 of FIG. 1, and the AI server 20-1 of FIG. 1. The AI server 20-1 includes the processing module 50-1 (e.g., associated with the servers) of FIG. 2, where the processing module 50-1 includes a collections module 120, an identigen entigen intelligence (IEI) module 122, and a query module 124. Alternatively, the collections module 120, the IEI module 122, and the query module 124 may be implemented by the processing module 50-1 (e.g., associated with the various user devices) of FIG. 3. The computing system functions to interpret content to produce a response to a query.

FIG. 4A illustrates an example of the interpreting of the content to produce the response to the query where the collections module 120 interprets (e.g., based on an interpretation approach such as rules) at least one of a collections request 132 from the query module 124 and a collections request within collections information 130 from the IEI module 122 to produce content request information (e.g., potential sources, content descriptors of desired content). Alternatively, or in addition to, the collections module 120 may facilitate gathering further content based on a plurality of collection requests from a plurality of devices of the computing system 10 of FIG. 1.

The collections request 132 is utilized to facilitate collection of content, where the content may be received in a real-time fashion once or at desired intervals, or in a static fashion from previous discrete time frames. For instance, the query module 124 issues the collections request 132 to facilitate collection of content as a background activity to support a long-term query (e.g., how many domestic airline flights over the next seven days include travelers between the age of 18 and 35 years old). The collections request 132 may include one or more of a requester identifier (ID), a content type (e.g., language, dialect, media type, topic, etc.), a content source indicator, security credentials (e.g., an authorization level, a password, a user ID, parameters utilized for encryption, etc.), a desired content quality level, trigger information (e.g., parameters under which to collect content based on a pre-event, an event (i.e., content quality level reaches a threshold to cause the trigger, trueness), or a timeframe), a desired format, and a desired timing associated with the content.

Having interpreted the collections request 132, the collections module 120 selects a source of content based on the content request information. The selecting includes one or more of identifying one or more potential sources based on the content request information, selecting the source of content from the potential sources utilizing a selection approach (e.g., favorable history, a favorable security level, favorable accessibility, favorable cost, favorable performance, etc.). For example, the collections module 120 selects the content source 16-1 when the content source 16-1 is known to provide a favorable content quality level for a domain associated with the collections request 132.

Having selected the source of content, the collections module 120 issues a content request 126 to the selected source of content. The issuing includes generating the content request 126 based on the content request information for the selected source of content and sending the content request 126 to the selected source of content. The content

request 126 may include one or more of a content type indicator, a requester ID, security credentials for content access, and any other information associated with the collections request 132. For example, the collections module 120 sends the content request 126, via the core network 24 of FIG. 1, to the content source 16-1. Alternatively, or in addition to, the collections module 120 may send a similar content request 126 to one or more of the user device 12-1, the wireless user device 14-1, and the transactional server 18-1 to facilitate collecting of further content.

In response to the content request 126, the collections module 120 receives one or more content responses 128. The content response 128 includes one or more of content associated with the content source, a content source identifier, security credential processing information, and any other information pertaining to the desired content. Having received the content response 128, the collections module 120 interprets the received content response 128 to produce collections information 130, where the collections information 130 further includes a collections response from the collections module 120 to the IEI module 122.

The collections response includes one or more of transformed content (e.g., completed sentences and paragraphs), timing information associated with the content, a content source ID, and a content quality level. Having generated the collections response of the collections information 130, the collections module 120 sends the collections information 130 to the IEI module 122. Having received the collections information 130 from the collections module 120, the IEI module 122 interprets the further content of the content response to generate further knowledge, where the further knowledge is stored in a memory associated with the IEI module 122 to facilitate subsequent answering of questions posed in received queries.

FIG. 4B further illustrates the example of the interpreting of the content to produce the response to the query where, the query module 124 interprets a received query request 136 from a requester to produce an interpretation of the query request. For example, the query module 124 receives the query request 136 from the user device 12-2, and/or from one or more of the wireless user device 14-2 and the transactional server 18-2. The query request 136 includes one or more of an identifier (ID) associated with the request (e.g., requester ID, ID of an entity to send a response to), a question, question constraints (e.g., within a timeframe, within a geographic area, within a domain of knowledge, etc.), and content associated with the question (e.g., which may be analyzed for new knowledge itself).

The interpreting of the query request 136 includes determining whether to issue a request to the IEI module 122 (e.g., a question, perhaps with content) and/or to issue a request to the collections module 120 (e.g., for further background content). For example, the query module 124 produces the interpretation of the query request to indicate to send the request directly to the IEI module 122 when the question is associated with a simple non-time varying function answer (e.g., question: "how many hydrogen atoms does a molecule of water have?").

Having interpreted the query request 136, the query module 124 issues at least one of an IEI request as query information 138 to the IEI module 122 (e.g., when receiving a simple new query request) and a collections request 132 to the collections module 120 (e.g., based on two or more query requests 136 requiring more substantive content gathering). The IEI request of the query information 138 includes one or more of an identifier (ID) of the query module 124, an ID of the requester (e.g., the user device

12-2), a question (e.g., with regards to content for analysis, with regards to knowledge minded by the AI server from general content), one or more constraints (e.g., assumptions, restrictions, etc.) associated with the question, content for analysis of the question, and timing information (e.g., a date range for relevance of the question).

Having received the query information 138 that includes the IEI request from the query module 124, the IEI module 122 determines whether a satisfactory response can be generated based on currently available knowledge, including that of the query request 136. The determining includes indicating that the satisfactory response cannot be generated when an estimated quality level of an answer falls below a minimum quality threshold level. When the satisfactory response cannot be generated, the IEI module 122 facilitates collecting more content. The facilitating includes issuing a collections request to the collections module 120 of the AI server 20-1 and/or to another server or user device, and interpreting a subsequent collections response 134 of collections information 130 that includes further content to produce further knowledge to enable a more favorable answer.

When the IEI module 122 indicates that the satisfactory response can be generated, the IEI module 122 issues an IEI response as query information 138 to the query module 124. The IEI response includes one or more of one or more answers, timing relevance of the one or more answers, an estimated quality level of each answer, and one or more assumptions associated with the answer. The issuing includes generating the IEI response based on the collections response 134 of the collections information 130 and the IEI request, and sending the IEI response as the query information 138 to the query module 124. Alternatively, or in addition to, at least some of the further content collected by the collections module 120 is utilized to generate a collections response 134 issued by the collections module 120 to the query module 124. The collections response 134 includes one or more of further content, a content availability indicator (e.g., when, where, required credentials, etc.), a content freshness indicator (e.g., timestamps, predicted time availability), content source identifiers, and a content quality level.

Having received the query information 138 from the IEI module 122, the query module 124 issues a query response 140 to the requester based on the IEI response and/or the collections response 134 directly from the collections module 120, where the collection module 120 generates the collections response 134 based on collected content and the collections request 132. The query response 140 includes one or more of an answer, answer timing, an answer quality level, and answer assumptions.

FIG. 4C is a logic diagram of an embodiment of a method for interpreting content to produce a response to a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4B, and also FIG. 4C. The method includes step 150 where a collections module of a processing module of one or more computing devices (e.g., of one or more servers) interprets a collections request to produce content request information. The interpreting may include one or more of identifying a desired content source, identifying a content type, identifying a content domain, and identifying content timing requirements.

The method continues at step 152 where the collections module selects a source of content based on the content request information. For example, the collections module

identifies one or more potential sources based on the content request information and selects the source of content from the potential sources utilizing a selection approach (e.g., based on one or more of favorable history, a favorable security level, favorable accessibility, favorable cost, favorable performance, etc.). The method continues at step 154 where the collections module issues a content request to the selected source of content. The issuing includes generating a content request based on the content request information for the selected source of content and sending the content request to the selected source of content.

The method continues at step 156 where the collections module issues collections information to an identigen entigen intelligence (IEI) module based on a received content response, where the IEI module extracts further knowledge from newly obtained content from the one or more received content responses. For example, the collections module generates the collections information based on newly obtained content from the one or more received content responses of the selected source of content.

The method continues at step 158 where a query module interprets a received query request from a requester to produce an interpretation of the query request. The interpreting may include determining whether to issue a request to the IEI module (e.g., a question) or to issue a request to the collections module to gather further background content. The method continues at step 160 where the query module issues a further collections request. For example, when receiving a new query request, the query module generates a request for the IEI module. As another example, when receiving a plurality of query requests for similar questions, the query module generates a request for the collections module to gather further background content.

The method continues at step 162 where the IEI module determines whether a satisfactory query response can be generated when receiving the request from the query module. For example, the IEI module indicates that the satisfactory query response cannot be generated when an estimated quality level of an answer is below a minimum answer quality threshold level. The method branches to step 166 when the IEI module determines that the satisfactory query response can be generated. The method continues to step 164 when the IEI module determines that the satisfactory query response cannot be generated. When the satisfactory query response cannot be generated, the method continues at step 164 where the IEI module facilitates collecting more content. The method loops back to step 150.

When the satisfactory query response can be generated, the method continues at step 166 where the IEI module issues an IEI response to the query module. The issuing includes generating the IEI response based on the collections response and the IEI request, and sending the IEI response to the query module. The method continues at step 168 where the query module issues a query response to the requester. For example, the query module generates the query response based on the IEI response and/or a collections response from the collections module and sends the query response to the requester, where the collections module generates the collections response based on collected content and the collections request.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second

13

memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. 5A is a schematic block diagram of an embodiment of the collections module 120 of FIG. 4A that includes a content acquisition module 180, a content selection module 182, a source selection module 184, a content security module 186, an acquisition timing module 188, a content transformation module 190, and a content quality module 192. Generally, an embodiment of this invention presents solutions where the collections module 120 supports collecting content.

In an example of operation of the collecting of the content, the content acquisition module 180 receives a collections request 132 from a requester. The content acquisition module 180 obtains content selection information 194 based on the collections request 132. The content selection information 194 includes one or more of content requirements, a desired content type indicator, a desired content source identifier, a content type indicator, a candidate source identifier (ID), and a content profile (e.g., a template of typical parameters of the content). For example, the content acquisition module 180 receives the content selection information 194 from the content selection module 182, where the content selection module 182 generates the content selection information 194 based on a content selection information request from the content acquisition module 180 and where the content acquisition module 180 generates the content selection information request based on the collections request 132.

The content acquisition module 180 obtains source selection information 196 based on the collections request 132. The source selection information 196 includes one or more of candidate source identifiers, a content profile, selected sources, source priority levels, and recommended source access timing. For example, the content acquisition module 180 receives the source selection information 196 from the source selection module 184, where the source selection module 184 generates the source selection information 196 based on a source selection information request from the content acquisition module 180 and where the content acquisition module 180 generates the source selection information request based on the collections request 132.

The content acquisition module 180 obtains acquisition timing information 200 based on the collections request 132. The acquisition timing information 200 includes one or more of recommended source access timing, confirmed source access timing, source access testing results, estimated velocity of content update's, content precious, timestamps, predicted time availability, required content acquisition triggers, content acquisition trigger detection indicators, and a duplicative indicator with a pending content request. For example, the content acquisition module 180 receives the acquisition timing information 200 from the acquisition timing module 188, where the acquisition timing module 188 generates the acquisition timing information 200 based on an acquisition timing information request from the content acquisition module 180 and where the content acquisition module 180 generates the acquisition timing information request based on the collections request 132.

Having obtained the content selection information 194, the source selection information 196, and the acquisition timing information 200, the content acquisition module 180

14

issues a content request 126 to a content source utilizing security information 198 from the content security module 186, where the content acquisition module 180 generates the content request 126 in accordance with the content selection information 194, the source selection information 196, and the acquisition timing information 200. The security information 198 includes one or more of source priority requirements, requester security information, available security procedures, and security credentials for trust and/or encryption. For example, the content acquisition module 180 generates the content request 126 to request a particular content type in accordance with the content selection information 194 and to include security parameters of the security information 198, initiates sending of the content request 126 in accordance with the acquisition timing information 200, and sends the content request 126 to a particular targeted content source in accordance with the source selection information 196.

In response to receiving a content response 128, the content acquisition module 180 determines the quality level of received content extracted from the content response 128. For example, the content acquisition module 180 receives content quality information 204 from the content quality module 192, where the content quality module 192 generates the quality level of the received content based on receiving a content quality request from the content acquisition module 180 and where the content acquisition module 180 generates the content quality request based on content extracted from the content response 128. The content quality information includes one or more of a content reliability threshold range, a content accuracy threshold range, a desired content quality level, a predicted content quality level, and a predicted level of trust.

When the quality level is below a minimum desired quality threshold level, the content acquisition module 180 facilitates acquisition of further content. The facilitating includes issuing another content request 126 to a same content source and/or to another content source to receive and interpret further received content. When the quality level is above the minimum desired quality threshold level, the content acquisition module 180 issues a collections response 134 to the requester. The issuing includes processing the content in accordance with a transformation approach to produce transformed content, generating the collections response 134 to include the transformed content, and sending the collections response 134 to the requester. The processing of the content to produce the transformed content includes receiving content transformation information 202 from the content transformation module 190, where the content transformation module 190 transforms the content in accordance with the transformation approach to produce the transformed content. The content transformation information includes a desired format, available formats, recommended formatting, the received content, transformation instructions, and the transformed content.

FIG. 5B is a logic diagram of an embodiment of a method for obtaining content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5A, and also FIG. 5B. The method includes step 210 where a processing module of one or more processing modules of one or more computing devices of the computing system receives a collections request from the requester. The method continues at step 212 where the processing module determines content selection information. The determining includes interpreting the collections request to identify requirements of the content.

15

The method continues at step 214 where the processing module determines source selection information. The determining includes interpreting the collections request to identify and select one or more sources for the content to be collected. The method continues at step 216 where the processing module determines acquisition timing information. The determining includes interpreting the collections request to identify timing requirements for the acquisition of the content from the one or more sources. The method continues at step 218 where the processing module issues a content request utilizing security information and in accordance with one or more of the content selection information, the source selection information, and the acquisition timing information. For example, the processing module issues the content request to the one or more sources for the content in accordance with the content requirements, where the sending of the request is in accordance with the acquisition timing information.

The method continues at step 220 where the processing module determines a content quality level for received content area the determining includes receiving the content from the one or more sources, obtaining content quality information for the received content based on a quality analysis of the received content. The method branches to step 224 when the content quality level is favorable and the method continues to step 222 when the quality level is unfavorable. For example, the processing module determines that the content quality level is favorable when the content quality level is equal to or above a minimum quality threshold level and determines that the content quality level is unfavorable when the content quality level is less than the minimum quality threshold level.

When the content quality level is unfavorable, the method continues at step 222 where the processing module facilitates acquisition and further content. For example, the processing module issues further content requests and receives further content for analysis. When the content quality level is favorable, the method continues at step 224 where the processing module issues a collections response to the requester. The issuing includes generating the collections response and sending the collections response to the requester. The generating of the collections response may include transforming the received content into transformed content in accordance with a transformation approach (e.g., reformatting, interpreting absolute meaning and translating into another language in accordance with the absolute meaning, etc.).

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. 5C is a schematic block diagram of an embodiment of the query module 124 of FIG. 4A that includes an answer acquisition module 230, a content requirements module 232, a source requirements module 234, a content security module 236, an answer timing module 238, an answer transformation module 240, and an answer quality module 242.

16

Generally, an embodiment of this invention presents solutions where the query module 124 supports responding to a query.

In an example of operation of the responding to the query, the answer acquisition module 230 receives a query request 136 from a requester. The answer acquisition module 230 obtains content requirements information 248 based on the query request 136. The content requirements information 248 includes one or more of content parameters, a desired content type, a desired content source if any, a content type if any, candidate source identifiers, a content profile, and a question of the query request 136. For example, the answer acquisition module 230 receives the content requirements information 248 from the content requirements module 232, where the content requirements module 232 generates the content requirements information 248 based on a content requirements information request from the answer acquisition module 230 and where the answer acquisition module 230 generates the content requirements information request based on the query request 136.

The answer acquisition module 230 obtains source requirements information 250 based on the query request 136. The source requirements information 250 includes one or more of candidate source identifiers, a content profile, a desired source parameter, recommended source parameters, source priority levels, and recommended source access timing. For example, the answer acquisition module 230 receives the source requirements information 250 from the source requirements module 234, where the source requirements module 234 generates the source requirements information 250 based on a source requirements information request from the answer acquisition module 230 and where the answer acquisition module 230 generates the source requirements information request based on the query request 136.

The answer acquisition module 230 obtains answer timing information 254 based on the query request 136. The answer timing information 254 includes one or more of requested answer timing, confirmed answer timing, source access testing results, estimated velocity of content updates, content freshness, timestamps, predicted time available, requested content acquisition trigger, and a content acquisition trigger detected indicator. For example, the answer acquisition module 230 receives the answer timing information 254 from the answer timing module 238, where the answer timing module 238 generates the answer timing information 254 based on an answer timing information request from the answer acquisition module 230 and where the answer acquisition module 230 generates the answer timing information request based on the query request 136.

Having obtained the content requirements information 248, the source requirements information 250, and the answer timing information 254, the answer acquisition module 230 determines whether to issue an IEI request 244 and/or a collections request 132 based on one or more of the content requirements information 248, the source requirements information 250, and the answer timing information 254. For example, the answer acquisition module 230 selects the IEI request 244 when an immediate answer to a simple query request 136 is required and is expected to have a favorable quality level. As another example, the answer acquisition module 230 selects the collections request 132 when a longer-term answer is required as indicated by the answer timing information to before and/or when the query request 136 has an unfavorable quality level.

When issuing the IEI request 244, the answer acquisition module 230 generates the IEI request 244 in accordance

17

with security information 252 received from the content security module 236 and based on one or more of the content requirements information 248, the source requirements information 250, and the answer timing information 254. Having generated the IEI request 244, the answer acquisition module 230 sends the IEI request 244 to at least one IEI module.

When issuing the collections request 132, the answer acquisition module 230 generates the collections request 132 in accordance with the security information 252 received from the content security module 236 and based on one or more of the content requirements information 248, the source requirements information 250, and the answer timing information 254. Having generated the collections request 132, the answer acquisition module 230 sends the collections request 132 to at least one collections module. Alternatively, the answer acquisition module 230 facilitate sending of the collections request 132 to one or more various user devices (e.g., to access a subject matter expert).

The answer acquisition module 230 determines a quality level of a received answer extracted from a collections response 134 and/or an IEI response 246. For example, the answer acquisition module 230 extracts the quality level of the received answer from answer quality information 258 received from the answer quality module 242 in response to an answer quality request from the answer acquisition module 230. When the quality level is unfavorable, the answer acquisition module 230 facilitates obtaining a further answer. The facilitation includes issuing at least one of a further IEI request 244 and a further collections request 132 to generate a further answer for further quality testing. When the quality level is favorable, the answer acquisition module 230 issues a query response 140 to the requester. The issuing includes generating the query response 140 based on answer transformation information 256 received from the answer transformation module 240, where the answer transformation module 240 generates the answer transformation information 256 to include a transformed answer based on receiving the answer from the answer acquisition module 230. The answer transformation information 250 6A further include the question, a desired format of the answer, available formats, recommended formatting, received IEI responses, transformation instructions, and transformed IEI responses into an answer.

FIG. 5D is a logic diagram of an embodiment of a method for providing a response to a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5C, and also FIG. 5D. The method includes step 270 where a processing module of one or more processing modules of one or more computing devices of the computing system receives a query request (e.g., a question) from a requester. The method continues at step 272 where the processing module determines content requirements information. The determining includes interpreting the query request to produce the content requirements. The method continues at step 274 where the processing module determines source requirements information. The determining includes interpreting the query request to produce the source requirements. The method continues at step 276 where the processing module determines answer timing information. The determining includes interpreting the query request to produce the answer timing information.

The method continues at step 278 the processing module determines whether to issue an IEI request and/or a collections request. For example, the determining includes select-

18

ing the IEI request when the answer timing information indicates that a simple one-time answer is appropriate. As another example, the processing module selects the collections request when the answer timing information indicates that the answer is associated with a series of events over an event time frame.

When issuing the IEI request, the method continues at step 280 where the processing module issues the IEI request to an IEI module. The issuing includes generating the IEI request in accordance with security information and based on one or more of the content requirements information, the source requirements information, and the answer timing information.

When issuing the collections request, the method continues at step 282 where the processing module issues the collections request to a collections module. The issuing includes generating the collections request in accordance with the security information and based on one or more of the content requirements information, the source requirements information, and the answer timing information. Alternatively, the processing module issues both the IEI request and the collections request when a satisfactory partial answer may be provided based on a corresponding IEI response and a further more generalized and specific answer may be provided based on a corresponding collections response and associated further IEI response.

The method continues at step 284 where the processing module determines a quality level of a received answer. The determining includes extracting the answer from the collections response and/or the IEI response and interpreting the answer in accordance with one or more of the content requirements information, the source requirements information, the answer timing information, and the query request to produce the quality level. The method branches to step 288 when the quality level is favorable and the method continues to step 286 when the quality level is unfavorable. For example, the processing module indicates that the quality level is favorable when the quality level is equal to or greater than a minimum answer quality threshold level. As another example, the processing module indicates that the quality level is unfavorable when the quality level is less than the minimum answer quality threshold level.

When the quality level is unfavorable, the method continues at step 286 where the processing module obtains a further answer. The obtaining includes at least one of issuing a further IEI request and a further collections request to facilitate obtaining of a further answer for further answer quality level testing as the method loops back to step 270. When the quality level is favorable, the method continues at step 288 where the processing module issues a query response to the requester. The issuing includes transforming the answer into a transformed answer in accordance with an answer transformation approach (e.g., formatting, further interpretations of the virtual question in light of the answer and further knowledge) and sending the transformed answer to the requester as the query response.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or

more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. 5E is a schematic block diagram of an embodiment of the identigen entigen intelligence (IEI) module 122 of FIG. 4A that includes a content ingestion module 300, an element identification module 302, and interpretation module 304, and answer resolution module 306, and an IEI control module 308. Generally, an embodiment of this invention presents solutions where the IEI module 122 supports interpreting content to produce knowledge that may be utilized to answer questions.

In an example of operation of the producing and utilizing of the knowledge, the content ingestion module 300 generates formatted content 314 based on question content 312 and/or source content 310, where the IEI module 122 receives an IEI request 244 that includes the question content 312 and the IEI module 122 receives a collections response 134 that includes the source content 310. The source content 310 includes content from a source extracted from the collections response 134. The question content 312 includes content extracted from the IEI request 244 (e.g., content paired with a question). The content ingestion module 300 generates the formatted content 314 in accordance with a formatting approach (e.g., creating proper sentences from words of the content). The formatted content 314 includes modified content that is compatible with subsequent element identification (e.g., complete sentences, combinations of words and interpreted sounds and/or inflection cues with temporal associations of words).

The element identification module 302 processes the formatted content 314 based on element rules 318 and an element list 332 to produce identified element information 340. Rules 316 includes the element rules 318 (e.g., match, partial match, language translation, etc.). Lists 330 includes the element list 332 (e.g., element ID, element context ID, element usage ID, words, characters, symbols etc.). The IEI control module 308 may provide the rules 316 and the lists 330 by accessing stored data 360 from a memory associated with the IEI module 122. Generally, an embodiment of this invention presents solutions where the stored data 360 may further include one or more of a descriptive dictionary, categories, representations of element sets, element list, sequence data, pending questions, pending request, recognized elements, unrecognized elements, errors, etc.

The identified element information 340 includes one or more of identifiers of elements identified in the formatted content 314, may include ordering and/or sequencing and grouping information. For example, the element identification module 302 compares elements of the formatted content 314 to known elements of the element list 332 to produce identifiers of the known elements as the identified element information 340 in accordance with the element rules 318. Alternatively, the element identification module 302 outputs un-identified element information 342 to the IEI control module 308, where the un-identified element information 342 includes temporary identifiers for elements not identifiable from the formatted content 314 when compared to the element list 332.

The interpretation module 304 processes the identified element information 340 in accordance with interpretation rules 320 (e.g., potentially valid permutations of various combinations of identified elements), question information 346 (e.g., a question extracted from the IEI request to hundred 44 which may be paired with content associated with the question), and a groupings list 334 (e.g., representations of associated groups of representations of things, a

set of element identifiers, valid element usage IDs in accordance with similar, an element context, permutations of sets of identifiers for possible interpretations of a sentence or other) to produce interpreted information 344. The interpreted information 344 includes potentially valid interpretations of combinations of identified elements. Generally, an embodiment of this invention presents solutions where the interpretation module 304 supports producing the interpreted information 344 by considering permutations of the identified element information 340 in accordance with the interpretation rules 320 and the groupings list 334.

The answer resolution module 306 processes the interpreted information 344 based on answer rules 322 (e.g., guidance to extract a desired answer), the question information 346, and inferred question information 352 (e.g., posed by the IEI control module or analysis of general collections of content or refinement of a stated question from a request) to produce preliminary answers 354 and an answer quality level 356. The answer generally lies in the interpreted information 344 as both new content received and knowledge based on groupings list 334 generated based on previously received content. The preliminary answers 354 includes an answer to a stated or inferred question that subject further refinement. The answer quality level 356 includes a determination of a quality level of the preliminary answers 354 based on the answer rules 322. The inferred question information 352 may further be associated with time information 348, where the time information includes one or more of current real-time, a time reference associated with entity submitting a request, and a time reference of a collections response.

When the IEI control module 308 determines that the answer quality level 356 is below an answer quality threshold level, the IEI control module 308 facilitates collecting of further content (e.g., by issuing a collections request 132 and receiving corresponding collections responses 134 for analysis). When the answer quality level 356 compares favorably to the answer quality threshold level, the IEI control module 308 issues an IEI response 246 based on the preliminary answers 354. When receiving training information 358, the IEI control module 308 facilitates updating of one or more of the lists 330 and the rules 316 and stores the updated list 330 and the updated rules 316 in the memories as updated stored data 360.

FIG. 5F is a logic diagram of an embodiment of a method for analyzing content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5E, and also FIG. 5F. The method includes step 370 where a processing module of one or more processing modules of one or more computing devices of the computing system facilitates updating of one or more rules and lists based on one or more of received training information and received content. For example, the processing module updates rules with received rules to produce updated rules and updates element lists with received elements to produce updated element lists. As another example, the processing module interprets the received content to identify a new word for at least temporary inclusion in the updated element list.

The method continues at step 372 where the processing module transforms at least some of the received content into formatted content. For example, the processing module processes the received content in accordance with a transformation approach to produce the formatted content, where

21

the formatted content supports compatibility with subsequent element identification (e.g., typical sentence structures of groups of words).

The method continues at step 374 where the processing module processes the formatted content based on the rules and the lists to produce identified element information and/or an identified element information. For example, the processing module compares the formatted content to element lists to identify a match producing identifiers for identified elements or new identifiers for unidentified elements when there is no match.

The method continues at step 376 with a processing module processes the identified element information based on rules, the lists, and question information to produce interpreted information. For example, the processing module compares the identified element information to associated groups of representations of things to generate potentially valid interpretations of combinations of identified elements.

The method continues at step 378 where the processing module processes the interpreted information based on the rules, the question information, and inferred question information to produce preliminary answers. For example, the processing module matches the interpreted information to one or more answers (e.g., embedded knowledge based on a fact base built from previously received content) with highest correctness likelihood levels that is subject to further refinement.

The method continues at step 380 where the processing module generates an answer quality level based on the preliminary answers, the rules, and the inferred question information. For example, the processing module predicts the answer correctness likelihood level based on the rules, the inferred question information, and the question information. The method branches to step 384 when the answer quality level is favorable and the method continues to step 382 when the answer quality level is unfavorable. For example, the generating of the answer quality level further includes the processing module indicating that the answer quality level is favorable when the answer quality level is greater than or equal to a minimum answer quality threshold level. As another example, the generating of the answer quality level further includes the processing module indicating that the answer quality level is unfavorable when the answer quality level is less than the minimum answer quality threshold level.

When the answer quality level is unfavorable, the method continues at step 382 where the processing module facilitates gathering clarifying information. For example, the processing module issues a collections request to facilitate receiving further content and or request question clarification from a question requester. When the answer quality level is favorable, the method continues at step 384 where the processing module issues a response that includes one or more answers based on the preliminary answers and/or further updated preliminary answers based on gathering further content. For example, the processing module generates a response that includes one or more answers and the answer quality level and issues the response to the requester.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element

22

section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. 6A is a schematic block diagram of an embodiment of the element identification module 302 of FIG. 5A and the interpretation module 304 of FIG. 5A. The element identification module 302 includes an element matching module 400 and an element grouping module 402. The interpretation module 304 includes a grouping matching module 404 and a grouping interpretation module 406. Generally, an embodiment of this invention presents solutions where the element identification module 302 supports identifying potentially valid permutations of groupings of elements while the interpretation module 304 interprets the potentially valid permutations of groupings of elements to produce interpreted information that includes the most likely of groupings based on a question.

In an example of operation of the identifying of the potentially valid permutations of groupings of elements, when matching elements of the formatted content 314, the element matching module 400 generates matched elements 412 (e.g., identifiers of elements contained in the formatted content 314) based on the element list 332. For example, the element matching module 400 matches a received element to an element of the element list 332 and outputs the matched elements 412 to include an identifier of the matched element. When finding elements that are unidentified, the element matching module 400 outputs un-recognized words information 408 (e.g., words not in the element list 332, may temporarily add) as part of un-identified element information 342. For example, the element matching module 400 indicates that a match cannot be made between a received element of the formatted content 314, generates the unrecognized words info 408 to include the received element and/or a temporary identifier, and issues and updated element list 414 that includes the temporary identifier and the corresponding unidentified received element.

The element grouping module 402 analyzes the matched elements 412 in accordance with element rules 318 to produce grouping error information 410 (e.g., incorrect sentence structure indicators) when a structural error is detected. The element grouping module 402 produces identified element information 340 when favorable structure is associated with the matched elements in accordance with the element rules 318. The identified element information 340 may further include grouping information of the plurality of permutations of groups of elements (e.g., several possible interpretations), where the grouping information includes one or more groups of words forming an associated set and/or super-group set of two or more subsets when subsets share a common core element.

In an example of operation of the interpreting of the potentially valid permutations of groupings of elements to produce the interpreted information, the grouping matching module 404 analyzes the identified element information 340 in accordance with a groupings list 334 to produce validated groupings information 416. For example, the grouping matching module 404 compares a grouping aspect of the identified element information 340 (e.g., for each permutation of groups of elements of possible interpretations), generates the validated groupings information 416 to include identification of valid permutations aligned with the groupings list 334. Alternatively, or in addition to, the grouping matching module 404 generates an updated groupings list

418 when determining a new valid grouping (e.g., has favorable structure and interpreted meaning) that is to be added to the groupings list **334**.

The grouping interpretation module **406** interprets the validated groupings information **416** based on the question information **346** and in accordance with the interpretation rules **320** to produce interpreted information **344** (e.g., most likely interpretations, next most likely interpretations, etc.). For example, the grouping interpretation module **406** obtains context, obtains favorable historical interpretations, processes the validated groupings based on interpretation rules **320**, where each interpretation is associated with a correctness likelihood level.

FIG. 6B is a logic diagram of an embodiment of a method for interpreting information within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5E-5F, 6A, and also FIG. 6B. The method includes step **430** where a processing module of one or more processing modules of one or more computing devices of the computing system analyzes formatted content. For example, the processing module attempt to match a received element of the formatted content to one or more elements of an elements list. When there is no match, the method branches to step **434** and when there is a match, the method continues to step **432**. When there is a match, the method continues at step **432** where the processing module outputs matched elements (e.g., to include the matched element and/or an identifier of the matched element). When there is no match, the method continues at step **434** where the processing module outputs unrecognized words (e.g., elements and/or a temporary identifier for the unmatched element).

The method continues at step **436** where the processing module analyzes matched elements. For example, the processing module attempt to match a detected structure of the matched elements (e.g., chained elements as in a received sequence) to favorable structures in accordance with element rules. The method branches to step **440** when the analysis is unfavorable and the method continues to step **438** when the analysis is favorable. When the analysis is favorable matching a detected structure to the favorable structure of the element rules, the method continues at step **438** where the processing module outputs identified element information (e.g., an identifier of the favorable structure, identifiers of each of the detected elements). When the analysis is unfavorable matching a detected structure to the favorable structure of the element rules, the method continues at step **440** where the processing module outputs grouping error information (e.g., a representation of the incorrect structure, identifiers of the elements of the incorrect structure, a temporary new identifier of the incorrect structure).

The method continues at step **442** where the processing module analyzes the identified element information to produce validated groupings information. For example, the processing module compares a grouping aspect of the identified element information and generates the validated groupings information to include identification of valid permutations that align with the groupings list. Alternatively, or in addition to, the processing module generates an updated groupings list when determining a new valid grouping.

The method continues at step **444** where the processing module interprets the validated groupings information to produce interpreted information. For example, the processing module obtains one or more of context and historical interpretations and processes the validated groupings based

on interpretation rules to generate the interpreted information, where each interpretation is associated with a correctness likelihood level (e.g., a quality level).

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. 6C is a schematic block diagram of an embodiment of the answer resolution module **306** of FIG. 5A that includes an interim answer module **460**, and answer prioritization module **462**, and a preliminary answer quality module **464**. Generally, an embodiment of this invention presents solutions where the answer resolution module **306** supports producing an answer for interpreted information **344**.

In an example of operation of the providing of the answer, the interim answer module **460** analyzes the interpreted information **344** based on question information **346** and inferred question information **352** to produce interim answers **466** (e.g., answers to stated and/or inferred questions without regard to rules that is subject to further refinement). The answer prioritization module **462** analyzes the interim answers **466** based on answer rules **322** to produce preliminary answer **354**. For example, the answer prioritization module **462** identifies all possible answers from the interim answers **466** that conform to the answer rules **322**.

The preliminary answer quality module **464** analyzes the preliminary answers **354** in accordance with the question information **346**, the inferred question information **352**, and the answer rules **322** to produce an answer quality level **356**. For example, for each of the preliminary answers **354**, the preliminary answer quality module **464** may compare a fit of the preliminary answer **354** to a corresponding previous answer and question quality level, calculate the answer quality level **356** based on a level of conformance to the answer rules **322**, calculate the answer quality level **356** based on alignment with the inferred question information **352**, and determine the answer quality level **356** based on an interpreted correlation with the question information **346**.

FIG. 6D is a logic diagram of an embodiment of a method for producing an answer within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5E-5F, 6C, and also FIG. 6D. The method includes step **480** where a processing module of one or more processing modules of one or more computing devices of the computing system analyzes received interpreted information based on question information and inferred question information to produce one or more interim answers. For example, the processing module generates potential answers based on patterns consistent with previously produced knowledge and likelihood of correctness.

The method continues at step **482** where the processing module analyzes the one or more interim answers based on answer rules to produce preliminary answers. For example,

the processing module identifies all possible answers from the interim answers that conform to the answer rules. The method continues at step 484 where the processing module analyzes the preliminary answers in accordance with the question information, the inferred question information, and the answer rules to produce an answer quality level. For example, for each of the elementary answers, the processing module may compare a fit of the preliminary answer to a corresponding previous answer-and-answer quality level, calculate the answer quality level based on performance to the answer rules, calculate answer quality level based on alignment with the inferred question information, and determine the answer quality level based on interpreted correlation with the question information.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. 7A is an information flow diagram for interpreting information within a computing system, where sets of entigens 504 are interpreted from sets of identigens 502 which are interpreted from sentences of words 500. Such identigen entigen intelligence (IP processing of the words (e.g., to IEI process) includes producing one or more of interim knowledge, a preliminary answer, and an answer quality level. For example, the IEI processing includes identifying permutations of identigens of a phrase of a sentence (e.g., interpreting human expressions to produce identigen groupings for each word of ingested content), reducing the permutations of identigens (e.g., utilizing rules to eliminate unfavorable permutations), mapping the reduced permutations of identigens to at least one set of entigens (e.g., most likely identigens become the entigens) to produce the interim knowledge, processing the knowledge in accordance with a knowledge database (e.g., comparing the set of entigens to the knowledge database) to produce a preliminary answer, and generating the answer quality level based on the preliminary answer for a corresponding domain.

Human expressions are utilized to portray facts and fiction about the real world. The real-world includes items, actions, and attributes. The human expressions include textual words, textual symbols, images, and other sensorial information (e.g., sounds). It is known that many words, within a given language, can mean different things based on groupings and orderings of the words. For example, the sentences of words 500 can include many different forms of sentences that mean vastly different things even when the words are very similar.

The present invention presents solutions where the computing system 10 supports producing a computer-based representation of a truest meaning possible of the human expressions given the way that multitudes of human expressions relate to these meanings. As a first step of the flow diagram to transition from human representations of things to a most precise computer representation of the things, the computer identifies the words, phrases, sentences, etc. from the human expressions to produce the sets of identigens 502.

Each identigen includes an identifier of their meaning and an identifier of an instance for each possible language, culture, etc. For example, the words car and automobile share a common meaning identifier but have different instance identifiers since they are different words and are spelled differently. As another example, the word duck is associated both with a bird and an action to elude even though they are spelled the same. In this example the bird duck has a different meaning than the elude duck and as such each has a different meaning identifier of the corresponding identigens.

As a second step of the flow diagram to transition from human representations of things to the most precise computer representation of the things, the computer extracts meaning from groupings of the identified words, phrases, sentences, etc. to produce the sets of entigens 504. Each entigen includes an identifier of a single conceivable and perceivable thing in space and time (e.g., independent of language and other aspects of the human expressions). For example, the words car and automobile are different instances of the same meaning and point to a common shared entigen. As another example, the word duck for the bird meaning has an associated unique entigen that is different than the entigen for the word duck for the elude meaning.

As a third step of the flow diagram to transition from human expressions of things to the most precise computer representation of the things, the computer reasons facts from the extracted meanings. For example, the computer maintains a fact-based of the valid meanings from the valid groupings or sets of entigens so as to support subsequent inferences, deductions, rationalizations of posed questions to produce answers that are aligned with a most factual view. As time goes on, and as an entigen has been identified, it can encounter an experience transformations in time, space, attributes, actions, and words which are used to identify it without creating contradictions or ever losing its identity.

FIG. 7B is a relationship block diagram illustrating an embodiment of relationships between things 510 and representations of things 512 within a computing system. The things 510 includes conceivable and perceivable things including actions 522, items 524, and attributes 526. The representation of things 512 includes representations of things used by humans 514 and representation of things used by computing devices 516 of embodiments of the present invention. The things 510 relates to the representations of things used by humans 514 where the invention presents solutions where the computing system 10 supports mapping the representations of things used by humans 514 to the representations of things used by computing devices 516, where the representations of things used by computing devices 516 map back to the things 510.

The representations of things used by humans 514 includes textual words 528, textual symbols 530, images (e.g., non-textual) 532, and other sensorial information 534 (e.g., sounds, sensor data, electrical fields, voice inflections, emotion representations, facial expressions, whistles, etc.). The representations of things used by computing devices 516 includes identigens 518 and entigens 520. The representations of things used by humans 514 maps to the identigens 518 and the identigens 518 map to the entigens 520. The entigens 520 uniquely maps back to the things 510 in space and time, a truest meaning the computer is looking for to create knowledge and answer questions based on the knowledge.

To accommodate the mapping of the representations of things used by humans 514 to the identigens 518, the

identigens **518** is partitioned into actenymys **544** (e.g., actions), itenymys **546** (e.g., items), attrenymys **548** (e.g., attributes), and functionals **550** (e.g., that join and/or describe). Each of the actenymys **544**, itenymys **546**, and attrenymys **548** may be further classified into singulatums **552** (e.g., identify one unique entigen) and pluratums **554** (e.g., identify a plurality of entigens that have similarities).

Each identigen **518** is associated with an identigens identifier (IDN) **536**. The IDN **536** includes a meaning identifier (ID) **538** portion, an instance ID **540** portion, and a type ID **542** portion. The meaning ID **538** includes an identifier of common meaning. The instance ID **540** includes an identifier of a particular word and language. The type ID **542** includes one or more identifiers for actenymys, itenymys, attrenymys, singulatums, pluratums, a time reference, and any other reference to describe the IDN **536**. The mapping of the representations of things used by humans **514** to the identigens **518** by the computing system of the present invention includes determining the identigens **518** in accordance with logic and instructions for forming groupings of words.

Generally, an embodiment of this invention presents solutions where the identigens **518** map to the entigens **520**. Multiple identigens may map to a common unique entigen. The mapping of the identigens **518** to the entigens **520** by the computing system of the present invention includes determining entigens in accordance with logic and instructions for forming groupings of identigens.

FIG. 7C is a diagram of an embodiment of a synonym words table **570** within a computing system, where the synonym words table **570** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538** and the instance ID **540**. The computing system of the present invention may utilize the synonym words table **570** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the words car, automobile, auto, bil (Swedish), carro (Spanish), and bil (Danish) all share a common meaning but are different instances (e.g., different words and languages). The words map to a common meaning ID but to individual unique instant identifiers. Each of the different identigens map to a common entigen since they describe the same thing.

FIG. 7D is a diagram of an embodiment of a polysemous words table **576** within a computing system, where the polysemous words table **576** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538** and the instance ID **540**. The computing system of the present invention may utilize the polysemous words table **576** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the word duck maps to four different identigens since the word duck has four associated different meanings (e.g., bird, fabric, to submerge, to elude) and instances. Each of the identigens represent different things and hence map to four different entigens.

FIG. 7E is a diagram of an embodiment of transforming words into groupings within a computing system that includes a words table **580**, a groupings of words section to validate permutations of groupings, and a groupings table **584** to capture the valid groupings. The words table **580** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538**, the instance ID **540**, and the type ID **542**. The computing system of the present invention may utilize the words table **580** to map textual words **572** to identigens **518** and map the identigens

518 to entigens **520**. For example, the word pilot may refer to a flyer and the action to fly. Each meaning has a different identigen and different entigen.

The computing system the present invention may apply rules to the fields of the words table **580** to validate various groupings of words. Those that are invalid are denoted with a "X" while those that are valid are associated with a check mark. For example, the grouping "pilot Tom" is invalid when the word pilot refers to flying and Tom refers to a person. The identigen combinations for the flying pilot and the person Tom are denoted as invalid by the rules. As another example, the grouping "pilot Tom" is valid when the word pilot refers to a flyer and Tom refers to the person. The identigen combinations for the flyer pilot and the person Tom are denoted as valid by the rules.

The groupings table **584** includes multiple fields including grouping ID **586**, word strings **588**, identigens **518**, and entigens **520**. The computing system of the present invention may produce the groupings table **584** as a stored fact base for valid and/or invalid groupings of words identified by their corresponding identigens. For example, the valid grouping "pilot Tom" referring to flyer Tom the person is represented with a grouping identifier of 3001 and identity and identifiers 150.001 and 457.001. The entigen field **520** may indicate associated entigens that correspond to the identigens. For example, entigen e717 corresponds to the flyer pilot meaning and entigen e61 corresponds to the time the person meaning. Alternatively, or in addition to, the entigen field **520** may be populated with a single entigen identifier (ENI).

The word strings field **588** may include any number of words in a string. Different ordering of the same words can produce multiple different strings and even different meanings and hence entigens. More broadly, each entry (e.g., role) of the groupings table **584** may refer to groupings of words, two or more word strings, an idiom, just identigens, just entigens, and/or any combination of the preceding elements. Each entry has a unique grouping identifier. An idiom may have a unique grouping ID and include identifiers of original word identigens and replacing identigens associated with the meaning of the idiom not just the meaning of the original words. Valid groupings may still have ambiguity on their own and may need more strings and/or context to select a best fit when interpreting a truest meaning of the grouping.

FIG. 8A is a data flow diagram for accumulating knowledge within a computing system, where a computing device, at a time= t_0 , ingests and processes facts **598** at a step **590** based on rules **316** and fact base information **600** to produce groupings **602** for storage in a fact base **592** (e.g., words, phrases, word groupings, identigens, entigens, quality levels). The facts **598** may include information from books, archive data, Central intelligence agency (CIA) world fact book, trusted content, etc. The ingesting may include filtering to organize and promote better valid groupings detection (e.g., considering similar domains together). The groupings **602** includes one or more of groupings identifiers, identigen identifiers, entigen identifiers, and estimated fit quality levels. The processing step **590** may include identifying identigens from words of the facts **598** in accordance with the rules **316** and the fact base info **600** and identifying groupings utilizing identigens in accordance with rules **316** and fact base info **600**.

Subsequent to ingestion and processing of the facts **598** to establish the fact base **592**, at a time= t_1 +, the computing device ingests and processes new content **604** at a step **594** in accordance with the rules **316** and the fact base informa-

tion 600 to produce preliminary grouping 606. The new content may include updated content (e.g., timewise) from periodicals, newsfeeds, social media, etc. The preliminary grouping 606 includes one or more of preliminary groupings identifiers, preliminary identigen identifiers, preliminary entigen identifiers, estimated fit quality levels, and representations of unidentified words.

The computing device validates the preliminary groupings 606 at a step 596 based on the rules 316 and the fact base info 600 to produce updated fact base info 608 for storage in the fact base 592. The validating includes one or more of reasoning a fit of existing fact base info 600 with the new preliminary grouping 606, discarding preliminary groupings, updating just time frame information associated with an entry of the existing fact base info 600 (e.g., to validate knowledge for the present), creating new entigens, and creating a median entigen to summarize portions of knowledge within a median indicator as a quality level indicator (e.g., suggestive not certain).

Storage of the updated fact base information 608 captures patterns that develop by themselves instead of searching for patterns as in prior art artificial intelligence systems. Growth of the fact base 592 enables subsequent reasoning to create new knowledge including deduction, induction, inference, and inferential sentiment (e.g., a chain of sentiment sentences). Examples of sentiments includes emotion, beliefs, convictions, feelings, judgments, notions, opinions, and views.

FIG. 8B is a diagram of an embodiment of a groupings table 620 within a computing system. The groupings table 620 includes multiple fields including grouping ID 586, word strings 588, an IF string 622 and a THEN string 624. Each of the fields for the IF string 622 and the THEN string 624 includes fields for an identigen (IDN) string 626, and an entigen (ENI) string 628. The computing system of the present invention may produce the groupings table 620 as a stored fact base to enable IF THEN based inference to generate a new knowledge inference 630.

As a specific example, grouping 5493 points out the logic of IF someone has a tumor, THEN someone is sick and the grouping 5494 points of the logic that IF someone is sick, THEN someone is sad. As a result of utilizing inference, the new knowledge inference 630 may produce grouping 5495 where IF someone has a tumor, THEN someone is possibly sad (e.g., or is sad).

FIG. 8C is a data flow diagram for answering questions utilizing accumulated knowledge within a computing system, where a computing device ingests and processes question information 346 at a step 640 based on rules 316 and fact base info 600 from a fact base 592 to produce preliminary grouping 606. The ingesting and processing questions step 640 includes identifying identigens from words of a question in accordance with the rules 316 and the fact base information 600 and may also include identifying groupings from the identified identigens in accordance with the rules 316 and the fact base information 600.

The computing device validates the preliminary grouping 606 at a step 596 based on the rules 316 and the fact base information 600 to produce identified element information 340. For example, the computing device reasons fit of existing fact base information with new preliminary groupings 606 to produce the identified element information 340 associated with highest quality levels. The computing device interprets a question of the identified element information 340 at a step 642 based on the rules 316 and the fact base information 600. The interpreting of the question may

include separating new content from the question and reducing the question based on the fact base information 600 and the new content.

The computing device produces preliminary answers 354 from the interpreted information 344 at a resolve answer step 644 based on the rules 316 and the fact base information 600. For example, the computing device compares the interpreted information 344 two the fact base information 600 to produce the preliminary answers 354 with highest quality levels utilizing one or more of deduction, induction, inferencing, and applying inferential sentiments logic. Alternatively, or in addition to, the computing device may save new knowledge identified from the question information 346 to update the fact base 592.

FIG. 8D is a data flow diagram for answering questions utilizing interference within a computing system that includes a groupings table 648 and the resolve answer step 644 of FIG. 8C. The groupings table 648 includes multiple fields including fields for a grouping (GRP) identifier (ID) 586, word strings 588, an identigen (IDN) string 626, and an entigen (ENI) 628. The groupings table 648 may be utilized to build a fact base to enable resolving a future question into an answer. For example, the grouping 8356 notes knowledge that Michael sleeps eight hours and grouping 8357 notes that Michael usually starts to sleep at 11 PM.

In a first question example that includes a question “Michael sleeping?”, the resolve answer step 644 analyzes the question from the interpreted information 344 in accordance with the fact base information 600, the rules 316, and a real-time indicator that the current time is 1 AM to produce a preliminary answer of “possibly YES” when inferring that Michael is probably sleeping at 1 AM when Michael usually starts sleeping at 11 PM and Michael usually sleeps for a duration of eight hours.

In a second question example that includes the question “Michael sleeping?”, the resolve answer step 644 analyzes the question from the interpreted information 344 in accordance with the fact base information 600, the rules 316, and a real-time indicator that the current time is now 11 AM to produce a preliminary answer of “possibly NO” when inferring that Michael is probably not sleeping at 11 AM when Michael usually starts sleeping at 11 PM and Michael usually sleeps for a duration of eight hours.

FIG. 8E is a relationship block diagram illustrating another embodiment of relationships between things and representations of things within a computing system. While things in the real world are described with words, it is often the case that a particular word has multiple meanings in isolation. Interpreting the meaning of the particular word may hinge on analyzing how the word is utilized in a phrase, a sentence, multiple sentences, paragraphs, and even whole documents or more. Describing and stratifying the use of words, word types, and possible meanings help in interpreting a true meaning.

Humans utilize textual words 528 to represent things in the real world. Quite often a particular word has multiple instances of different grammatical use when part of a phrase of one or more sentences. The grammatical use 649 of words includes the nouns and the verbs, and also includes adverbs, adjectives, pronouns, conjunctions, prepositions, determiners, exclamations, etc.

As an example of multiple grammatical use, the word “bat” in the English language can be utilized as a noun or a verb. For instance, when utilized as a noun, the word “bat” may apply to a baseball bat or may apply to a flying “bat.”

As another instance, when utilized as a verb, the word “bat” may apply to the action of hitting or batting an object, i.e., “bat the ball.”

To stratify word types by use, the words are associated with a word type (e.g., type identifier **542**). The word types include objects (e.g., items **524**), characteristics (e.g., attributes **526**), actions **522**, and the functionals **550** for joining other words and describing words. For example, when the word “bat” is utilized as a noun, the word is describing the object of either the baseball bat or the flying bat. As another example, when the word “bat” is utilized as a verb, the word is describing the action of hitting.

To determine possible meanings, the words, by word type, are mapped to associative meanings (e.g., identigens **518**). For each possible associative meaning, the word type is documented with the meaning and further with an identifier (ID) of the instance (e.g., an identigen identifier).

For the example of the word “bat” when utilized as a noun for the baseball bat, a first identigen identifier **536-1** includes a type ID **542-1** associated with the object **524**, an instance ID **540-1** associated with the first identigen identifier (e.g., unique for the baseball bat), and a meaning ID **538-1** associated with the baseball bat. For the example of the word “bat” when utilized as a noun for the flying bat, a second identigen identifier **536-2** includes a type ID **542-1** associated with the object **524**, an instance ID **540-2** associated with the second identigen identifier (e.g., unique for the flying bat), and a meaning ID **538-2** associated with the flying bat. For the example of the word “bat” when utilized as a verb for the bat that hits, a third identigen identifier **536-2** includes a type ID **542-2** associated with the actions **522**, an instance ID **540-3** associated with the third identigen identifier (e.g., unique for the bat that hits), and a meaning ID **538-3** associated with the bat that hits.

With the word described by a type and possible associative meanings, a combination of full grammatical use of the word within the phrase etc., application of rules, and utilization of an ever-growing knowledge database that represents knowledge by linked entigens, the absolute meaning (e.g., entigen **520**) of the word is represented as a unique entigen. For example, a first entigen e1 represents the absolute meaning of a baseball bat (e.g., a generic baseball bat not a particular baseball bat that belongs to anyone), a second entigen e2 represents the absolute meaning of the flying bat (e.g., a generic flying bat not a particular flying bat), and a third entigen e3 represents the absolute meaning of the verb bat (e.g., to hit).

An embodiment of methods to ingest text to produce absolute meanings for storage in a knowledge database are discussed in greater detail with reference to FIGS. **8F-H**. Those embodiments further discuss the discerning of the grammatical use, the use of the rules, and the utilization of the knowledge database to definitively interpret the absolute meaning of a string of words.

Another embodiment of methods to respond to a query to produce an answer based on knowledge stored in the knowledge database are discussed in greater detail with reference to FIGS. **8J-L**. Those embodiments further discuss the discerning of the grammatical use, the use of the rules, and the utilization of the knowledge database to interpret the query. The query interpretation is utilized to extract the answer from the knowledge database to facilitate forming the query response.

FIGS. **8F** and **8G** are schematic block diagrams of another embodiment of a computing system that includes the content ingestion module **300** of FIG. **5E**, the element identification module **302** of FIG. **5E**, the interpretation module **304** of

FIG. **5E**, the IEI control module **308** of FIG. **5E**, and the SS memory **96** of FIG. **2**. Generally, an embodiment of this invention provides presents solutions where the computing system **10** supports processing content to produce knowledge for storage in a knowledge database.

The processing of the content to produce the knowledge includes a series of steps. For example, a first step includes identifying words of an ingested phrase to produce tokenized words. As depicted in FIG. **8F**, a specific example of the first step includes the content ingestion module **300** comparing words of source content **310** to dictionary entries to produce formatted content **314** that includes identifiers of known words. Alternatively, when a comparison is unfavorable, the temporary identifier may be assigned to an unknown word. For instance, the content ingestion module **300** produces identifiers associated with the words “the”, “black”, “bat”, “eats”, and “fruit” when the ingested phrase includes “The black bat eats fruit”, and generates the formatted content **314** to include the identifiers of the words.

A second step of the processing of the content to produce the knowledge includes, for each tokenized word, identifying one or more identigens that correspond the tokenized word, where each identigen describes one of an object, a characteristic, and an action. As depicted in FIG. **8F**, a specific example of the second step includes the element identification module **302** performing a look up of identigen identifiers, utilizing an element list **332** and in accordance with element rules **318**, of the one or more identigens associated with each tokenized word of the formatted content **314** to produce identified element information **340**.

A unique identifier is associated with each of the potential object, the characteristic, and the action (OCA) associated with the tokenized word (e.g. sequential identigens). For instance, the element identification module **302** identifies a functional symbol for “the”, identifies a single identigen for “black”, identifies two identigens for “bat” (e.g., baseball bat and flying bat), identifies a single identigen for “eats”, and identifies a single identigen for “fruit.” When at least one tokenized word is associated with multiple identigens, two or more permutations of sequential combinations of identigens for each tokenized word result. For example, when “bat” is associated with two identigens, two permutations of sequential combinations of identigens result for the ingested phrase.

A third step of the processing of the content to produce the knowledge includes, for each permutation of sequential combinations of identigens, generating a corresponding equation package (i.e., candidate interpretation), where the equation package includes a sequential linking of pairs of identigens (e.g., relationships), where each sequential linking pairs a preceding identigen to a next identigen, and where an equation element describes a relationship between paired identigens (OCAs) such as describes, acts on, is a, belongs to, did, did to, etc. Multiple OCAs occur for a common word when the word has multiple potential meanings (e.g., a baseball bat, a flying bat).

As depicted in FIG. **8F**, a specific example of the third step includes the interpretation module **304**, for each permutation of identigens of each tokenized word of the identified element information **340**, the interpretation module **304** generates, in accordance with interpretation rules **320** and a groupings list **334**, an equation package to include one or more of the identifiers of the tokenized words, a list of identifiers of the identigens of the equation package, a list of pairing identifiers for sequential pairs of identigens, and a quality metric associated with each sequential pair of identigens (e.g., likelihood of a proper interpretation). For

instance, the interpretation module **304** produces a first equation package that includes a first identigen pairing of a black bat (e.g., flying bat with a higher quality metric level), the second pairing of bat eats (e.g., the flying bat eats, with a higher quality metric level), and a third pairing of eats fruit, and the interpretation module **304** produces a second equation package that includes a first pairing of a black bat (e.g., baseball bat, with a neutral quality metric level), the second pairing of bat eats (e.g., the baseball bat eats, with a lower quality metric level), and a third pairing of eats fruit.

A fourth step of the processing of the content to produce the knowledge includes selecting a surviving equation package associated with a most favorable confidence level. As depicted in FIG. **8F**, a specific example of the fourth step includes the interpretation module **304** applying interpretation rules **320** (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce a number of permutations of the sequential combinations of identigens to produce interpreted information **344** that includes identification of at least one equation package as a surviving interpretation SI (e.g., higher quality metric level).

Non-surviving equation packages are eliminated that compare unfavorably to pairing rules and/or are associated with an unfavorable quality metric levels to produce a non-surviving interpretation NSI **2** (e.g., lower quality metric level), where an overall quality metric level may be assigned to each equation package based on quality metric levels of each pairing, such that a higher quality metric level of an equation package indicates a higher probability of a most favorable interpretation. For instance, the interpretation module **304** eliminates the equation package that includes the second pairing indicating that the “baseball bat eats” which is inconsistent with a desired quality metric level of one or more of the groupings list **334** and the interpretation rules **320** and selects the equation package associated with the “flying bat eats” which is favorably consistent with the one or more of the quality metric levels of the groupings list **334** and the interpretation rules **320**.

A fifth step of the processing of the content to produce the knowledge utilizing the confidence level includes integrating knowledge of the surviving equation package into a knowledge database. For example, integrating at least a portion of the reduced OCA combinations into a graphical database to produce updated knowledge. As another example, the portion of the reduced OCA combinations may be translated into rows and columns entries when utilizing a rows and columns database rather than a graphical database. When utilizing the rows and columns approach for the knowledge database, subsequent access to the knowledge database may utilize structured query language (SQL) queries.

As depicted in FIG. **8G**, a specific example of the fifth step includes the IEI control module **308** recovering fact base information **600** from SS memory **96** to identify a portion of the knowledge database for potential modification utilizing the OCAs of the surviving interpretation SI **1** (i.e., compare a pattern of relationships between the OCAs of the surviving interpretation SI **1** from the interpreted information **344** to relationships of OCAs of the portion of the knowledge database including potentially new quality metric levels).

The fifth step further includes determining modifications (e.g., additions, subtractions, further clarifications required when information is complex, etc.) to the portion of the knowledge database based on the new quality metric levels. For instance, the IEI control module **308** causes adding the

element “black” as a “describes” relationship of an existing bat OCA and adding the element “fruit” as an eats “does to” relationship to implement the modifications to the portion of the fact base information **600** to produce updated fact base information **608** for storage in the SS memory **96**.

FIG. **8H** is a logic diagram of an embodiment of a method for processing content to produce knowledge for storage within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8E**, **8F**, and also FIG. **8G**. The method includes step **650** where a processing module of one or more processing modules of one or more computing devices of the computing system identifies words of an ingested phrase to produce tokenized words. The identified includes comparing words to known words of dictionary entries to produce identifiers of known words.

For each tokenized word, the method continues at step **651** where the processing module identifies one or more identigens that corresponds to the tokenized word, where each identigen describes one of an object, a characteristic, and an action (e.g., OCA). The identifying includes performing a lookup of identifiers of the one or more identigens associated with each tokenized word, where the different identifiers associated with each of the potential object, the characteristic, and the action associated with the tokenized word.

The method continues at step **652** where the processing module, for each permutation of sequential combinations of identigens, generates a plurality of equation elements to form a corresponding equation package, where each equation element describes a relationship between sequentially linked pairs of identigens, where each sequential linking pairs a preceding identigen to a next identigen. For example, for each permutation of identigens of each tokenized word, the processing module generates the equation package to include a plurality of equation elements, where each equation element describes the relationship (e.g., describes, acts on, is, belongs to, did, did too, etc.) between sequentially adjacent identigens of a plurality of sequential combinations of identigens. Each equation element may be further associated with a quality metric to evaluate a favorability level of an interpretation in light of the sequence of identigens of the equation package.

The method continues at step **653** where the processing module selects a surviving equation package associated with most favorable interpretation. For example, the processing module applies rules (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens), to reduce the number of permutations of the sequential combinations of identigens to identify at least one equation package, where non-surviving equation packages are eliminated the compare unfavorably to pairing rules and/or are associated with an unfavorable quality metric levels to produce a non-surviving interpretation, where an overall quality metric level is assigned to each equation package based on quality metric levels of each pairing, such that a higher quality metric level indicates an equation package with a higher probability of favorability of correctness.

The method continues at step **654** where the processing module integrates knowledge of the surviving equation package into a knowledge database. For example, the processing module integrates at least a portion of the reduced OCA combinations into a graphical database to produce updated knowledge. The integrating may include recovering fact base information from storage of the knowledge data-

35

base to identify a portion of the knowledge database for potential modifications utilizing the OCAs of the surviving equation package (i.e., compare a pattern of relationships between the OCAs of the surviving equation package to relationships of the OCAs of the portion of the knowledge database including potentially new quality metric levels). The integrating further includes determining modifications (e.g., additions, subtractions, further clarifications required when complex information is presented, etc.) to produce the updated knowledge database that is based on fit of acceptable quality metric levels, and implementing the modifications to the portion of the fact base information to produce the updated fact base information for storage in the portion of the knowledge database.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

FIGS. 8J and 8K are schematic block diagrams of another embodiment of a computing system that includes the content ingestion module 300 of FIG. 5E, the element identification module 302 of FIG. 5E, the interpretation module 304 of FIG. 5E, the answer resolution module 306 of FIG. 5E, and the SS memory 96 of FIG. 2. Generally, an embodiment of this invention provides solutions where the computing system 10 supports for generating a query response to a query utilizing a knowledge database.

The generating of the query response to the query includes a series of steps. For example, a first step includes identifying words of an ingested query to produce tokenized words. As depicted in FIG. 8J, a specific example of the first step includes the content ingestion module 300 comparing words of query info 138 to dictionary entries to produce formatted content 314 that includes identifiers of known words. For instance, the content ingestion module 300 produces identifiers for each word of the query "what black animal flies and eats fruit and insects?"

A second step of the generating of the query response to the query includes, for each tokenized word, identifying one or more identigens that correspond the tokenized word, where each identigen describes one of an object, a characteristic, and an action (OCA). As depicted in FIG. 8J, a specific example of the second step includes the element identification module 302 performing a look up of identifiers, utilizing an element list 332 and in accordance with element rules 318, of the one or more identigens associated with each tokenized word of the formatted content 314 to produce identified element information 340. A unique identifier is associated with each of the potential object, the characteristic, and the action associated with a particular tokenized word. For instance, the element identification module 302 produces a single identigen identifier for each of the black color, an animal, flies, eats, fruit, and insects.

A third step of the generating of the query response to the query includes, for each permutation of sequential combinations of identigens, generating a corresponding equation package (i.e., candidate interpretation). The equation pack-

36

age includes a sequential linking of pairs of identigens, where each sequential linking pairs a preceding identigen to a next identigen. An equation element describes a relationship between paired identigens (OCAs) such as describes, acts on, is a, belongs to, did, did to, etc.

As depicted in FIG. 8J, a specific example of the third step includes the interpretation module 304, for each permutation of identigens of each tokenized word of the identified element information 340, generating the equation packages in accordance with interpretation rules 320 and a groupings list 334 to produce a series of equation elements that include pairings of identigens. For instance, the interpretation module 304 generates a first pairing to describe a black animal, a second pairing to describe an animal that flies, a third pairing to describe flies and eats, a fourth pairing to describe eats fruit, and a fifth pairing to describe eats fruit and insects.

A fourth step of the generating the query response to the query includes selecting a surviving equation package associated with a most favorable interpretation. As depicted in FIG. 8J, a specific example of the fourth step includes the interpretation module 304 applying the interpretation rules 320 (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce the number of permutations of the sequential combinations of identigens to produce interpreted information 344. The interpreted information 344 includes identification of at least one equation package as a surviving interpretation SI 10, where non-surviving equation packages, if any, are eliminated that compare unfavorably to pairing rules to produce a non-surviving interpretation.

A fifth step of the generating the query response to the query includes utilizing a knowledge database, generating a query response to the surviving equation package of the query, where the surviving equation package of the query is transformed to produce query knowledge for comparison to a portion of the knowledge database. An answer is extracted from the portion of the knowledge database to produce the query response.

As depicted in FIG. 8K, a specific example of the fifth step includes the answer resolution module 306 interpreting the surviving interpretation SI 10 of the interpreted information 344 in accordance with answer rules 322 to produce query knowledge QK 10 (i.e., a graphical representation of knowledge when the knowledge database utilizes a graphical database). For example, the answer resolution module 306 accesses fact base information 600 from the SS memory 96 to identify the portion of the knowledge database associated with a favorable comparison of the query knowledge QK 10 (e.g., by comparing attributes of the query knowledge QK 10 to attributes of the fact base information 600), and generates preliminary answers 354 that includes the answer to the query. For instance, the answer is "bat" when the associated OCAs of bat, such as black, eats fruit, eats insects, is an animal, and flies, aligns with OCAs of the query knowledge.

FIG. 8L is a logic diagram of an embodiment of a method for generating a query response to a query utilizing knowledge within a knowledge database within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8D, 8J, and also FIG. 8K. The method includes step 655 where a processing module of one or more processing modules of one or more computing devices of the computing system identifies words of an ingested query to produce tokenized words. For

example, the processing module compares words to known words of dictionary entries to produce identifiers of known words.

For each tokenized word, the method continues at step 656 where the processing module identifies one or more identigens that correspond to the tokenized word, where each identigen describes one of an object, a characteristic, and an action. For example, the processing module performs a lookup of identifiers of the one or more identigens associated with each tokenized word, where different identifiers associated with each permutation of a potential object, characteristic, and action associated with the tokenized word.

For each permutation of sequential combinations of identigens, the method continues at step 657 where the processing module generates a plurality of equation elements to form a corresponding equation package, where each equation element describes a relationship between sequentially linked pairs of identigens. Each sequential linking pairs a preceding identigen to a next identigen. For example, for each permutation of identigens of each tokenized word, the processing module includes all other permutations of all other tokenized words to generate the equation packages. Each equation package includes a plurality of equation elements describing the relationships between sequentially adjacent identigens of a plurality of sequential combinations of identigens.

The method continues at step 658 where the processing module selects a surviving equation package associated with a most favorable interpretation. For example, the processing module applies rules (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce the number of permutations of the sequential combinations of identigens to identify at least one equation package. Non-surviving equation packages are eliminated the compare unfavorably to pairing rules.

The method continues at step 659 where the processing module generates a query response to the surviving equation package, where the surviving equation package is transformed to produce query knowledge for locating the portion of a knowledge database that includes an answer to the query. As an example of generating the query response, the processing module interprets the surviving the equation package in accordance with answer rules to produce the query knowledge (e.g., a graphical representation of knowledge when the knowledge database utilizes a graphical database format).

The processing module accesses fact base information from the knowledge database to identify the portion of the knowledge database associated with a favorable comparison of the query knowledge (e.g., favorable comparison of attributes of the query knowledge to the portion of the knowledge database, aligning favorably comparing entigens without conflicting entigens). The processing module extracts an answer from the portion of the knowledge database to produce the query response.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing

modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

FIGS. 9A and 9B are schematic block diagrams of another embodiment of a computing system illustrating an embodiment of a method for interpreting an image within the computing system. The computing system includes the content ingestion module 300 of FIG. 5E, the element identification module 302 of FIG. 5E, the interpretation module 304 of FIG. 5E, the answer resolution module 306 of FIG. 5E, and a knowledge database 664. The knowledge database 664 may be implemented utilizing one or more of the memories of FIG. 2.

FIG. 9A illustrates an example of operation of the method for interpreting the image where the interpretation module 304 obtains a set of image segment identigens for each of a plurality of image segments of an image to produce a plurality of sets of image segment identigens. A first set of image segment identigens is a set of possible interpretations of a first image segment of the plurality of image segments. The obtaining of the set of image segment identigens for each of the plurality of image segments of the image to produce the plurality of sets of image segment identigens includes a variety of approaches.

A first approach includes the interpretation module 304 receiving the plurality of sets of image segment identigens with the image. For instance, the interpretation module 304 receives sets of image segment identigens 674 along with image 660.

A second approach includes the interpretation module 304 accessing, using an identifier of the image, the knowledge database 664 to recover the plurality of sets of image segment identigens. For instance, the interpretation module 304 utilizes the identifier of the image 660 to access the knowledge database 664 to recover the sets of image segment identigens 674 when the knowledge database 664 stores the sets of image segment identigens 674 indexed by the identifier of the image.

A third approach includes the interpretation module 304 interpreting a response message to produce the plurality of sets of image segment identigens. The response message is in response to a query that includes a representation of the image. For instance, the interpretation module 304 generates the query to include the identifier of the image 660, sends the query to the knowledge database 664, receives the response message, and extracts the sets of image segment identigens 674 from the response message. Alternatively, the interpretation module 304 sends the query to another computing device.

A fourth approach includes interpreting the image to identify the plurality of image segments and accessing, for each image segment of the plurality of image segments, the knowledge database 664 to recover a corresponding set of image segment identigens. For instance, the content ingestion module 300 partitions the image 660 to produce image segments 662 (e.g., of a television, a bed, and an elephant). The element identification module 302 compares (e.g., substantially matches) the image segments 662 to image segment information 668 from the knowledge database 664 to produce identified image segments 670 (e.g., representing the television, the bed, and the elephant). The image segment information 668 includes representations of stored images.

The interpretation module accesses the knowledge database 664 utilizing the identified image segment 670 to recover image segment identigens 672 to produce the sets of

image segment identigens **674**. The image segment identigens **672** includes a list of one or more identigens that is associated with each of the identified image segments. For example, the first set of image segment identigens includes identigen #1 for the television, the second set of image segment identigens includes identigen #4 for the bed, and the third set of image segment identigens includes identigen #7 for the elephant.

FIG. 9B further illustrates the example of operation of the method for interpreting the image where the answer resolution module **306** identifies a subset of valid image segment identigens of each set of image segment identigens of the plurality of sets of image segment identigens by applying identigen rules to the plurality of sets of image segment identigens to produce a plurality of subsets of valid image segment identigens. As an example, each valid image segment identigen of a first subset of valid image segment identigens represents a most likely interpretation of the first image segment of the plurality of image segments.

The identifying the subset of valid image segment identigens of each set of image segment identigens of the plurality of sets of image segment identigens includes selecting a first image segment identigen of a first set of image segment identigens of the plurality of sets of image segment identigens for inclusion in the first subset of valid image segment identigens when the identigen rules indicates favorable association between the first image segment identigen and a second image segment identigen of a second set of image segment identigens of the plurality of sets of image segment identigens. For example, the answer resolution module **306** accesses the knowledge database **664** to recover identigen rules **676** and applies the rules to image segment identigens of adjacent sets of image segment identigens to identify valid combinations of the image segment identigens. For instance, the answer resolution module **306** identifies identigen #1 for the television as valid and identifies identigen #4 for the bed as valid when the identigen rule **676** include a favorable association between identigens #1 and #4.

Having identified the subset of valid image segment identigens for each set of image segment identigens, the answer resolution module **306** generates an image entigen group **678** utilizing the plurality of subsets of valid image segment identigens. The image entigen group **678** represents a most likely interpretation of the image **660**.

The generating the image entigen group **678** utilizing the plurality of subsets of valid image segment identigens includes selecting at least one plurality of valid image segment identigens from the plurality of subsets of valid image segment identigens to produce at least one permutation of a plurality of image entigens of the image entigen group. One valid image segment identigen is selected from each subset of valid image segment identigens to produce each of the at least one permutation of the plurality of image entigens of the image entigen group. For instance, the answer resolution module **306** generates the image entigen group **678** to include entigens #1 and #4 when the permutation of entigens #1 and #4 compares favorably to the identigen rule **676**. Having produced the image entigen group **678**, the answer resolution module **306** identifies an invalid set of image segment identigens of the plurality of sets of image segment identigens by applying the identigen rules **676** to the plurality of sets of image segment identigens. For instance, the answer resolution module **306** identifies the third identigen set as invalid when the image segment identigen #7 for the elephant does not favorably pair with any of the other image segment identigens of the

other sets of image segment identigens as indicated by the identigen rule **676** (e.g., an elephant is generally not found with a bed or a television).

Having identified the invalid set of image segment identigens, the answer resolution module **306** generates an updated image **680** based on the image **660** and in accordance with the invalid set of image segment identigens. The updated image **680** excludes an image segment that corresponds to the invalid set of image segment identigens. For example, the answer resolution module **306** excludes the third image segment of the elephant from the image **660** to produce an output the updated image **680**.

The method described above can alternatively be performed by other modules of the computing system **10** of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. 10A is a schematic block diagram of another embodiment of a computing system in accordance that includes query content sources **700**, the AI server **20-1** of FIG. 1, and the user device **12-1** of FIG. 1. The query content sources **700** include the content sources **16-1** through **16-N** of FIG. 1. The AI server **20-1** includes the processing module **50-1** of FIG. 2 and the SS memory **96** of FIG. 2. The processing module **50-1** includes the collections module **120** of FIG. 4A, the IEI module **122** of FIG. 4A, and the query module **124** of FIG. 4A. Generally, this invention presents solutions where the computing system **10** supports processing a query.

The processing of the query includes a series of steps. For example, a first step includes IEI processing an exclusion statement associated with the query to produce a knowledge base restriction when subsequently utilizing a knowledge base to produce a query response to the query. The query is with regards to a question and/or a search request. For instance, find drone sightings near a military base.

The exclusion statement includes a sentence that describes at least one parent attribute to exclude from the query response to the query. For instance, exclude military unmanned aircraft vehicles (UAVs) from the finding of the drone sightings. The knowledge base restriction excludes one or more of identifying attributes of entigens of the knowledge base when subsequently producing the query response. Alternatively, the knowledge base restriction identifies a portion of the knowledge base to exclude. Further alternatively, the knowledge base restriction generates an updated knowledge base that excludes the portion of the knowledge base to exclude, identifying another portion of the knowledge base that is associated with the query and is not excluded based on the exclusion statement.

The example of the first step further includes IEI processing the exclusion statement to produce an exclusion entigen group (e.g., associated with exclude military UAVs), comparing the exclusion entigen group to groups of entigens in the knowledge base to locate a portion of the knowledge base for exclusion, and identifying each entigen of the portion of the knowledge base for exclusion to produce the knowledge base restriction. As illustrated in FIG. 10A, a

specific example includes the IEI module 122 receiving an IEI request 244 from the query module 124, where the query module 124 receives a query with exclusion request 703 from the user device 12-1 that includes the query and the exclusion statement.

The query module 124 issues the IEI request 244 to the IEI module 122, where the IEI request 244 includes content of the query with exclusion request 703. The IEI module 122 IEI processes the exclusion statement utilizing fact base information 600 (e.g., knowledge base) from the SS memory 96 to produce the exclusion entigen group, compares the exclusion entigen group to groups of entigens of the knowledge base by accessing the fact base information 600 to locate the portion of the knowledge base for exclusion, and identifies each entigen of the portion of the knowledge base for exclusion to produce the knowledge base restriction.

A second step of the processing of the query includes obtaining sufficient knowledge to produce the query response with favorable results. For example, determine that the knowledge base has insufficient knowledge to facilitate producing the query response with favorable results (i.e., produce a preliminary query response, determine that a quality level of the preliminary query response falls below a minimum quality threshold level, the query is with respect to a future time frame thus requiring new knowledge). The obtaining the sufficient knowledge further includes identifying a content source, obtaining further content from the content sources, IEI processing the further content to produce incremental knowledge, and integrating the incremental knowledge with the knowledge base to produce an updated knowledge base.

As illustrated in FIG. 10A, a specific example of the second step includes, the IEI module 122 determining that the query is associated with a subsequent time frame, identifying content sources 16-1 through 16-N of the query content sources 700, and causing gathering of further content that is not excluded to produce further knowledge. The IEI module 122 issues a collections request 132 to the collections module 120 (e.g., including the knowledge base restriction), where the collections module 120 issues a query with exclusion content request 705 (e.g., to request content that is not restricted) to the query content sources 700.

The collections module 120 receives a query with exclusion content response 707, and issues a collections response 134 to the IEI module 122 including the further content. The IEI module 122 IEI processes the further content to produce the incremental knowledge and integrates the incremental knowledge with the fact base information 600 for storage in the SS memory 96.

A third step of the processing of the query includes IEI processing the query utilizing the knowledge base to produce the query response to the query in accordance with the knowledge base restriction. For example, IEI module 122 processes the query to produce a query entigen group, compares the query entigen group to the groups of entigens in the updated knowledge base in accordance with the knowledge base restriction to locate an unrestricted portion of the knowledge base associated with the query, and generates the query response based on the unrestricted portion of the knowledge base.

As illustrated in FIG. 10A, a specific example of the third step includes the IEI module 122 IEI processing the query to produce the query entigen group and accessing the fact base information 600 from the SS memory 96 to facilitate comparing the query entigen group to the groups of entigens in the updated knowledge base in accordance with the knowledge base restriction to locate the unrestricted portion of the

knowledge base associated with the query. The IEI processing further includes generating the query response based on the unrestricted portion of the knowledge base and issuing an IEI response 246 to the query module 124 that includes the query response. The query module 124 issues a query with exclusion response 709 to the user device 12-1 that includes the query response (e.g., and further includes identifiers of the portions of the knowledge base, etc.).

FIGS. 10B and 10C are schematic block diagrams of an embodiment of a portion of knowledge base that includes the fact base information 600 before processing the exclusion statement and the effective knowledge base entigen group 712 after processing the exclusion statement based on the query with exclusion request 703 (e.g., find UAV sightings around a military base that excludes military UAVs). With the military UAVs of the fact base information 600 of FIG. 10B excluded to produce the effective knowledge base entigen group 712 of FIG. 10C, the finding of the UAV sightings around the military base enjoys a performance enhancement by excluding a portion of the fact base information 600 associated with military UAVs.

FIG. 10D is a logic diagram of an embodiment of a method for processing a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8D, 10A, and also FIGS. 10B-10C. The method includes step 720 where a processing module of one or more processing modules of one or more computing devices of the computing system IEI processes an exclusion statement associated with a query to produce a knowledge base restriction when subsequently utilizing a knowledge base to produce a query response to the query. For example, the processing module IEI processes the exclusion statement to produce an exclusion entigen group (e.g., associated with excluded military UAVs), compares the exclusion entigen group to groups of entigens in the knowledge base to locate a portion of the knowledge base for exclusion, and identifies each entigen of the portion of the knowledge base for exclusion to produce the knowledge base restriction.

The method continues at step 722 where the processing module obtains sufficient knowledge to enable producing the query response with favorable results. For example, the processing module determines that the knowledge base has insufficient knowledge to facilitate producing the query response with favorable results (i.e., produce a preliminary query response, determine that a quality level of the preliminary query response falls below a minimum quality threshold level, the query is with respect to a future time frame thus requiring new knowledge). The processing module identifies a content source, causes gathering the further content that is not excluded to produce further knowledge that is not excluded, IEI processes the further content to produce incremental knowledge, and integrates the incremental knowledge with the knowledge base to produce an updated knowledge base.

The method continues at step 724 where the processing module IEI processes the query utilizing the knowledge base to produce the query response to the query in accordance with the knowledge base restriction. For example, the processing module IEI processes the query to produce a query entigen group, compares the query entigen group to the groups of entigens in the updated knowledge base in accordance with the knowledge base restriction to locate an unrestricted portion of the knowledge base associated with the query. The processing module generates the query response based on the unrestricted portion of the knowledge base.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

FIGS. **10E** and **10F** are schematic block diagrams of another embodiment of a computing system illustrating an embodiment of a method for generating a response to a query within the computing system. The computing system includes the content ingestion module **300** of FIG. **5E**, the element identification module **302** of FIG. **5E**, the interpretation module **304** of FIG. **5E**, the answer resolution module **306** of FIG. **5E**, and a knowledge database **701**. The knowledge database **701** may be implemented utilizing one or more of the memories of FIG. **2**.

FIG. **10E** illustrates an example of operation of the method for generating the response to the query where a first step includes the content ingestion module **300** obtaining a query **1000**. The query includes a question associated with an exclusion such as in Boolean logic “not” something. The obtaining includes at least one of receiving the query **1000** from a requesting entity (e.g., a user interface associated with the user, a computing device operably coupled to the computing system, etc.) and recovering the query **1000** from the knowledge database **701**. For example, the content ingestion module **300** receives the query **1000** from the requesting entity, where the query **1000** includes a string of words: “UAV-sightings exclude military-UAV-sightings?” In this example the requesting entity is inquiring about sightings of unmanned aerial vehicles that are not military style unmanned aerial vehicles. Asking the question this way utilizes the exclusion rather than specifically asking for alternative UAVs such as commercial and consumer models.

Having obtained the query **1000**, a second step of the example method of operation includes the content ingestion module **300** parsing the string of words of the query **1000** to produce words **1002** of the query. For example, the content ingestion module **300** produces the words **1002** to include the words: “UAV-sightings”, “exclude”, and “military-UAV-sightings” when the query **1000** includes “UAV-sightings exclude military-UAV-sightings?”

With the query **1000** parsed into words **1002**, a third step of the example method of operation includes the element identification module **302** determining a set of identigens for each word of a plurality of words **1002** of the query of a topic to produce a plurality of sets of identigens **708**. A set of identigens of the plurality of sets of identigens represents one or more different meanings of a word of the plurality of words. Each identigen of the set of identigens includes a meaning identifier, an instance identifier, and a time reference. Each meaning identifier associated with the set of identigens represents a different meaning of the one or more different meanings of the word of the plurality of words. Each time reference provides time information when a corresponding different meaning of the one or more different meanings is valid. A first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words.

As an example of the producing of the sets of identigens **708**, the element identification module **302** recovers identigen information **706** from the knowledge database **701** based on the words **1002**. For example, the element identification module **302** recovers identigen group #10-11 associated with a “UAV-sightings” meaning of the word “UAV-sightings” to form a first identigen set. The element identification module **302** further recovers identigens #7 and #8 associated with a “exclude” meaning of the word “exclude” to form a second identigen set. The element identification module **302** further recovers identigen group #9-10-11 associated with a “military-UAV-sightings” meaning of the word “military-UAV-sightings” to form a third identigen set.

Having produced the plurality of sets of identigens **708**, a fourth step of the example method of operation includes the interpretation module **304** identifying an excluding entigen **1008** of an excluding query entigen group **1009**. The excluding entigen **1008** is associated with the excluded entigen group **1009** to be disassociated with the topic with regards to a response to the query. The identifying the excluding entigen **1008** of the excluding query entigen group **1009** includes one or more approaches.

A first approach includes matching a first query entigen of the excluding query entigen group to list of excluding category entigens. For example, the interpretation module **304** matches entigen #8 for the meaning “rule out” to an entry of the list of excluding category entigens to produce the excluding entigen **1008**.

A second approach includes detecting an indication of an excluding category from the knowledge database for the first query entigen of the excluding query entigen group. For example, the interpretation module **304** interprets a record recovered from the knowledge database **701** with regards to the entigen #8 that indicates the excluding category to produce the excluding entigen **1008**.

A third approach includes identifying a set of characteristics from the knowledge database for the first query entigen of the excluding query entigen group. At least one characteristic of the set of characteristics is associated with the excluding category. For example, the interpretation module **304** interprets the record recovered from the knowledge database **701** with regards to the entigen #8 that indicates the characteristic associated with the excluding category.

Having produced the sets of identigens **708** and identified the excluding entigen, a fifth step of the example method of operation includes the interpretation module **304** interpreting, in accordance with identigen pairing rules **710** of the knowledge database **701**, the plurality of sets of identigens **708** to determine a most likely meaning interpretation of the query **1000** and produce an excluding query entigen group **1006** comprising the excluding entigen **1008** of a plurality of query entigens. The knowledge database includes a multitude of entigen groups associated with a multitude of topics. The multitude of topics includes the topic.

Each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens. The excluding query entigen group **1006** represents the most likely meaning interpretation of the query. The excluding entigen **1008** functions to exclude at least a portion of an excluded entigen group to be disassociated with the topic with regards to a response to the query. Each query entigen of the excluding query entigen group corresponds to a selected identigen of

the set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words.

Each query entigen of the excluding query entigen group **1006** represents a single conceivable and perceivable thing in space and time that is independent of language and corresponds to a time reference of the selected identigen associated with the excluding query entigen group. The selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules of the knowledge database.

In an example of the interpreting the plurality of sets of identigens **708** to produce the excluding query entigen group **1006**, the interpretation module **304** receives identigen rules **710** from the knowledge database **7** one with regards to permutations of combinations of the identigens of each of the sets of identigens **708**. For instance, the interpretation module **304** pairs the identigen group #10-11 with identigen #8 in pairs the identigen #8 with the identigen group #9-10-11 in accordance with the identigen rules **710** to produce the excluding query entigen group **1006**.

FIG. **10F** further illustrates the example of operation of the method for generating the response to the query where a sixth step includes the answer resolution module **306** recovering a response entigen group **1010** for the query **1000** from the knowledge database **7** one utilizing the excluding query entigen group **1006** and based on the excluding entigen **1008** and the excluded entigen group **1009**. The response entigen group **1010** includes one or more response entigens and one or more response entigen relationships between at least some of the one or more response entigens. The set of response entigens includes a portion of the excluding query entigen group. The response entigen group provides the response to the query.

The recovering the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group includes a series of sub-steps. A first sub-step includes eliminating the excluding entigen and the excluded entigen group from the excluding query entigen group to produce a remaining subset of query entigens of the excluding query entigen group. For example, the answer resolution module **306** eliminates the entigen #8 for the meaning "rule out" and the entigen group #9-10-11 for the meaning "military-UAV-sightings" from the excluding query entigen group **1006** to produce the remaining subset of query entigens (e.g., entigen group #10-11 for UAV sightings).

A second sub-step includes accessing the knowledge database utilizing the remaining subset of query entigens to identify a superset of response entigens of the response entigen group. For example, the interpretation module **306** interprets entigen information **716** received from the knowledge database **701** with regards to all UAV sightings (e.g., nonmilitary and military) as the superset of response entigens in accordance with the remaining subset of query entigens.

A third sub-step includes removing conflicting response entigens associated with the excluded entigen group in accordance with the excluding entigen from the superset of response entigens to produce the response entigen group. For example, the interpretation module **306** eliminates conflicting entigens associated with sightings of military UAVs in accordance with the excluding entigen to exclude from the superset of response entigens conflicting entigens associated with military UAV sightings.

Alternatively, or in addition to, the recovering the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group further includes a series of alternates sub-steps. A first alternate sub-step includes accessing the knowledge database utilizing the excluded entigen group and in accordance with the excluding entigen to produce a search entigen group. For example, the answer resolution module **306** produces the search entigen group to include non-military UAV sightings. A second alternate sub-step includes accessing the knowledge database utilizing the search entigen group to identify the response entigen group. For example, the answer resolution module **306** interprets further entigen information **716** from the knowledge database **701** with regards to non-military UAV sightings (e.g., sightings 1-3).

Having produced the response entigen group **1010**, a seventh step of the example method of operation includes the answer resolution module **306** generating a query response phrase utilizing the response entigen group as a representation of the response entigen group. For example, the answer resolution module **306** generates the query response phrase to include "non-military UAV sightings include #1 and #2 of tello and #3 of dracon".

Having generated the query response phrase, the seventh step further includes the answer resolution module **306** outputting at least one of the response entigen group **1010** and the query response phrase to a requesting entity associated with the query.

Alternately, or in addition to, the seventh step further includes the answer resolution module **306** combining the excluding query entigen group with the response entigen group to produce an incremental entigen group associated with the topic. Having produced the incremental entigen group, the answer resolution module **306** facilitates storage of the incremental entigen group in the knowledge database **701** to provide expanded knowledge of the topic.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. **11A** is a schematic block diagram of another embodiment of a computing system that includes query content sources **730**, the AI server **20-1** of FIG. **1**, and the user device **12-1** of FIG. **1**. The query content sources **730** include the content sources **16-1** through **16-N** of FIG. **1**. The AI server **20-1** includes the processing module **50-1** of FIG. **2** and the SS memory **96** of FIG. **2**. The processing module **50-1** includes the collections module **120** of FIG. **4A**, the IEI module **122** of FIG. **4A**, and the query module **124** of FIG. **4A**. Generally, an embodiment of this invention presents solutions where the computing system **10** supports processing a query.

The processing of the query includes a series of steps. For example, a first step includes IEI processing an expansion statement associated with a query to produce a knowledge focus when subsequently utilizing a knowledge base to

produce a query response to the query (e.g., locate an unnamed the bomber suspect). The expansion statement includes a sentence that describes at least one attribute to expand upon when searching content and/or knowledge to produce the query response to the query (e.g., utilizing alias names connected to the unnamed bomber suspect).

The knowledge focus includes one or more of identified attributes of entigens of the knowledge base to include when subsequently producing the query response, identity of a portion of the knowledge base to include, generating an updated knowledge base that includes the portion of the knowledge base to include, and identifying further content to acquire to produce knowledge related to the expansion statement. For example, IEI process the expansion statement to produce an expansion entigen group (e.g., associated with alias names connected to the suspect), compare the expansion entigen group to groups of entigens in the knowledge base to locate a portion of the knowledge base for inclusion, and identify each entigen of the portion of the knowledge base for inclusion to produce the knowledge focus.

As a specific example of the first step, the IEI module 122 receives an IEI request 244 from the query module 124. The query module 124 receives a query with expansion request 732 from the user device 12-1 that includes the query and the expansion statement and issues the IEI request 244 to the IEI module 122. The IEI request 244 includes the query with expansion request 732.

The IEI module 122 IEI processes the expansion statement utilizing fact base information 600 (e.g., knowledge base) from the SS memory 96 to produce the expansion entigen group, compares the expansion entigen group to groups of entigens of the knowledge base by accessing the fact base information 600 to locate the portion of the knowledge base for expansion. Having located the portion, the IEI module 122 identifies each entigen of the portion of the knowledge base for expansion to produce the knowledge base focus.

A second step of the processing of the query includes obtaining further knowledge to produce the query response with favorable results. For example, determine that the knowledge base has insufficient knowledge to facilitate producing the query response with favorable results (i.e., produce a preliminary query response, determine that a quality level of the preliminary query response falls below a minimum quality threshold level; the query is with respect to a future time frame thus requiring new knowledge). Having identified the insufficient knowledge, content sources are identified, a content requirement is generated (e.g., hit on alias names from IEI processing the expansion statement), further content is obtained from the content sources utilizing the content requirement (e.g., alias trigger), the further content is IEI processed to produce incremental knowledge for integration with the knowledge base to produce an updated knowledge base.

As a specific example of the second step, the IEI module 122 determines that the query is associated with a trigger on substantially live content, identifies content sources 16-1 through 16-N of the query content sources 730, causes gathering of further content that is associated with the knowledge focus to produce further knowledge. The causing includes the IEI module 122 issuing a collections request 132 to the collections module 120 (e.g., including the knowledge focus of specific alias names). The collections module 120 issues a query with expansion content request 734 (e.g., to request content that is associated with aliases) to the query content sources 730, receives a query with expansion content response 736, and issues a collections

response 134 to the IEI module 122 including the further content. The IEI module 122 IEI processes the further content to produce the incremental knowledge and integrates the incremental knowledge with the fact base information 600 for storage in the SS memory 96 as the updated knowledge base.

A third step of processing the query includes IEI processing the query utilizing the knowledge base to produce the query response to the query in accordance with the knowledge focus. For example, IEI process the query to produce a query entigen group, compare the query entigen group to the groups of entigens in the updated knowledge base in accordance with the knowledge base focus to locate an area of focus portion of the knowledge base associated with the query, and generate the query response based on the area of focus portion of the knowledge base (e.g., put the puzzle together with inference and deduction rules).

As a specific example of the third step, the IEI module 122 IEI processes the query to produce the query entigen group, accesses the fact base information 600 from the SS memory 96 to facilitate comparing the query entigen group to the groups of entigens in the updated knowledge base in accordance with the knowledge base restriction to locate the unrestricted portion of the knowledge base associated with the query. The IEI module 122 generates the query response based on the unrestricted portion of the knowledge base and issues and IEI response 246 to the query module 124 that includes the query response. The query module 124 issues a query with exclusion response 708 to the user device 12-1 that includes the query response (e.g., and further includes identifiers of the portions of the knowledge base, etc.).

FIGS. 11B and 11C are schematic block diagrams of another embodiment of a portion of knowledge base that includes the fact base information 600 before processing the expansion statement and an effective knowledge base entigen group 742 after processing the expansion statement based on the query with expansion request 732 (e.g., locate a main bomber suspect utilizing alias names). With the alias name portions of the fact base information 600 of FIG. 11B expanded to produce the effective knowledge base entigen group 742 of FIG. 11C, the finding of the unnamed bomber suspect enjoys a performance enhancement by expanding upon the alias name portion of the fact base information 600 associated with the unnamed bomber suspect.

FIG. 11D is a logic diagram of another method for processing a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8D, 11A, and also FIGS. 11B-11C. The method includes step 750 where a processing module of one or more processing modules of one or more computing devices of the computing system IEI processes and expansion statement associated with a query to produce a knowledge base focus when subsequently utilizing a knowledge base to produce a query response to the query. For example, the processing module IEI processes the expansion statement to produce an expansion entigen group (e.g., associated with alias names connected to an unnamed bomber suspect), compares the expansion entigen group to groups of entigens in the knowledge base to locate a portion of the knowledge base for inclusion, and identifies each entigen of the portion of the knowledge base for inclusion to produce the knowledge focus.

The method continues at step 752 where the processing module obtains further knowledge to enable producing the query response with favorable results. For example, the processing module determines that the knowledge base has

insufficient knowledge to facilitate producing the query response with favorable results (i.e., produce a preliminary query response, determine that a quality level of the pulmonary query response falls below a minimum quality threshold level; the query is with respect to a future time frame thus requiring new knowledge). The processing module further identifies content sources, generates a content requirement (e.g., hit on alias names from IEI processing expansion statement), obtains further content from the content sources utilizing the content requirement (e.g., alias trigger), IEI processes the further content to produce incremental knowledge, and integrates the incremental knowledge with the knowledge base to produce an updated knowledge base.

The method continues at step **754** with a processing module IEI processes the query utilizing the knowledge base to produce the query response to the query in accordance with the knowledge focus. For example, the processing module IEI processes the query to produce a query entigen group, compares the query entigen group to the groups of entigens of the updated knowledge base in accordance with the knowledge base focus to locate an area of focus portion of the knowledge base associated with the query. The processing module generates the query response based on the area focus portion of the knowledge base (e.g., put the puzzle together with inference and deduction rules).

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. **12A** is a schematic block diagram of another embodiment of a computing system that includes the AI server **20-1** of FIG. **1** and the user device **12-1** of FIG. **1**. The AI server **20-1** includes the processing module **50-1** of FIG. **2** and the SS memory **96** of FIG. **2**. The processing module **50-1** includes the IEI module **122** of FIG. **4A** and the query module **124** of FIG. **4A**. Generally, an embodiment of this invention presents solutions where the computing system **10** supports processing a query.

The processing of the query includes a series of steps. For example, a first step includes determining mandatory inclusive and exclusive attributes associated with a query for subsequent processing with a knowledge base to produce a query response. Mandatory inclusive attributes include descriptors and values of characteristics of an entigen that must be present to satisfy the query. Mandatory exclusive attributes include descriptors and values of characteristics of an entigen that must not be present to satisfy the query. The determining includes extracting from the query and IEI processing the query utilizing the knowledge base to identify the mandatory inclusive and exclusive attributes.

As a specific example of the first step, the IEI module **122** receives an IEI request **244** from the query module **124** that includes a query with inclusive and exclusive attributes request **770** from the user device **12-1**. For instance, the request **770** includes a text string "identify a car line with a high-performance model that does not require all changes."

The IEI module **122** extracts the query (e.g., identify a car line) from the request and extracts a mandatory attribute statement from the request. The mandatory attribute statement identifies the mandatory inclusive attribute (e.g., a performance model) and the mandatory exclusive attribute (e.g., not associated with oil changes).

A second step of processing the query includes IEI processing the query utilizing the knowledge base to produce a query entigen group. For example, identify a plurality of permutations of entigen groupings. Each permutation pairs sequential entigens associated with possible interpretations of sequential words of the query. The example further includes applying entigen pairing rules, in accordance with the knowledge base, to the plurality of permutations to identify a surviving interpretation to produce the query entigen group.

As a specific example of the second step, the IEI module **122** IEI processes the query utilizing fact base information **600** from the SS memory **96** to produce the query entigen group that will be utilized in a subsequent search for a car line. For instance, the IEI module **122** extracts the query (e.g., identify the car line) from the query and IEI processes the text string of the query to produce the query entigen group.

A third step of the processing of the query includes identifying a portion of the knowledge base associated with the query entigen group. For example, compare the query entigen group to entigens of the knowledge base to identify a portion of the knowledge base associated with the query entigen group. As a specific example of the third step, the IEI module **122** recovers entigen groupings from the fact base information **600** of the SS memory **96** for comparison to the query entigen group and indicates a portion when the recovered entigen substantially matches at least some of the entigens of the query entigen group (e.g., entigens of car lines and attributes).

A fourth step of the processing of the query includes interpreting the portion of the knowledge base associated with the query entigen group in accordance with the mandatory inclusive and exclusive attributes to produce the query response. For example, apply logic (e.g., induction, deduction, inference, etc.) to the entigens of the portion of the knowledge base to locate an answer entigen as the query response. The locating includes excluding entigens and linked entigens associated with the exclusive attribute. The locating further includes including entigens and linked entigens associated with the inclusive attribute.

As a specific example of the fourth step, the IEI module **122** applies the logic to the entigens of the fact base information **600** from the SS memory **96** in accordance with the mandatory inclusive and exclusive attributes to identify the entigen associated with the query response and issues an IEI response **246** to the query module **124** that includes the query response. The query module **124** issues a query with inclusive and exclusive attributes response **772** to the user device **12-1**. For instance, the query response, as depicted in FIG. **12C**, identifies Tessy as a car line with the model that has high performance and does not require oil changes.

FIGS. **12B** and **12C** are schematic block diagrams of another embodiment of a portion of knowledge base within a computing system that includes the fact base information **600** before processing the mandatory attribute statement and an effective knowledge base entigen group **782** after processing the mandatory attribute statement based on the query with inclusive and exclusive attributes request **770** (e.g., identify a car line with a high-performance model that does not require oil changes). With the high-performance and oil

51

requirement portions of the fact base information **600** of FIG. **12B** expanded to produce the effective knowledge base entigen group **782** of FIG. **12C**, the finding of the car line enjoys a performance enhancement by expanding upon the high-performance and oil requirement portions of the fact base information **600** associated with the car line. For instance, while Tessy and Shevy both have high-performance cars, only Tessy as a high-performance model that does not require oil.

FIG. **12D** is a logic diagram of an embodiment of another method for processing a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8D**, **12A**, and also FIGS. **12B-12C**. The method includes step **800** where a processing module of one or more processing modules of one or more computing devices of the computing system determines mandatory inclusive and exclusive attributes associated with the query for subsequent processing utilizing a knowledge base to produce a query response. The determining includes one or more of extracting the mandatory inclusive and exclusive attributes from the query and IEL processing the query utilizing the knowledge base to identify the mandatory inclusive and exclusive attributes.

The method continues at step **802** where the processing module IEL processes the query utilizing the knowledge base to produce a query entigen group. For example, the processing module identifies a plurality of permutations of identigen groupings, where each permutation pairs sequential identigens associated with possible interpretations of sequential words of the query, and applies entigen pairing rules, in accordance with the knowledge base, to the plurality of permutations to identify a surviving interpretation. The processing module further transforms the surviving interpretation into the query entigen group.

The method continues at step **804** where the processing module identifies a portion of the knowledge base associated with the query entigen group. For example, the processing module compares the query entigen group to entigens of the knowledge base to identify a portion of the knowledge base associated with the query entigen group.

The method continues at step **806** where the processing module interprets a portion of the knowledge base associated with the query entigen group in accordance with the mandatory inclusive and exclusive attributes to produce the query response. For example, the processing module applies logic (e.g., induction, deduction, inference, etc.) to the entigens of the portion of the knowledge base to locate an answer entigen as the query response. The locating further includes excluding of exclusive entigens (e.g., associated with the mandatory exclusive attribute) and other related entigens linked to the exclusive entigens. The locating further includes including of inclusive entigens (e.g., associated with the mandatory inclusive attributes) and other related entigens linked to the inclusive entigens.

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing

52

system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

FIG. **13A** is a schematic block diagram of another embodiment of a computing system that includes the content ingestion module **300** of FIG. **5E**, the element identification module **302** of FIG. **5E**, the interpretation module **304** of FIG. **5E**, and the SS memory **96** of FIG. **2**. Generally, an embodiment of this invention presents solutions where the computing system **10** supports processing an image.

The processing of the image includes a series of steps. For example, a first step includes identifying image elements of an ingested image to produce tokenized image elements. As a specific example of the first step, the content ingestion module **300** processes an image **830** of source content **310** in accordance with element rules **318** and an element list **332** (of images) to produce formatted content **314**, where the image **830** includes a plurality of image elements.

As an instance of the identifying of the image elements, the content ingestion module **300** identifies the source content **310** as a face image (e.g., based on a file name, based on a quick interpretation), selects facial image related element rules **318** and a facial image related element list **332**, partitions the image into the plurality of image elements (e.g., pixel mapping, element identification rules), identifies element relationships between the plurality of image elements (e.g., proximity, relationship, orientation, actions, etc.) for each image element, and compares the image element to image elements of the element list **332** in accordance with the image element rules **318** to produce an identified image element.

In the instance, an image element **832-1** is produced when matching an image element of the image to a stored image of a pair of eyes, an image element **832-2** is produced when matching another image element of the image to a stored image of a nose, and an image element **832-3** is produced when matching another image element of the image to a stored image of a mouth. In addition, the content ingestion module **300** produces an image element related to the face itself (e.g., all other areas of the face, positioning information of the other image elements with respect to the other areas of the face).

A second step of the processing of the image includes, for each tokenized image element, identifying one or more identigens that correspond the tokenized image element including characteristics of each image element objects and characteristics describing relationships between the objects (e.g., distance between nose and mouth, position on the face itself). Each Identigen describes one of an object, a characteristic, and an action.

As a specific example of the second step to identify the one or more identigens, the element identification module **302** performs a look up of identifiers, utilizing an element list **332** and in accordance with element rules **318**, of the one or more identigens associated with each tokenized image element of the formatted content **314** to produce identified element information **340**. A unique identifier is associated with each of the potential object.

The element identification module **300** further performs estimations to produce the characteristic associated with a particular tokenized image element. For instance, the identified element information **340** is produced to include an eyes identigen **834-1** for the eyes image element **832-1**, a nose identigen **834-2** for the nose image element **832-2**, a mouth identigen **834-3** for the mouth image element **832-3**, an eyes dimension identigen **834-4** is estimated for a distance between the eyes, a nose dimension identigen **834-5** is estimated for a width of the nose, a mouth dimension

identigen **834-6** is estimated for a width of the mouth, and further characteristic identigens are produced for further dimensions (e.g., between objects, etc.), colors, shapes, skin texture, blemishes, facial lines, or any other distinguishing facial characteristic. There could be hundreds or more.

A third step of the processing of the image includes, for each permutation of sequential combinations of identigens (e.g., generally just one in this example, but dimensions could have ranges), generating a corresponding equation package (i.e., candidate interpretation). The equation package includes a sequential linking of pairs of the identigens. Each sequential linking pairs a preceding identigen to a next identigen. An equation element describes a relationship between paired identigens (OCA's) such as describes, acts on, is a, belongs to, did, did to, etc. Multiple OCAs occur for a common word when the word has multiple meanings (e.g., a baseball bat, a flying bat).

As a specific example of the third step, for each permutation of identigens of each tokenized image element of the identified element information **340**, the interpretation module **304** includes all other permutations of all other tokenized image elements to generate the equation packages in accordance with interpretation rules **320** and a groupings list **334**. For instance, the interpretation module **304** produces a first pairing of the eyes and nose identigens, a second pairing of the nose and mouth identigens, and further pairings to link dimensions and other characteristics. The fourth step of the processing of the image include selecting a surviving equation package associated with a most favorable interpretation as a surviving interpretation **836**. As a specific example of the fourth step, the interpretation module **304** applies the interpretation rules **320** (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens and characteristics of identigens and relationships of identigens) to reduce the number of permutations of the sequential combinations of identigens to produce interpreted information **344**.

The interpreted information **344** includes identification of at least one equation package as the surviving interpretation **836**, where non-surviving equation packages are eliminated that compare unfavorably to pairing rules and/or are associated with an unfavorable confidence level to produce a non-surviving interpretation. A confidence level may be assigned to each equation package such that a higher confidence level indicates that equation package with equation elements that are substantially the same in other equation packages, i.e., more consistency.

A fifth step of the processing of the image includes integrating the surviving interpretation **836** into a knowledge base to facilitate subsequent matching of a subject image to the ingested image. As a specific example of the first step, the interpretation module **304**, or another module, locates a portion of the knowledge base that compares favorably to the surviving interpretation **836**, interprets the surviving interpretation **836** (e.g., entigens, links based on characteristics), and updates the portion of the knowledge base with the interpretation to produce an updated knowledge base.

FIG. **13B** is schematic block diagram of another embodiment of a portion of knowledge base that pertains to the processing of the image to produce an effective knowledge-base entigen grouping **840** from the interpreted information **344**, when the interpreted information **344** includes an example text string of "a face with objects: to eyes, nose, a mouth, with characteristics (e.g., dimensions, color, etc.)." The effective knowledgebase entigen group **840** provides a portion of the updated knowledge base to enable subsequent

matching of another image of the same face to the stored representation of the face. The matching includes creating an entigen group for the other image in a similar manner as described and comparing the entigen group for the other image to the effective knowledgebase entigen group **840** to produce a positive match (e.g., matching when a substantial number of the characteristics of each object of the face are within a positive match tolerance).

FIG. **13C** is a logic diagram of an embodiment of another method for processing an image within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8D**, **13A**, and also FIG. **13B**. The method includes step **850** where a processing module of one or more processing modules of one or more computing devices of the computing system identifies image elements of an ingested image to produce tokenized image elements. For example, the processing module processes an ingested image as source content in accordance with element rules and an element list (of images) to produce formatted content.

The image includes a plurality of image elements. The processing module identifies the source content as a face image (e.g., based on a file name, based on an interpretation of the image), select facial image related element rules and a facial image related element list, partition the image into the plurality of image elements (e.g., pixel mapping, element identification rules). The processing module identifies element relationships between the plurality of image elements (e.g., proximity, relationship, orientation, actions, etc.) for each image element and compares the image element to image elements of the element list in accordance with the image element rules to produce an identified image element. For example, a first image element is produced when matching an image element of the image to a stored image of a pair of eyes, a second image element is produced when matching another image element of the image to a stored image of a nose, and a third image element is produced when matching yet another image element of the image to a stored image of a mouth.

For each tokenized image element, the method continues at step **852** where the processing module identifies one or more identigens that correspond to the tokenized image element including characteristics of the image element objects and characteristics describing relationships between the objects. For example, the processing module performs a look up of identifiers, utilizing an element list and in accordance with element rules, of the one or more identigens associated with each tokenized image element of the formatted content to produce identified element information.

A unique identifier is associated with each of the potential object. The processing module performs estimations to produce the characteristic associated with a particular tokenized image element. For instance, the identified element information is produced to include an eyes identigen for the eyes image element, a nose identigen for the nose image element, a mouth identigen for the mouth image element, an eyes dimension identigen is estimated for a distance between the eyes, a nose dimension identigen is estimated for a width of the nose, a mouth dimension identigen is estimated for a width of the mouth, and further characteristic identigens are produced for further dimensions (e.g., between objects, etc.), colors, shapes, skin texture, blemishes, facial lines, or any other distinguishing facial characteristic. There could be hundreds or more characteristic related identigens.

For each permutation of sequential combinations of identigens, the method continues at step **854** where the processing module generates a corresponding equation package,

where the equation package includes a sequential linking of pairs of identigens, where each sequential linking pairs a preceding identigen to a next identigen, and where the equation element describes a relationship between paired identigens. For example, the processing module, for each permutation of identigens of each tokenized image element of the identified element information, includes permutations of all other tokenized image elements to generate the equation packages in accordance with interpretation rules and the groupings list. For instance, the processing module produces a first pairing of the eyes and nose identigens, the second pairing of the nose and mouth identigens, and further pairings to link dimensions and other characteristics.

The method continues at step 856 for the processing module selects a surviving equation package associated with a most favorable interpretation as a surviving interpretation. For example, the processing module applies interpretation rules (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens and characteristics of identigens and relationships of identigens) to reduce the number of permutations of the sequential combinations of identigens to produce interpreted information that includes identification of at least one equation package as the surviving interpretation.

Non-surviving equation packages are eliminated that compare unfavorably to pairing rules and/or are associated with an unfavorable confidence level to produce a non-surviving interpretation. For example, a confidence level may be assigned to each equation package such that a higher confidence level indicates that equation package with equation elements that are substantially the same in other equation packages, i.e., more consistency.

The method continues at step 858 where the processing module integrates the surviving interpretation into a knowledge base to facilitate subsequent matching of a subject image (e.g., a new facial image of an unknown person) to the ingested image. For example, the processing module locates a portion of the knowledge base that compares favorably to the surviving interpretation. The processing module interprets the surviving interpretation (e.g., entigens, linkage between characteristic related entigens) to update the portion of the knowledge base with the interpretation (e.g., adding entigens to the knowledge base when new, replacing contradictory entigens associated with the knowledge base, etc.).

The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a number of desired types (e.g., data, video, speech, text, graphics, audio, etc. any of which may generally be referred to as 'data').

As may be used herein, the terms "substantially" and "approximately" provides an industry-accepted tolerance for its corresponding term and/or relativity between items. For some industries, an industry-accepted tolerance is less than one percent and, for other industries, the industry-accepted tolerance is 10 percent or more. Other examples of industry-accepted tolerance range from less than one percent to fifty percent. Industry-accepted tolerances correspond to, but are not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, thermal noise, dimensions, signaling errors, dropped packets, temperatures, pressures, material compositions, and/or performance metrics. Within an industry, tolerance variances of accepted tolerances may be more or less than a percentage level (e.g., dimension tolerance of less than $\pm 1\%$). Some relativity between items may range from a difference of less than a percentage level to a few percent. Other relativity between items may range from a difference of a few percent to magnitude of differences.

As may also be used herein, the term(s) "configured to", "operably coupled to", "coupled to", and/or "coupling" includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as "coupled to".

As may even further be used herein, the term "configured to", "operable to", "coupled to", or "operably coupled to" indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term "associated with", includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

As may be used herein, the term "compares favorably", indicates that a comparison between two or more items, signals, etc., provides a desired relationship. For example, when the desired relationship is that signal 1 has a greater magnitude than signal 2, a favorable comparison may be achieved when the magnitude of signal 1 is greater than that of signal 2 or when the magnitude of signal 2 is less than that of signal 1. As may be used herein, the term "compares unfavorably", indicates that a comparison between two or more items, signals, etc., fails to provide the desired relationship.

As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase "at least one of a, b, and c" or of this generic form "at least one of a, b, or c", with more or less elements than "a", "b", and "c". In either phrasing, the phrases are to be interpreted identically. In particular, "at least one of a, b, and c" is equivalent to "at least one of a, b, or c" and shall mean a, b, and/or c. As an example, it means: "a" only, "b" only, "c" only, "a" and "b", "a" and "c", "b" and "c", and/or "a", "b", and "c".

As may also be used herein, the terms "processing module", "processing circuit", "processor", "processing circuitry", and/or "processing unit" may be a single processing device or a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit,

field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, processing circuitry, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, processing circuitry, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, processing circuitry, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processing module, module, processing circuit, processing circuitry and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, processing circuitry and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture. One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.

To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate definitions of both functional building blocks and flow diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors executing appropriate software and the like or any combination thereof.

In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be

incorporated in or otherwise used in conjunction with one or more other routines. In addition, a flow diagram may include an “end” and/or “continue” indication. The “end” and/or “continue” indications reflect that the steps presented can end as described and shown or optionally be incorporated in or otherwise used in conjunction with one or more other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

Unless specifically stated to the contra, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.

The term “module” is used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules.

As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. The memory device may be in a form a solid-state memory, a hard drive memory, cloud memory, thumb drive, server memory, computing device memory, and/or other physical medium for storing digital information.

While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present

59

disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

What is claimed is:

1. A method for execution by a computing device, the method comprises:

determining a set of identigens for each word of a plurality of words of a query of a topic to produce a plurality of sets of identigens, wherein a set of identigens of the plurality of sets of identigens represents one or more different meanings of a word of the plurality of words, wherein each identigen of the set of identigens includes a meaning identifier, an instance identifier, and a time reference, wherein each meaning identifier associated with the set of identigens represents a different meaning of the one or more different meanings of the word of the plurality of words, wherein each time reference provides time information when a corresponding different meaning of the one or more different meanings is valid, wherein a first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words;

interpreting, in accordance with identigen pairing rules of a knowledge database, the plurality of sets of identigens to determine a most likely meaning interpretation of the query and produce an excluding query entigen group comprising an excluding entigen of a plurality of query entigens, wherein the knowledge database includes a multitude of entigen groups associated with a multitude of topics, wherein the multitude of topics includes the topic, wherein each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens, wherein the excluding query entigen group represents the most likely meaning interpretation of the query, wherein the excluding entigen functions to exclude at least a portion of an excluded entigen group to be disassociated with the topic with regards to a response to the query, wherein each query entigen of the excluding query entigen group corresponds to a selected identigen of the set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words, wherein each query entigen of the excluding query entigen group represents a single conceivable and perceivable thing in space and time that is independent of language and corresponds to a time reference of the selected identigen associated with the excluding query entigen group, wherein the selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules of the knowledge database; and

recovering a response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group, wherein the response entigen group includes one or more response entigens and one or more response entigen relationships between at least some of the one or more response entigens, wherein the set of response entigens includes a portion of the excluding query entigen group, wherein the response entigen group provides the response to the query.

2. The method of claim 1 further comprises:

identifying the excluding entigen of the excluding query entigen group, wherein the excluding entigen functions

60

is associated with the excluded entigen group to be disassociated with the topic with regards to the response to the query, wherein the identifying the excluding entigen of the excluding query entigen group includes one or more of:

matching a first query entigen of the excluding query entigen group to list of excluding category entigens; detecting an indication of an excluding category from the knowledge database for the first query entigen of the excluding query entigen group; and identifying a set of characteristics from the knowledge database for the first query entigen of the excluding query entigen group, wherein at least one characteristic of the set of characteristics is associated with the excluding category.

3. The method of claim 1 further comprises:

generating a query response phrase utilizing the response entigen group as a representation of the response entigen group; and

outputting, via a user interface of the computing device, at least one of the response entigen group and the query response phrase to a requesting entity associated with the query.

4. The method of claim 1 further comprises:

combining the excluding query entigen group with the response entigen group to produce an incremental entigen group associated with the topic; and

storing the incremental entigen group in the knowledge database to provide expanded knowledge of the topic.

5. The method of claim 1, wherein the recovering the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group comprises:

eliminating the excluding entigen and the excluded entigen group from the excluding query entigen group to produce a remaining subset of query entigens of the excluding query entigen group;

accessing the knowledge database utilizing the remaining subset of query entigens to identify a superset of response entigens of the response entigen group; and removing conflicting response entigens associated with the excluded entigen group in accordance with the excluding entigen from the superset of response entigens to produce the response entigen group.

6. The method of claim 1, wherein the recovering the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group further comprises:

accessing the knowledge database utilizing the excluded entigen group and in accordance with the excluding entigen to produce a search entigen group; and

accessing the knowledge database utilizing the search entigen group to identify the response entigen group.

7. A computing device of a computing system, the computing device comprises:

an interface;

a local memory; and

a processing module operably coupled to the interface and the local memory, wherein the processing module functions to:

determine a set of identigens for each word of a plurality of words of a query of a topic to produce a plurality of sets of identigens, wherein a set of identigens of the plurality of sets of identigens represents one or more different meanings of a word

61

of the plurality of words, wherein each identigen of the set of identigens includes a meaning identifier, an instance identifier, and a time reference, wherein each meaning identifier associated with the set of identigens represents a different meaning of the one or more different meanings of the word of the plurality of words, wherein each time reference provides time information when a corresponding different meaning of the one or more different meanings is valid, wherein a first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words;

interpret, in accordance with identigen pairing rules of a knowledge database, the plurality of sets of identigens to determine a most likely meaning interpretation of the query and produce an excluding query entigen group comprising an excluding entigen of a plurality of query entigens, wherein the knowledge database includes a multitude of entigen groups associated with a multitude of topics, wherein the multitude of topics includes the topic, wherein each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens, wherein the excluding query entigen group represents the most likely meaning interpretation of the query, wherein the excluding entigen functions to exclude at least a portion of an excluded entigen group to be disassociated with the topic with regards to a response to the query, wherein each query entigen of the excluding query entigen group corresponds to a selected identigen of the set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words, wherein each query entigen of the excluding query entigen group represents a single conceivable and perceivable thing in space and time that is independent of language and corresponds to a time reference of the selected identigen associated with the excluding query entigen group, wherein the selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules of the knowledge database; and

recover, via the interface, a response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group, wherein the response entigen group includes one or more response entigens and one or more response entigen relationships between at least some of the one or more response entigens, wherein the set of response entigens includes a portion of the excluding query entigen group, wherein the response entigen group provides the response to the query.

8. The computing device of claim 7, wherein the processing module further functions to:

identify the excluding entigen of the excluding query entigen group, wherein the excluding entigen functions is associated with the excluded entigen group to be disassociated with the topic with regards to the response to the query, wherein the processing module identifies the excluding entigen of the excluding query entigen group by one or more of:

62

matching a first query entigen of the excluding query entigen group to list of excluding category entigens; detecting an indication of an excluding category from the knowledge database for the first query entigen of the excluding query entigen group; and identifying a set of characteristics from the knowledge database for the first query entigen of the excluding query entigen group, wherein at least one characteristic of the set of characteristics is associated with the excluding category.

9. The computing device of claim 7, wherein the processing module further functions to:

generate a query response phrase utilizing the response entigen group as a representation of the response entigen group; and

output, via the interface, at least one of the response entigen group and the query response phrase to a requesting entity associated with the query.

10. The computing device of claim 7, wherein the processing module further functions to:

combine the excluding query entigen group with the response entigen group to produce an incremental entigen group associated with the topic; and

store, via the interface, the incremental entigen group in the knowledge database to provide expanded knowledge of the topic.

11. The computing device of claim 7, wherein the processing module functions to recover the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group by:

eliminating the excluding entigen and the excluded entigen group from the excluding query entigen group to produce a remaining subset of query entigens of the excluding query entigen group;

accessing, via the interface, the knowledge database utilizing the remaining subset of query entigens to identify a superset of response entigens of the response entigen group; and

removing conflicting response entigens associated with the excluded entigen group in accordance with the excluding entigen from the superset of response entigens to produce the response entigen group.

12. The computing device of claim 7, wherein the processing module further functions to recover the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group by:

accessing, via the interface, the knowledge database utilizing the excluded entigen group and in accordance with the excluding entigen to produce a search entigen group; and

accessing, via the interface, the knowledge database utilizing the search entigen group to identify the response entigen group.

13. A non-transitory computer readable memory comprises:

a first memory element that stores operational instructions that, when executed by a processing module, causes the processing module to:

determine a set of identigens for each word of a plurality of words of a query of a topic to produce a plurality of sets of identigens, wherein a set of identigens of the plurality of sets of identigens represents one or more different meanings of a word of the plurality of words, wherein each identigen of the set of identigens includes a meaning identifier, an

63

instance identifier, and a time reference, wherein each meaning identifier associated with the set of identigens represents a different meaning of the one or more different meanings of the word of the plurality of words, wherein each time reference provides time information when a corresponding different meaning of the one or more different meanings is valid, wherein a first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words;

a second memory element that stores operational instructions that, when executed by the processing module, causes the processing module to:

interpret, in accordance with identigen pairing rules of a knowledge database, the plurality of sets of identigens to determine a most likely meaning interpretation of the query and produce an excluding query entigen group comprising an excluding entigen of a plurality of query entigens, wherein the knowledge database includes a multitude of entigen groups associated with a multitude of topics, wherein the multitude of topics includes the topic, wherein each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens, wherein the excluding query entigen group represents the most likely meaning interpretation of the query, wherein the excluding entigen functions to exclude at least a portion of an excluded entigen group to be disassociated with the topic with regards to a response to the query, wherein each query entigen of the excluding query entigen group corresponds to a selected identigen of the set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words, wherein each query entigen of the excluding query entigen group represents a single conceivable and perceivable thing in space and time that is independent of language and corresponds to a time reference of the selected identigen associated with the excluding query entigen group, wherein the selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules of the knowledge database; and

a third memory element that stores operational instructions that, when executed by the processing module, causes the processing module to:

recover a response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group, wherein the response entigen group includes one or more response entigens and one or more response entigen relationships between at least some of the one or more response entigens, wherein the set of response entigens includes a portion of the excluding query entigen group, wherein the response entigen group provides the response to the query.

14. The non-transitory computer readable memory of claim 13 further comprises:

a fourth memory element stores operational instructions that, when executed by the processing module, causes the processing module to:

64

identify the excluding entigen of the excluding query entigen group, wherein the excluding entigen functions is associated with the excluded entigen group to be disassociated with the topic with regards to the response to the query, wherein the processing module identifies the excluding entigen of the excluding query entigen group by one or more of:

matching a first query entigen of the excluding query entigen group to list of excluding category entigens;

detecting an indication of an excluding category from the knowledge database for the first query entigen of the excluding query entigen group; and

identifying a set of characteristics from the knowledge database for the first query entigen of the excluding query entigen group, wherein at least one characteristic of the set of characteristics is associated with the excluding category.

15. The non-transitory computer readable memory of claim 13 further comprises:

a fifth memory element stores operational instructions that, when executed by the processing module, causes the processing module to:

generate a query response phrase utilizing the response entigen group as a representation of the response entigen group; and

output at least one of the response entigen group and the query response phrase to a requesting entity associated with the query.

16. The non-transitory computer readable memory of claim 13 further comprises:

a sixth memory element stores operational instructions that, when executed by the processing module, causes the processing module to:

combine the excluding query entigen group with the response entigen group to produce an incremental entigen group associated with the topic; and

store the incremental entigen group in the knowledge database to provide expanded knowledge of the topic.

17. The non-transitory computer readable memory of claim 13, wherein the processing module functions to execute the operational instructions stored by the third memory element to cause the processing module to recover the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group by:

eliminating the excluding entigen and the excluded entigen group from the excluding query entigen group to produce a remaining subset of query entigens of the excluding query entigen group;

accessing the knowledge database utilizing the remaining subset of query entigens to identify a superset of response entigens of the response entigen group; and removing conflicting response entigens associated with the excluded entigen group in accordance with the excluding entigen from the superset of response entigens to produce the response entigen group.

18. The non-transitory computer readable memory of claim 13, wherein the processing module further functions to execute the operational instructions stored by the third memory element to cause the processing module to recover the response entigen group for the query from the knowledge database utilizing the excluding query entigen group and based on the excluding entigen and the excluded entigen group by:

65

accessing the knowledge database utilizing the excluded
entigen group and in accordance with the excluding
entigen to produce a search entigen group; and
accessing the knowledge database utilizing the search
entigen group to identify the response entigen group. 5

* * * * *

66